

Manoeuvre Pattern-of-Life Prediction for Space Domain Awareness

Learning a station-keeping control law from noisy tracking data:
drag estimation, Gaussian-mixture burn detection, SINDy/BINDy and Gaussian-process
models, and an end-to-end assessment

UROP project, Imperial College London

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Abstract

A Starlink-like satellite keeps its mean semi-major axis (SMA) inside a 500 m deadband by firing a low-thrust engine whenever the SMA decays to the lower edge. We simulate 400 days of this behaviour with two-body gravity, J_2 , drag, lunar and solar attraction and solar radiation pressure, and then try to recover everything an observer would need to *predict* the next manoeuvre from noisy state measurements alone. The analysis is a four-stage chain: (1) estimate the drag coefficient c_d (finite-difference regression, energy method, shooting, extended Kalman filter, spectral “force spectroscopy”); (2) detect the burns with a Gaussian mixture model (GMM) on the unmodelled acceleration; (3) learn the natural and thrust dynamics and the switching law with sparse identification (SINDy), its Bayesian variant (BINDy) and Gaussian processes (GP); (4) forecast future burns. Each stage is scored against the truth, an error budget swaps the truth into one stage at a time, every stage is re-run at sampling steps from 6 s to 10 min, and 16 headline numbers are checked against closed-form formulae (all agree within 1%, or obey their bound). The GMM finds all 92 test burns with no false events from 6-s to 10-min data; the drag coefficient is recovered to 0.03–0.08% at every sampling step; learning the thrust law and the natural law *separately* cuts the year-ahead forecast error from 23 hours (joint regression) to about one hour; the band edges, not the physics coefficients, dominate the remaining error. With variable space weather a learned law with a drag proxy halves the error of a periodic schedule, and the forecasts reduce missed conjunctions, lost tracks and sensor time. Methods are ranked for each stage of the pipeline.

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1 Introduction

1.1 The brief

The UK Space Agency challenge “Manoeuvre Pattern of Life Prediction for SDA” [1] starts from the growth of the active satellite population: of $\sim 50\,000$ tracked objects about 15 000 are active and may manoeuvre, and mega-constellations will multiply this by 2035. Unpredicted manoeuvres break three core Space Domain Awareness (SDA) functions:

1. **close-approach (conjunction) screening**, which assumes predicted trajectories are accurate over the screening window — in the brief’s Figure 1 a Starlink satellite keeping its SMA inside a 500 m deadband moves its miss distance by *hundreds of km in a week* [2];
2. **track association**, because a manoeuvre makes the predicted position wrong, so new observations cannot be linked to the right catalogue object [4];
3. **sensor scheduling**, because the uncertainty of an unpredicted manoeuvre has to be removed by tasking extra observations [3].

Detecting *past* manoeuvres is well studied; the gap the brief identifies is predicting *future* manoeuvres by **reconstructing the control law from time series of measurements** [5].

1.2 The question and the approach

Question

From noisy position and velocity data, can the station-keeping law of a satellite (“burn when the mean SMA reaches the lower edge, stop at the upper edge”) be learned well enough to forecast its future burns — and which method is best at each step of that chain?

The approach is a controlled experiment: a simulator whose truth is known, a chain of estimators each of which can be scored against that truth, and studies of how the chain degrades with noise, data length and sampling. Figure 1 shows the chain; Sections 4–8 follow it stage by stage.

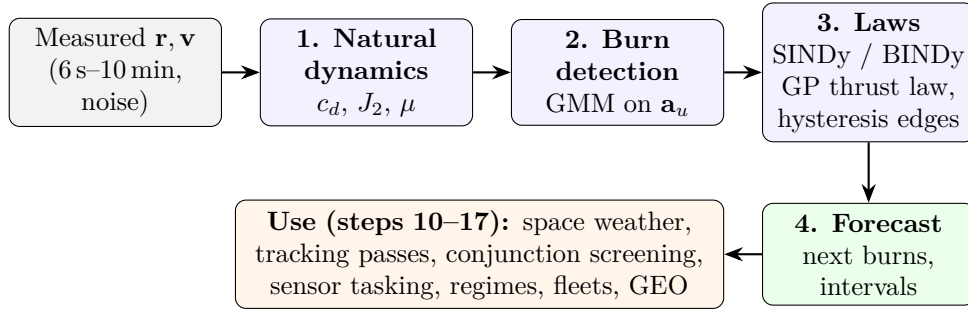


Figure 1: The analysis chain. Stage 1 removes the known physics so that what remains is the thrust; stage 2 finds the burns in that residual; stage 3 learns the drag, thrust and switching laws from the labelled data; stage 4 runs the learned law forward. Steps 10–17 test the forecasts in operational settings.

1.3 Contributions

1. A 400-day, 6-s simulation of a smooth (ODE-friendly) deadband controller with memory, built on the J_2 -corrected mean SMA, reproducing the brief’s Figure 1 (Section 2).
2. A comparison of five drag estimators against noise, data length and sampling (Section 4).
3. A GMM burn detector on the unmodelled acceleration, explained component by component, with a sampling stress test and an alternative feature for coarse data (Section 5).

4. SINDy/BINDy and GP models of the thrust and natural laws, learned together and apart, plus the hysteresis switching law (Sections 6–7).
5. An end-to-end assessment: stage accuracy, an error budget, sampling robustness of the whole chain, and closed-form checks of the headline numbers (Sections 8 and 11).
6. Extensions to the operational questions of the brief (Section 9) and a ranking of all methods (Section 10).

1.4 Conventions

Lengths are in Earth radii ($\text{DU} = 6371 \text{ km}$) and time in $\text{TU} = 1/\omega_{\oplus} = 13\,713.44 \text{ s}$ inside the code, so $\mu = \mu_{\oplus} \text{ TU}^2/\text{DU}^3 = 289.873$; all results are given in SI units. TRAIN is days 0–40, TEST days 40–400; forecast horizons are 15, 30, 60, 120, 240 and 360 days after day 40. The reference noise is 0.1 m in position and 0.5 mm/s in velocity. Every number quoted comes from the latest run of `python main.py (outputs/results.json)`.

2 Simulation: physics, controller and data

2.1 Equations of motion

The state is $\mathbf{x} = (\mathbf{r}, \mathbf{v})$, plus a thruster state s for the controlled runs:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = -\frac{\mu \mathbf{r}}{r^3} + \mathbf{a}_{J_2} + \mathbf{a}_{\text{drag}} + \mathbf{a}_{\text{Moon}} + \mathbf{a}_{\text{Sun}} + \mathbf{a}_{\text{SRP}} + \mathbf{a}_{\text{thrust}}. \quad (1)$$

Oblateness. With the *equatorial* radius $R_e = 6378.137 \text{ km}$ and $J_2 = 1.08263 \times 10^{-3}$ [6],

$$\mathbf{a}_{J_2} = \frac{3}{2} \frac{J_2 \mu R_e^2}{r^4} \begin{pmatrix} \frac{x}{r} (5 \frac{z^2}{r^2} - 1) \\ \frac{y}{r} (5 \frac{z^2}{r^2} - 1) \\ \frac{z}{r} (5 \frac{z^2}{r^2} - 3) \end{pmatrix} = -\nabla U_{J_2}, \quad U_{J_2} = \frac{\mu J_2 R_e^2}{2r^3} (3 \sin^2 \phi - 1). \quad (2)$$

Drag. A lumped ballistic coefficient, $\mathbf{a}_{\text{drag}} = -c_d |\mathbf{v}| \mathbf{v}$, with constant density in steps 1–9 (time-varying in steps 10–16). Its value gives 5.3 m/hour of SMA decay at 550 km.

Third bodies and SRP. Tidal attraction $\mu_b ((\mathbf{r}_b - \mathbf{r})/|\mathbf{r}_b - \mathbf{r}|^3 - \mathbf{r}_b/r_b^3)$ for the Moon and the Sun on circular orbits, and a Sun-facing flat plate for SRP with $PC_{RA}/m = 4.56 \mu\text{N}/\text{m}^2 \times 1.3 \times 5/260 \text{ m}^2/\text{kg}$ [7].

Integration. DOP853 at tolerance 10^{-12} for the free runs; LSODA [8] for the controlled runs, because the tanh switches make the equations stiff for a few seconds at each burn edge.

2.2 Force budget and the J_2 check

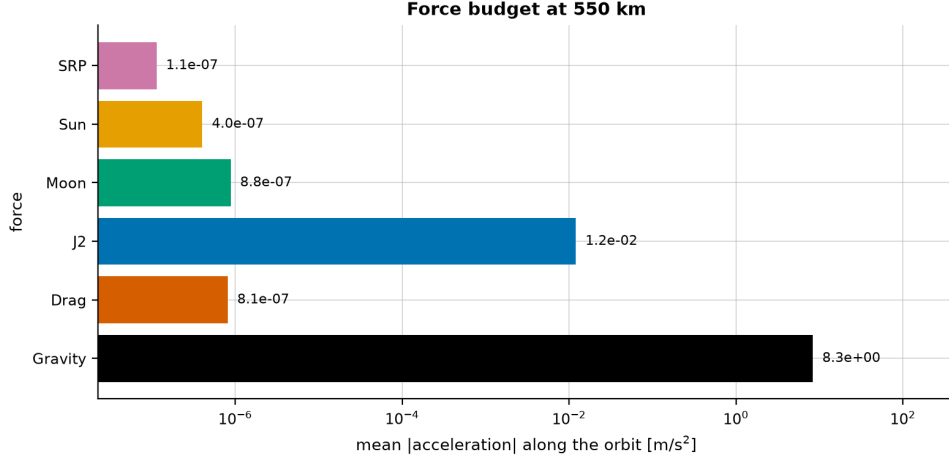


Figure 2: Mean magnitude of each acceleration along the 550 km orbit. J_2 ($1.2 \times 10^{-2} \text{ m/s}^2$) is four orders above the Moon (8.8×10^{-7}), drag (8.1×10^{-7}), Sun (4.0×10^{-7}) and SRP (1.1×10^{-7}); the thruster (2×10^{-4}) sits between J_2 and drag, which is what makes it detectable once J_2 is removed.

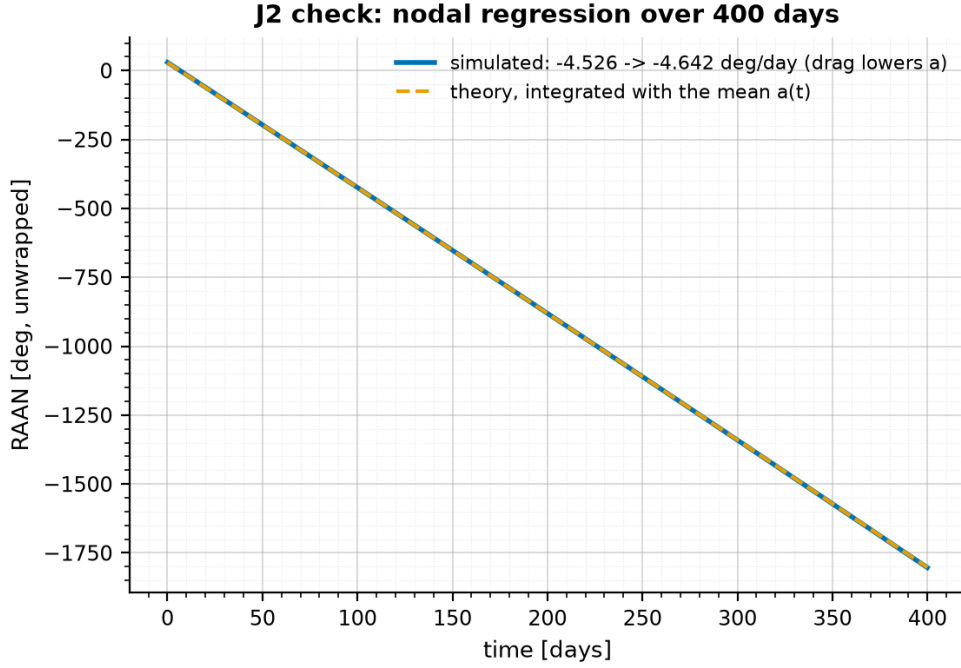


Figure 3: Right ascension of the ascending node of the natural run against the theoretical nodal regression integrated with the run's own drag-decaying $a(t)$. They agree to 0.04% after 400 days.

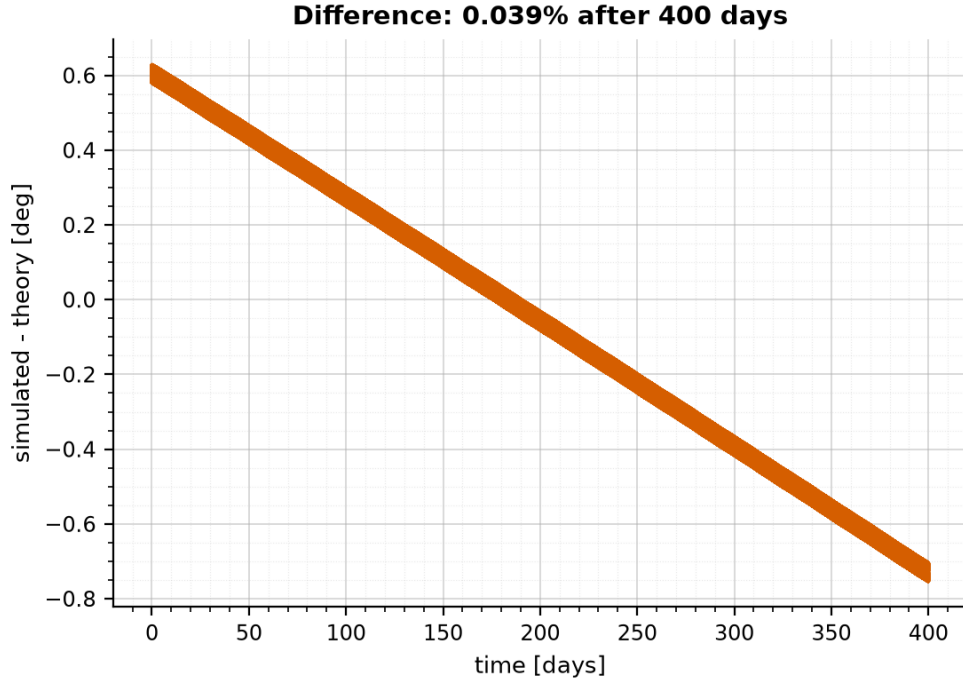


Figure 4: Simulated minus theoretical node angle: 0.039% of the total regression after 400 days.

Calculation check. $\dot{\Omega} = -\frac{3}{2}nJ_2(R_e/a)^2 \cos i = -4.503^\circ/\text{day}$ at the mean SMA of the controlled run, against $-4.505^\circ/\text{day}$ measured (0.04%). The rate in Figure 3 grows from -4.526 to $-4.642^\circ/\text{day}$ because the uncontrolled orbit decays by 51 km. A first version used the mean Earth radius in the J_2 term, making it $(6371/6378.137)^2 - 1 = -0.22\%$ too weak; this check found and fixed it.

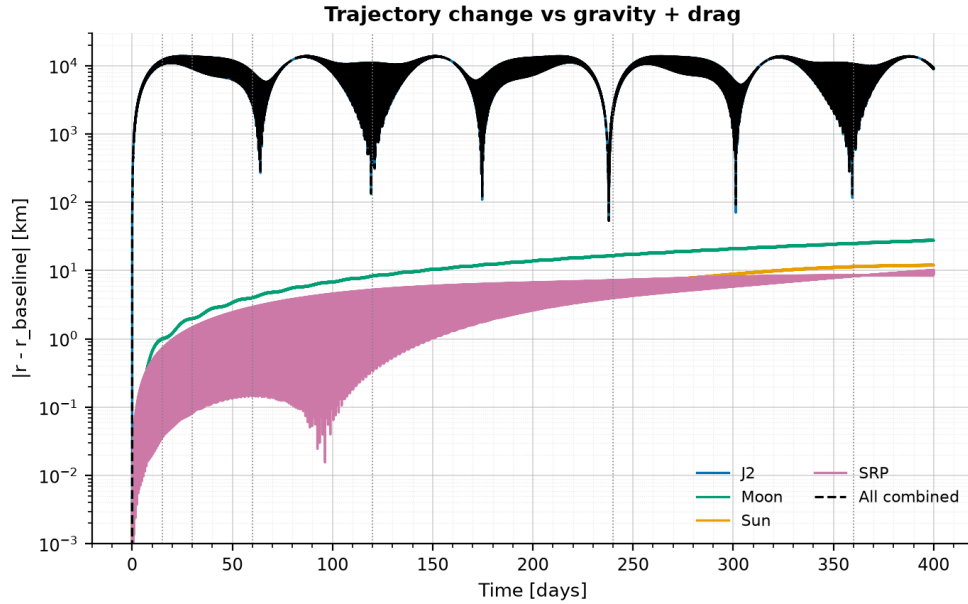


Figure 5: Distance between an orbit with and without each force, against time. Leaving out J_2 gives an orbit-sized error within days; the Moon, Sun and SRP each move the satellite by 4–12 km in a year.

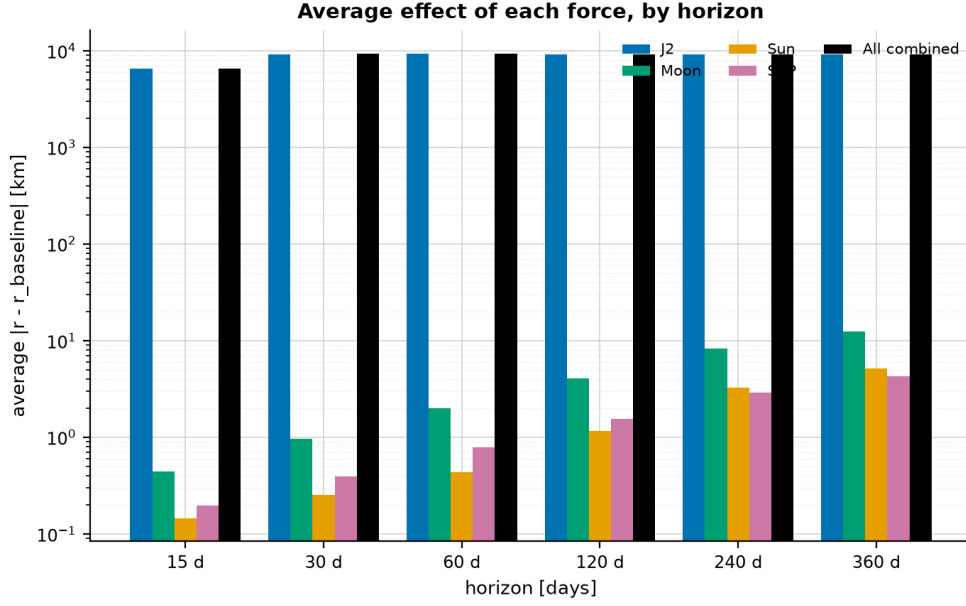


Figure 6: Average position change caused by each force, by horizon (15–360 days).

2.3 The station-keeping controller

Why the mean SMA. The osculating SMA $a_{\text{osc}} = -\mu/(2\varepsilon)$ with $\varepsilon = v^2/2 - \mu/r$ swings by 12 km per orbit under J_2 , 25 times the band. Adding the J_2 potential to the energy removes the swing,

$$a_{\text{mean}} = -\frac{\mu}{2\varepsilon_{J_2}}, \quad \varepsilon_{J_2} = \frac{v^2}{2} - \frac{\mu}{r} + U_{J_2}(\mathbf{r}), \quad (3)$$

and ε_{J_2} changes only through drag and thrust.

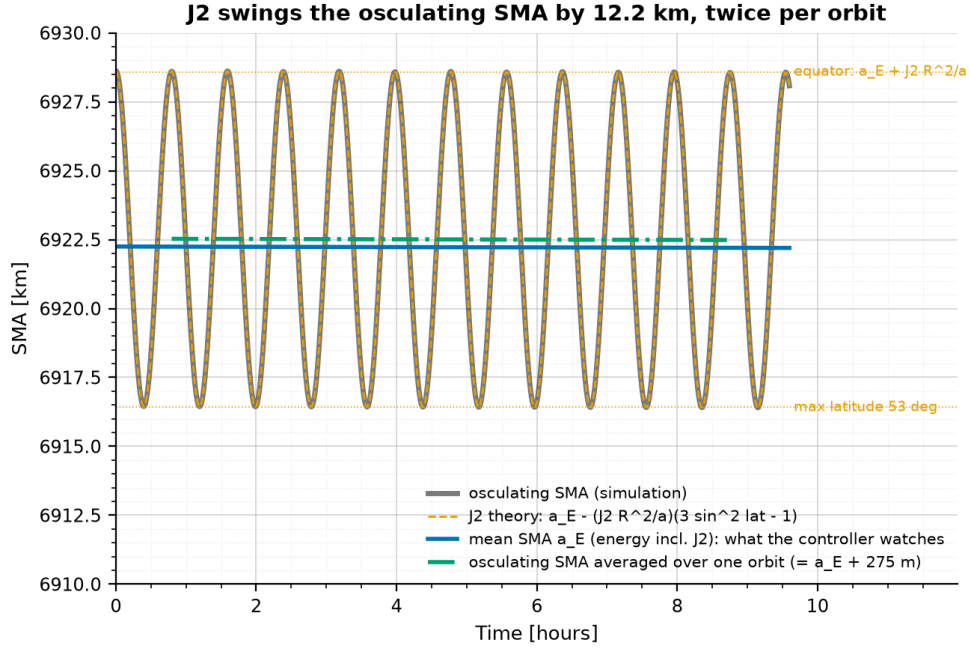


Figure 7: Osculating SMA (12 km swing, twice per orbit) and the mean SMA used by the controller (flat to centimetres between burns).

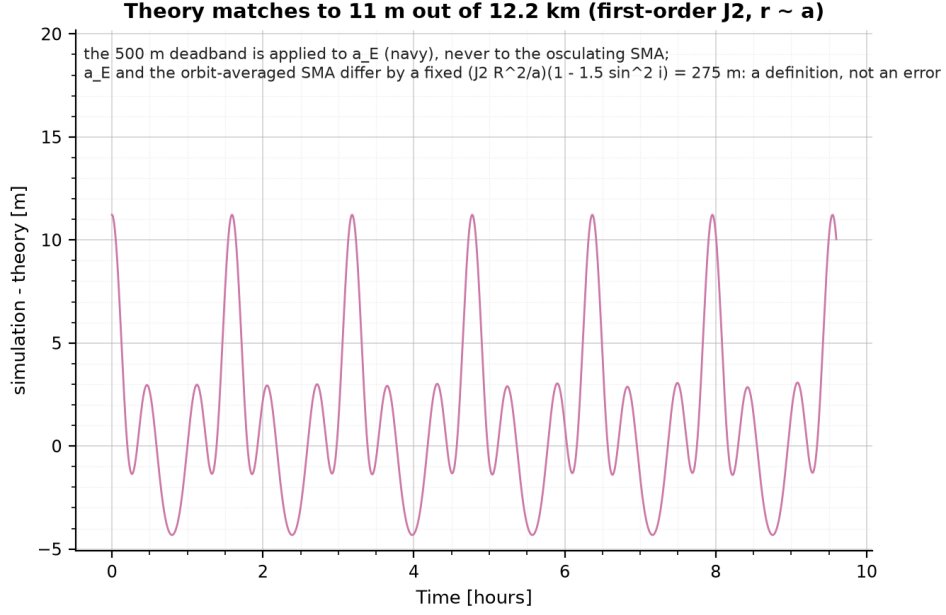


Figure 8: Simulated minus first-order theory for the osculating swing: 11 m out of 12.2 km.

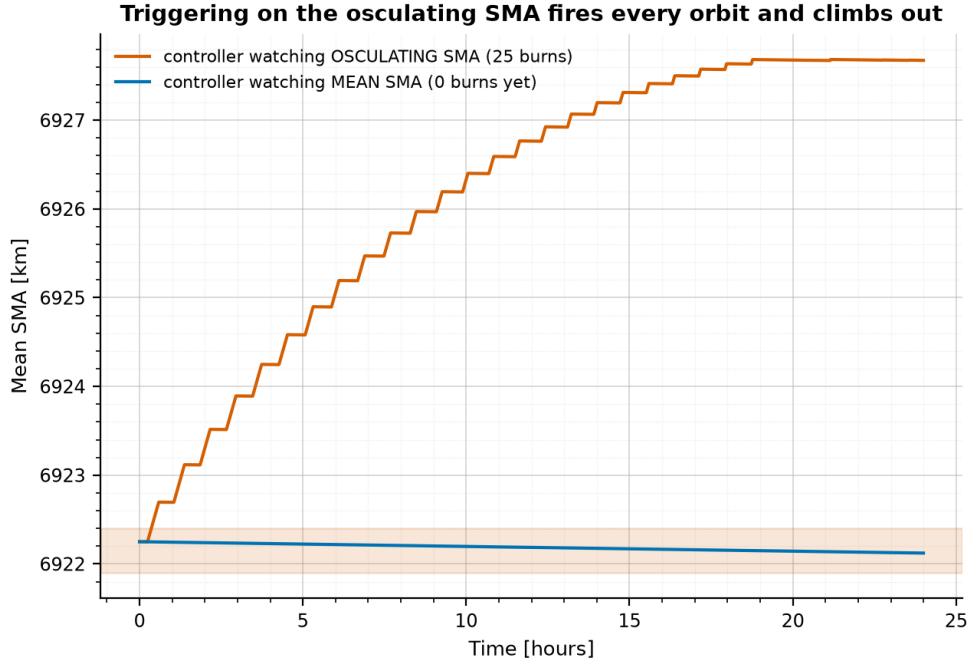


Figure 9: A controller that triggers on the osculating SMA fires every orbit and climbs out of the band: why the mean SMA is needed.

Calculation check. Short-period swing of the osculating SMA: peak to peak $3J_2 R_e^2 \sin^2 i/a = 12.17$ km, against 12.22 km in the simulation (0.4%; the formula is first order in J_2).

The smooth law with memory. With $\sigma(x) = \frac{1}{2}(1 + \tanh(x/w))$, $w = 5$ m,

$$\tau \dot{s} = \sigma(a_L - a_{\text{mean}})(1 - s) - \sigma(a_{\text{mean}} - a_U)s + \text{latch}(s), \quad \mathbf{a}_{\text{thrust}} = T \theta(s) \hat{\mathbf{v}}, \quad \theta(s) = \sigma\left(\frac{s - 0.9}{0.02}\right), \quad (4)$$

with $a_L = 6921.9$ km, $a_U = 6922.4$ km, $\tau = 60$ s and $T = 2 \times 10^{-4}$ m/s². Between the edges both switches are off and s keeps its value: a **hysteresis**, so the same SMA can belong to a coast or a

burn.

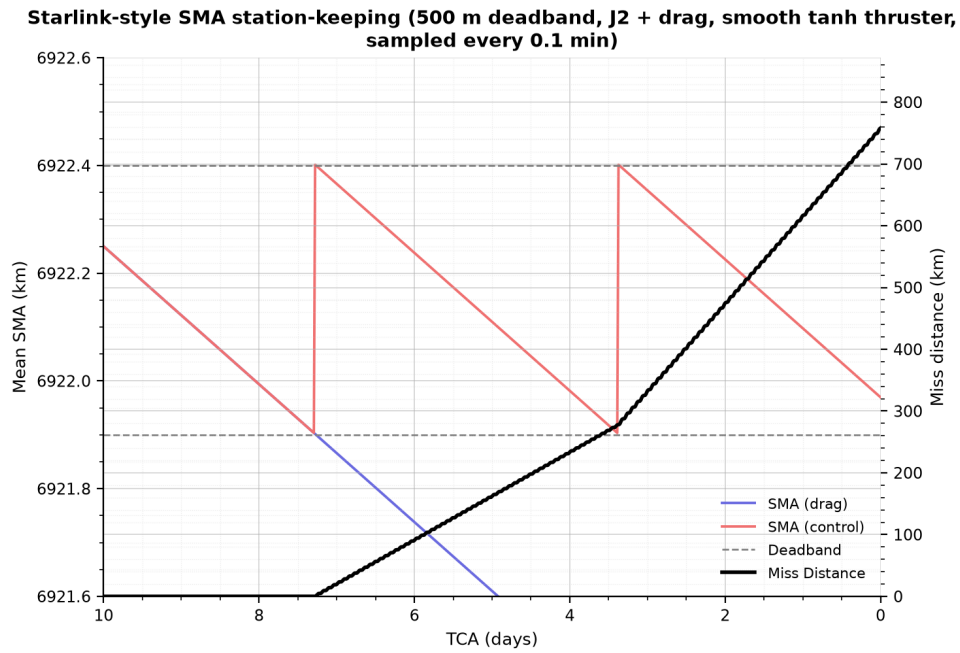


Figure 10: Reproduction of the brief's Figure 1: mean SMA of the controlled satellite and of its drag-only twin, and the distance between them.

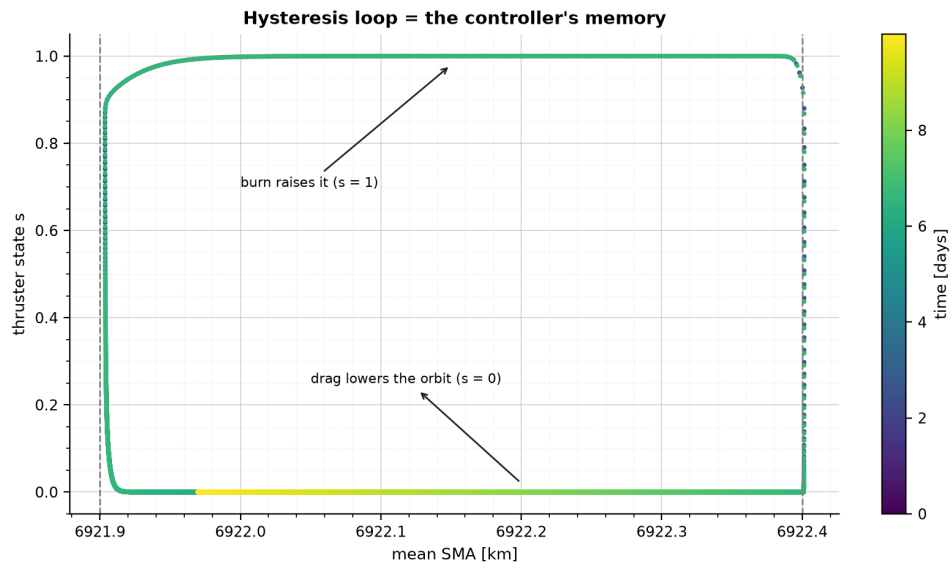


Figure 11: Thruster state against mean SMA: the burn starts at the lower edge and stops at the upper edge, tracing a loop (memory).

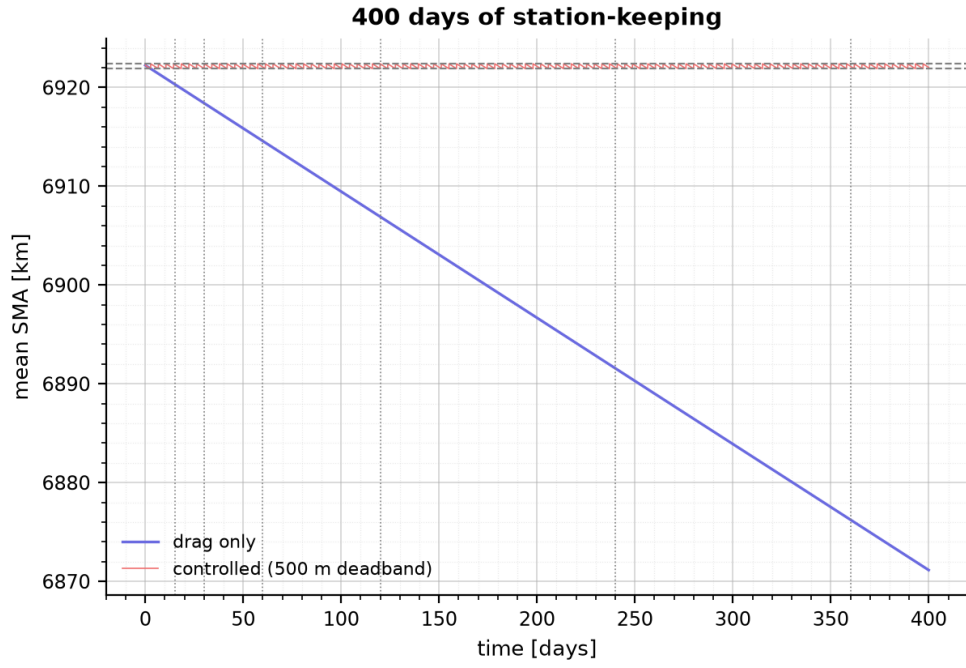


Figure 12: 400 days of station keeping: the mean SMA stays inside $[6921.904, 6922.402]$ km, with 102 burns.

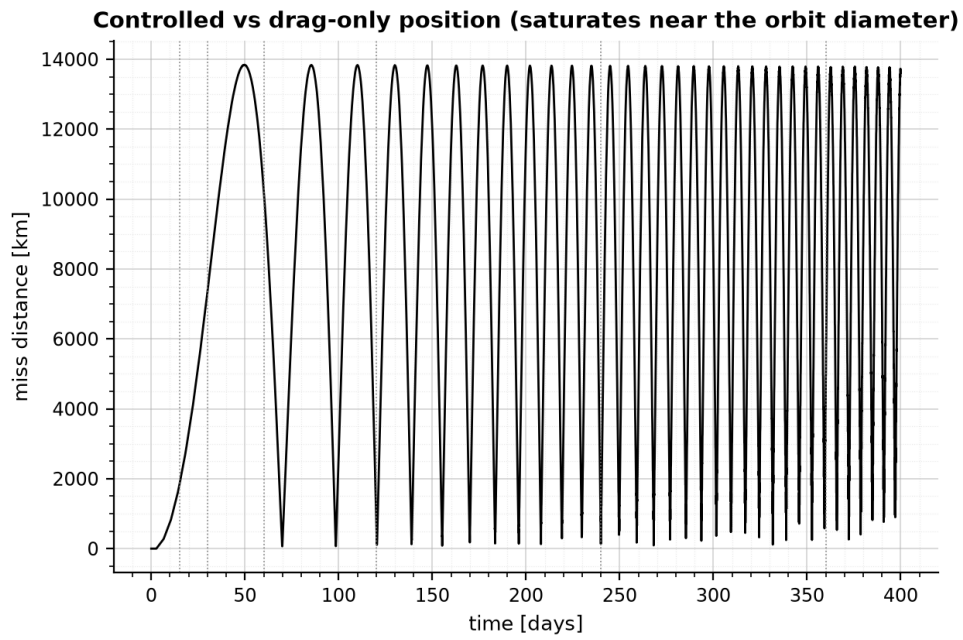


Figure 13: Distance between the controlled satellite and its drag-only twin: it saturates near the orbit diameter as the two drift to different phases.

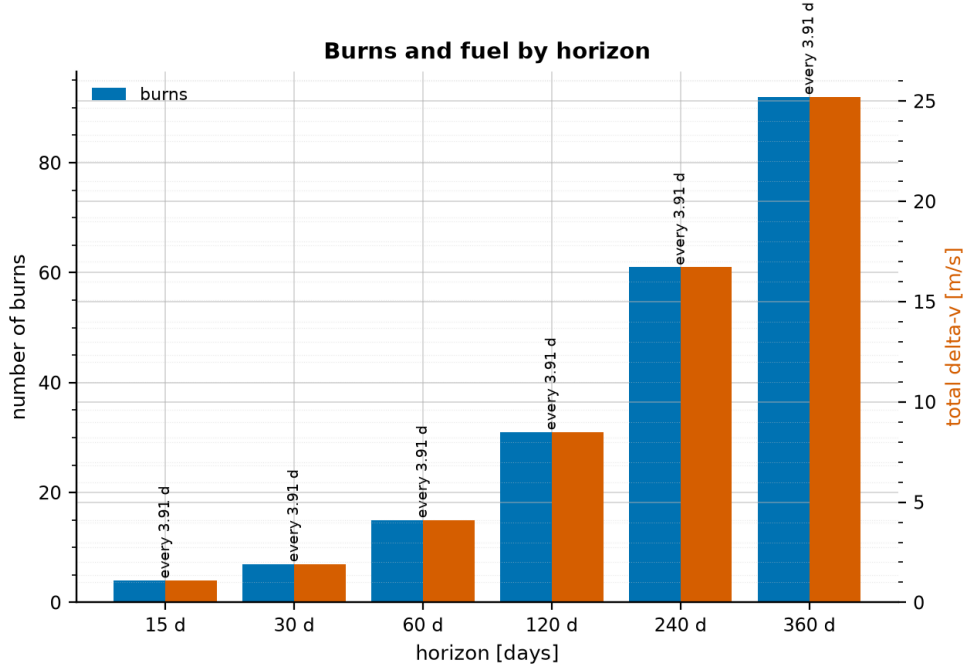


Figure 14: Number of burns and total Δv by horizon.

2.4 What the simulation produced

102 burns, one every **3.9051 days**, each **22.8 min** long, $\Delta v = 0.273$ m/s per burn and 27.8 m/s in 400 days. Fuel hardly depends on the band width (3.8–4.2 m/s per 60 days for 250–1000 m bands): the thruster replaces what drag removes; a narrower band only means more, smaller burns.

Calculation check. All from first principles with $a = 6922.15$ km, $v = \sqrt{\mu/a} = 7.588$ km/s, $n = \sqrt{\mu/a^3}$:

- coast decay $\dot{a} = -2c_d\sqrt{\mu a}$ (circular orbit, $a_t = -c_d v^2$) = 5.333 m/h vs 5.335 m/h measured (0.05%);
- burn climb $\dot{a} = 2aT/v - k = 1308$ m/h vs 1311 m/h measured (0.2%; the ramps);
- $\Delta v = v \Delta a / (2a)$ with $\Delta a = \text{band} + \text{decay during the burn}$: 0.2741 vs 0.2730 m/s (0.4%);
- burn duration $\Delta v / T = 1370$ s vs 1368 s (0.2%);
- period $P = w/k + t_b = 3.9051$ d vs 3.9051 d (0.00%), with w the effective band (498 m);
- Δv budget $400 \text{ d} / P \times \Delta v = 28.1$ vs 27.8 m/s (0.8%).

2.5 The data sets

Run	Content	Used by
natural run	all forces, no thruster, 1-min samples, 400 days	stage 1 (steps 1–4)
divergence runs	gravity + drag, plus each extra force on its own	step 1
controlled run	smooth deadband, J_2 + drag, 6-s samples, 400 days, 102 burns	stages 2–4 (steps 5–9, 19, 20)
drag-only run	the controlled orbit without a thruster	step 5
mean-element runs	calibrated fast model (Section 9)	steps 10–16

3 Method selection: bias–variance tuning (step 0)

Every adjustable setting (stencil order, smoothing window, number of mixture components, sparsity threshold) is chosen from data rather than by hand. Over repeated noise draws the mean squared error of an estimate splits into [18]

$$\mathbb{E}[(\hat{y} - y)^2] = (\mathbb{E}[\hat{y}] - y)^2 + \mathbb{E}[(\hat{y} - \mathbb{E}[\hat{y}])^2] = \text{bias}^2 + \text{variance}, \quad (5)$$

and the tuned setting minimises the sum (Table 1). The choices are saved to a settings file in the cache folder and read by every later step.

Table 1: Settings chosen by bias–variance tuning.

Setting	Default	Tuned	Objective
FD stencil order	6	per step: 1 min→4, 2 min→2, 5/10 min→6	MSE of c_d from 10 m-noise positions
Savitzky–Golay window / order	21 / 5	61 / 5	MSE of the classifier’s $\dot{\mathbf{v}}$
stacking window (step 8)	21 / 5	41 / 5	$\text{bias}^2 + \text{variance}/N$ for N stacked burns
GMM components K	4	4	Brier score of $P(\text{burn} \mid x)$, one-standard-error rule
SMA derivative window	41	41	MSE of the learned coast and burn rates
STLSQ threshold	0.05	0.05	MSE of predicted $\dot{a}$ on TEST (sparsest within 1%)

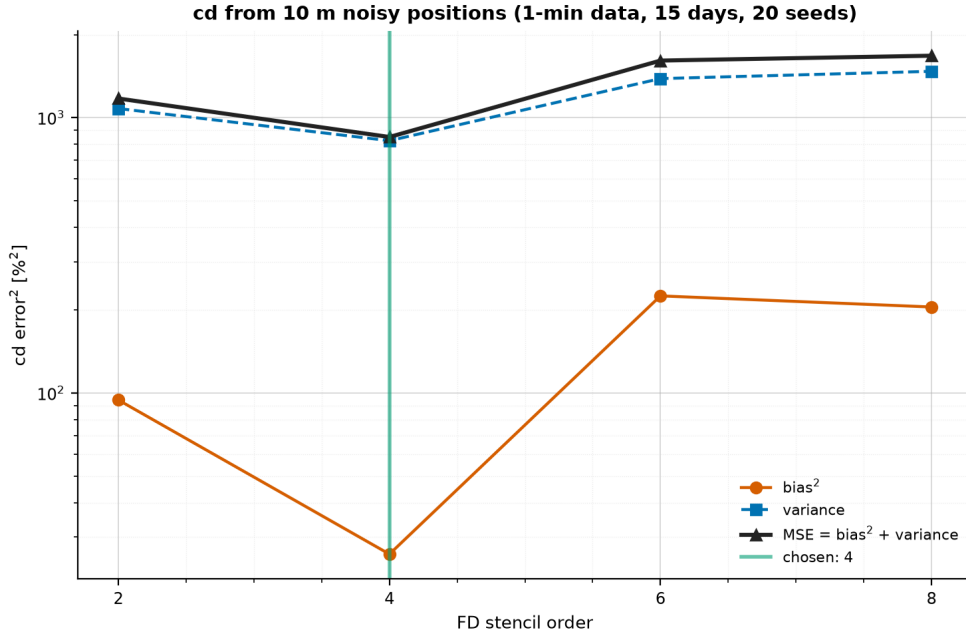


Figure 15: Bias², variance and MSE of the finite-difference c_d against the stencil order. Higher orders reduce truncation bias but amplify noise.

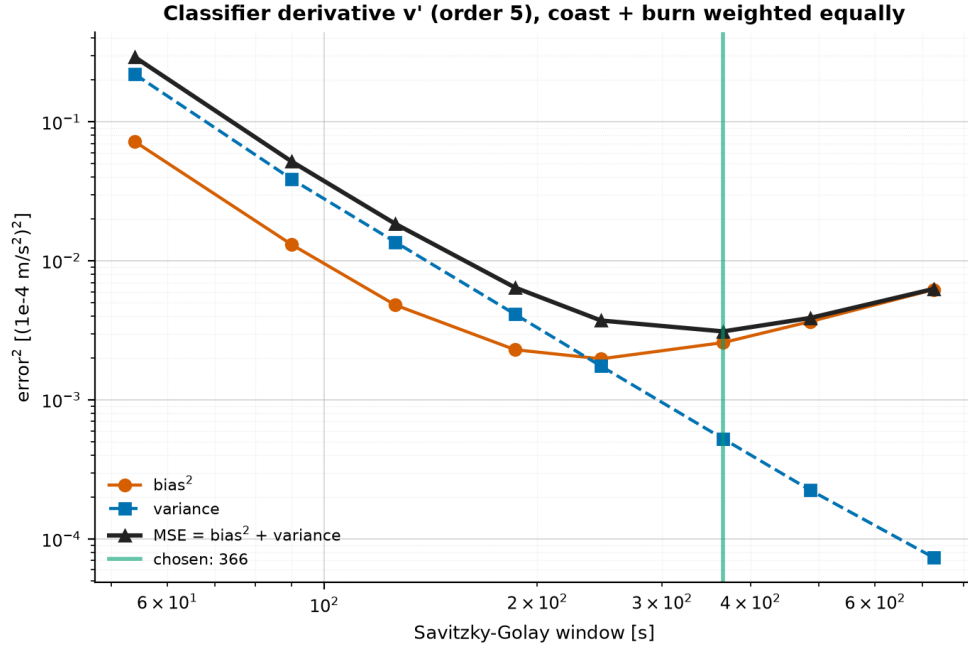


Figure 16: The same decomposition for the Savitzky–Golay window of the classifier’s velocity derivative: short windows are noisy, long windows smear the burn edges; 61 samples (6 min) is best.

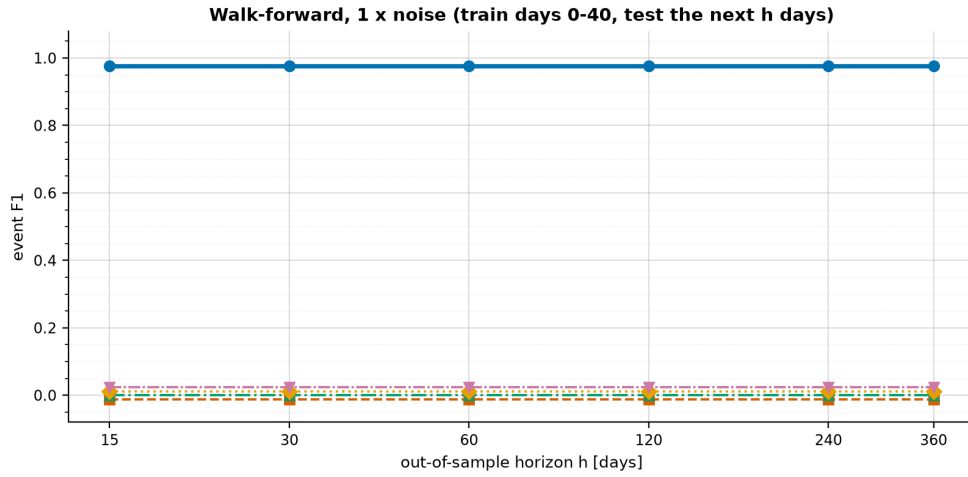


Figure 17: Walk-forward and walk-backward test of the derivative methods: a GMM trained on one 40-day window and tested on the following (or preceding) 15–360 days. Only Savitzky–Golay keeps $F1 = 1$ in all 12 test periods at the reference noise.

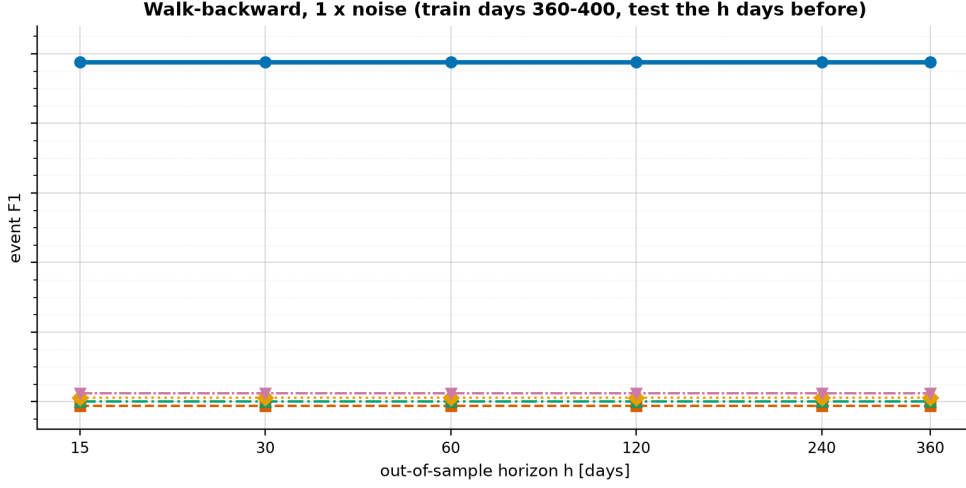


Figure 18: Walk-backward test at the reference noise: train on days 360–400, test on the h days before.

4 Stage 1: estimating the natural dynamics and c_d (steps 2–4)

Before a burn can be seen, the known physics has to be removed, and the one unknown that matters most for the forecast is the drag coefficient. Five estimators were compared.

4.1 Methods

Finite-difference (FD) regression. Differentiate the positions twice with a central stencil of order p (truncation error $O(h^p)$, noise amplification $\propto \sigma/h^2$) and solve a linear least-squares problem in which each unknown multiplies a known shape:

$$\ddot{\mathbf{r}} + \frac{\mu \mathbf{r}}{r^3} = [c_d, J_2, \mu_{\text{Moon}}, \mu_{\text{Sun}}, PCRA/m] \cdot [\text{shapes}], \quad (6)$$

with the columns scaled to unit norm (they differ by ten orders of magnitude).

Energy method. One derivative only: the mean SMA decays at $\dot{a} = -(2a^2/\mu)c_d|\mathbf{v}|^3$, so c_d follows from the slope of a straight line through the whole SMA history, which averages all the noise.

Shooting. Propagate with trial parameters and fit the whole trajectory by Levenberg–Marquardt [28], started from the energy answer.

Extended Kalman filter (EKF). State $(\mathbf{r}, \mathbf{v}, c_d)$, propagated with its state-transition matrix and updated by every measurement [9].

Force spectroscopy. Identify each force by the frequency band where it holds $\geq 90\%$ of its power and read its strength from the band area (Parseval) [29].

4.2 Results

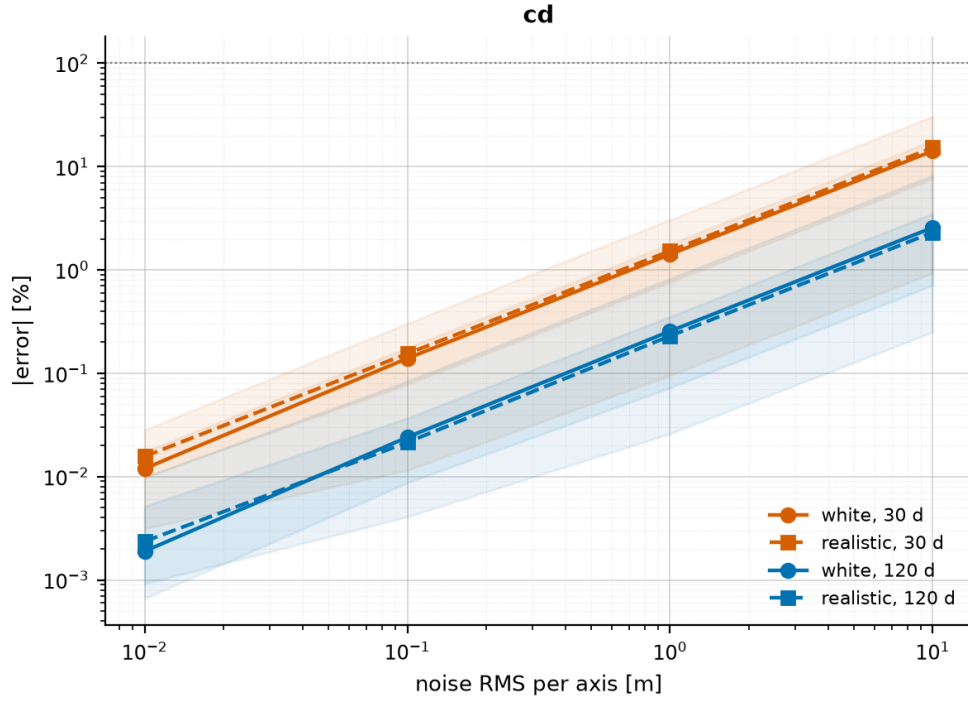


Figure 19: c_d error of the FD regression and the energy method against white position noise (15 days of 1-min data). FD: 0.009 / 0.64 / 6.3 / 64% at 0 / 1 / 10 / 100 m; energy: 0.006 / 0.008 / 0.035 / 0.35%.

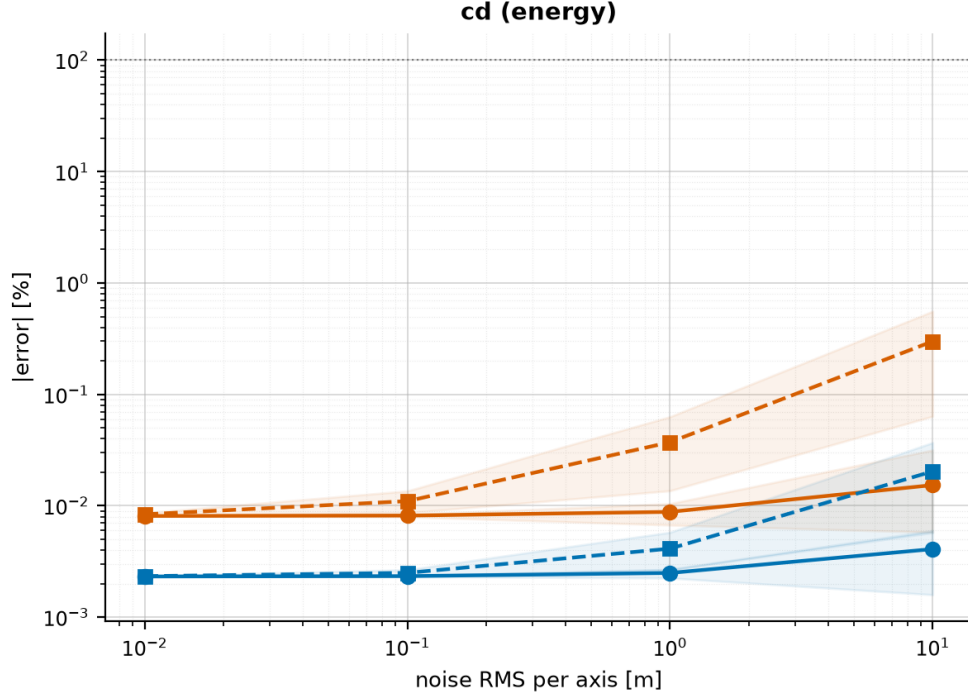


Figure 20: c_d error of the energy method against white position noise: two orders of magnitude below FD regression.

Calculation check. A second difference turns white noise σ_r into $\sigma_a = \sqrt{6} \sigma_r / h^2$, so the FD c_d error should be proportional to the noise: 10 times the 1 m error (30 days) is 14.14%, the measured 10 m error is 14.16% (0.1%). The energy method fits one slope through N samples, so its error falls as $\sigma/\sqrt{N}$ and is 180 times smaller.

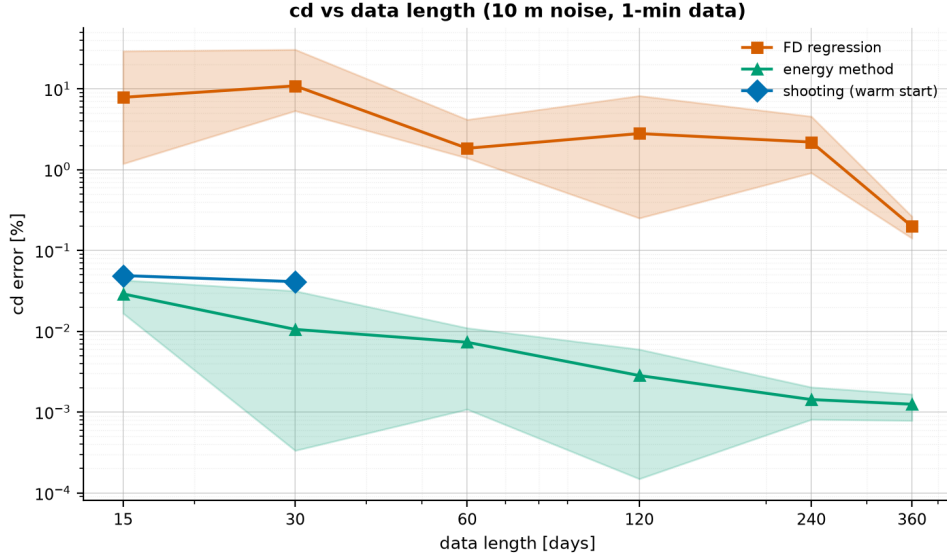


Figure 21: c_d error against data length at 10 m noise: the energy method improves from 0.03% (15 days) to 0.001% (360 days), FD from 7.9% to 0.2%.

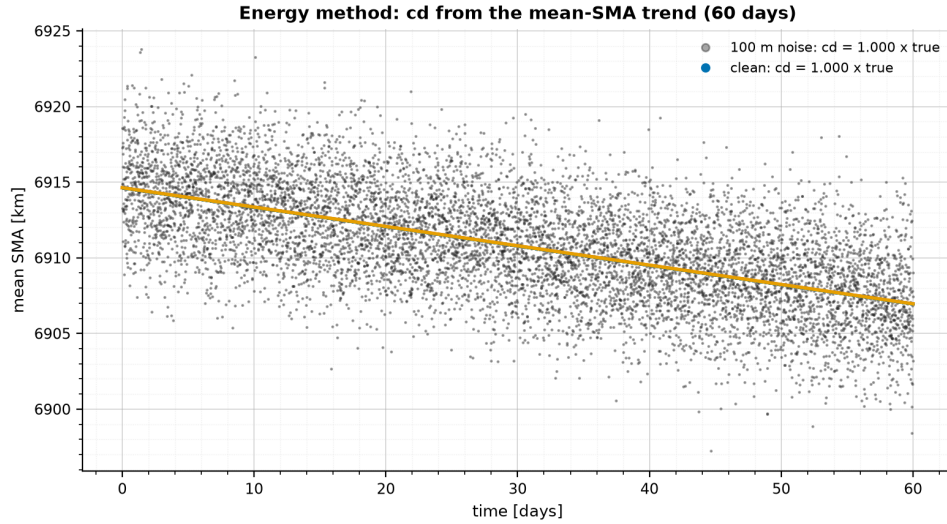


Figure 22: The energy method: the mean SMA of the natural run and the fitted decay line whose slope gives c_d .

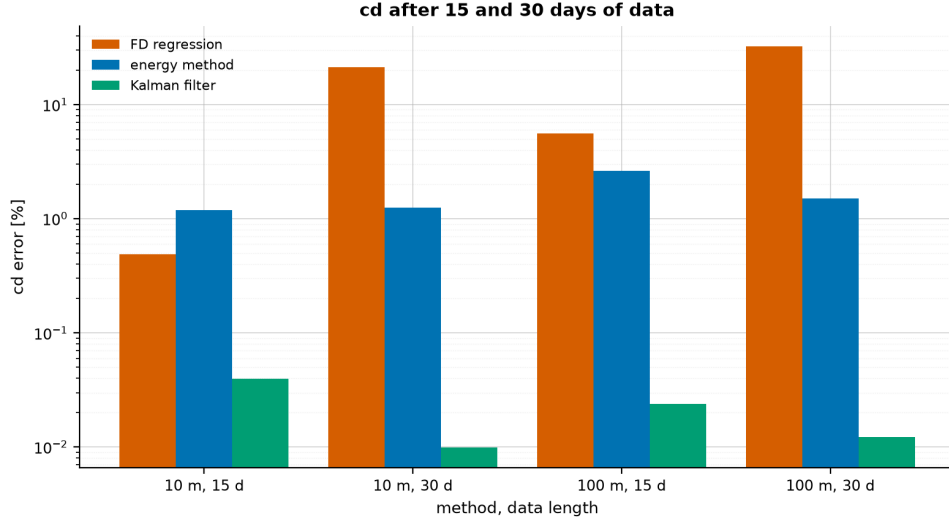


Figure 23: On coarse (10-min) data with 10 m noise the EKF reaches 0.04% after 15 days and 0.01% after 30, the energy method 1.2–2.6%, FD 0.5–32%.

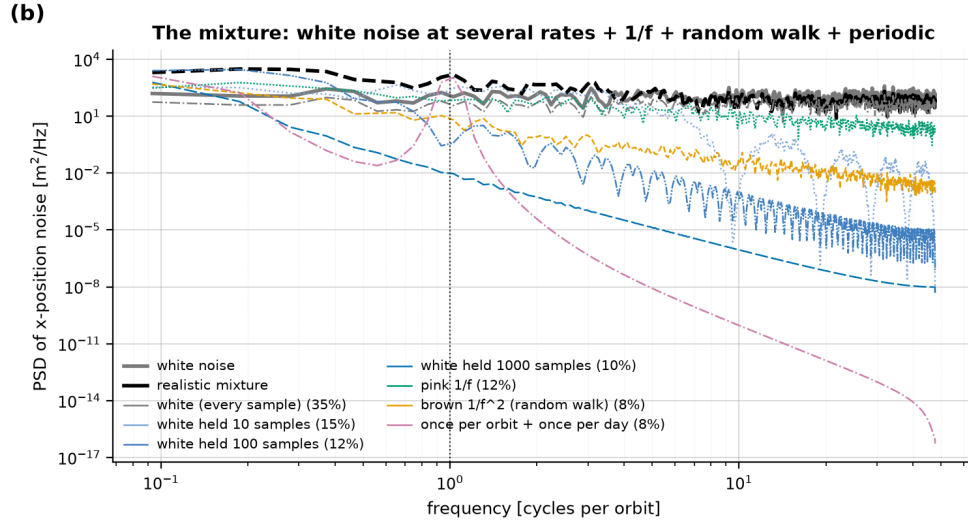


Figure 24: Realistic (coloured) noise at the same RMS: slow noise (random walk, once-per-day terms) mimics a drag trend and makes the energy method 4 times worse; FD barely changes.

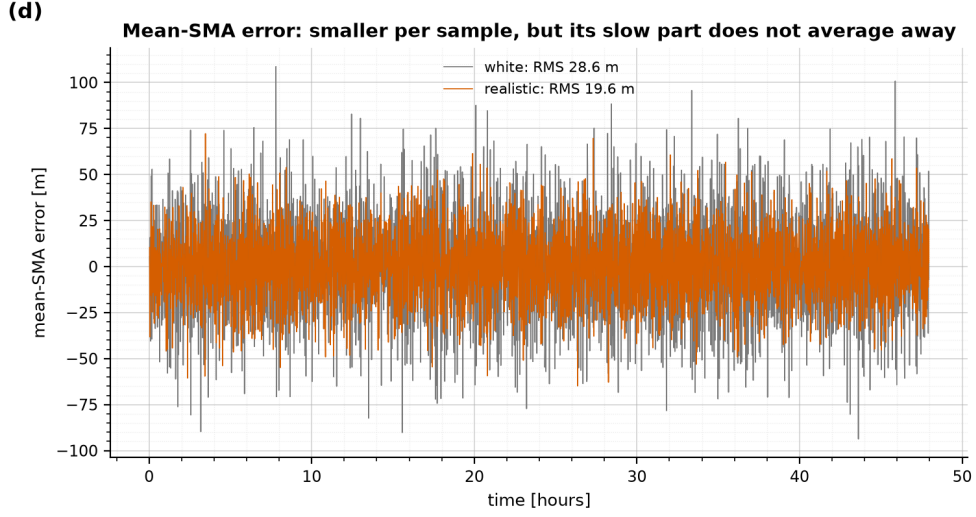


Figure 25: Mean-SMA error from the realistic noise: smaller per sample, but its slow part does not average away, so it biases the energy method.

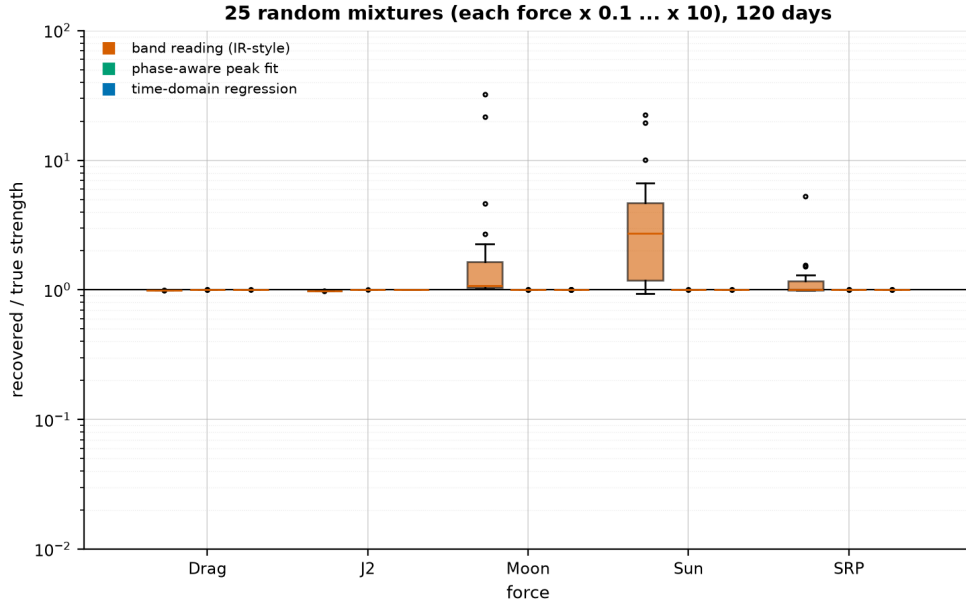


Figure 26: Force spectroscopy: median error of the band reading in random force mixtures. Drag 0.5%, J_2 2%, SRP 0.9%, Moon 8%; the Sun (172%) is not separable because its lines lie under the Moon's.

Sampling robustness.

- **FD regression** needs a fine step: μ is recovered to $2 \times 10^{-5}\%$ at 1 min but 70% at 60 min, and the stencil order must change with the step (Table 1).
- **J_2 detectability:** the J_2 signature lies at twice the orbital frequency, so it vanishes once the step exceeds $T_{\text{orb}}/4 = 23.9$ min (SNR 55 dB at 1 min, -3 dB at 90 min, Figure 27).
- **In the full chain** (Section 8) the drag is learned from the mean SMA on clean coast arcs: 0.03% at 6 s, 0.07% at 1 min, 0.08% at 10 min — sampling hardly matters because the mean SMA averages the noise of every sample.

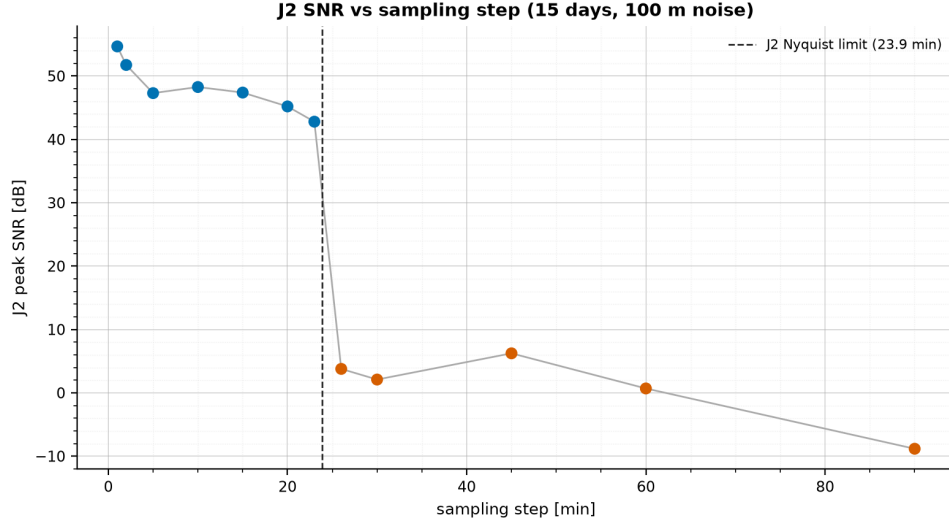


Figure 27: J_2 signal-to-noise ratio against the sampling step; the dashed line is the Nyquist limit $T_{\text{orb}}/4$.

Calculation check. $T_{\text{orb}} = 2\pi\sqrt{a^3/\mu} = 95.53$ min; the J_2 line at $2/T_{\text{orb}}$ needs a step below $T_{\text{orb}}/4 = 23.88$ min, the limit used in the code (23.88 min, 0.03%).

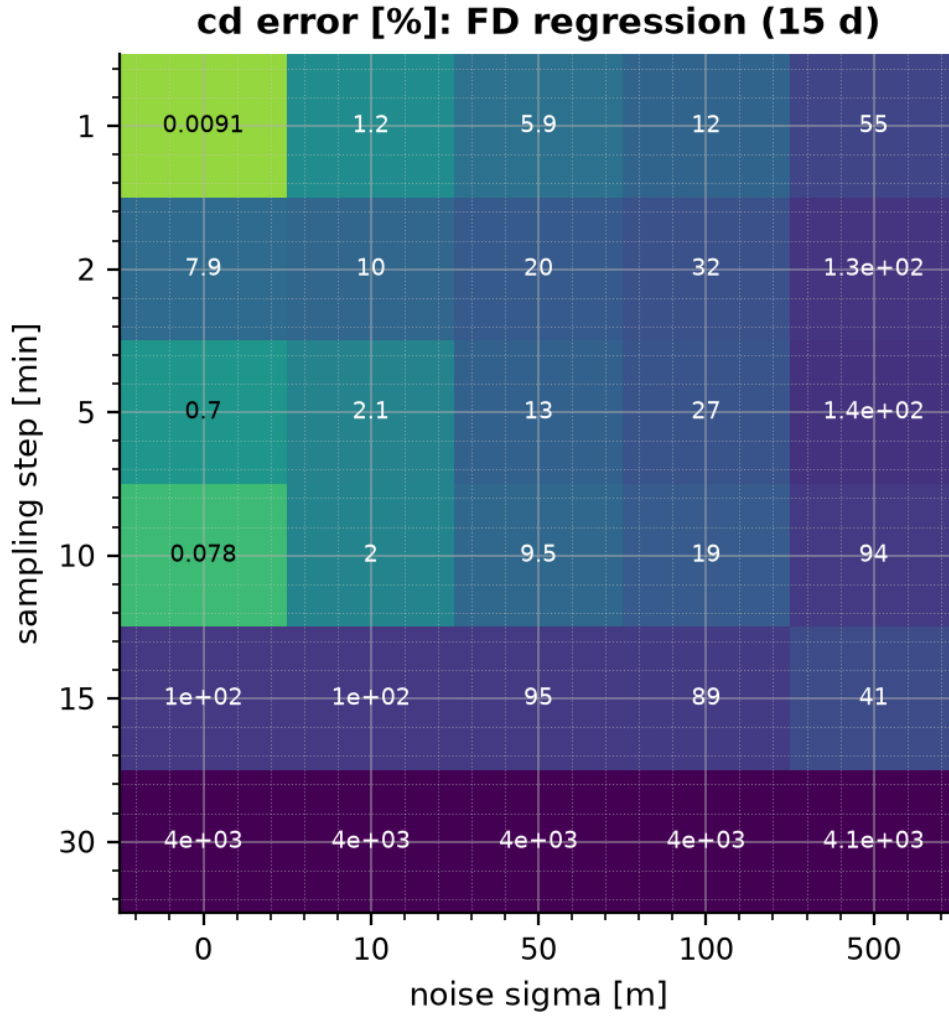


Figure 28: FD c_d error over sampling step $\times$ noise: the valley of good steps moves to coarser sampling as the noise grows.

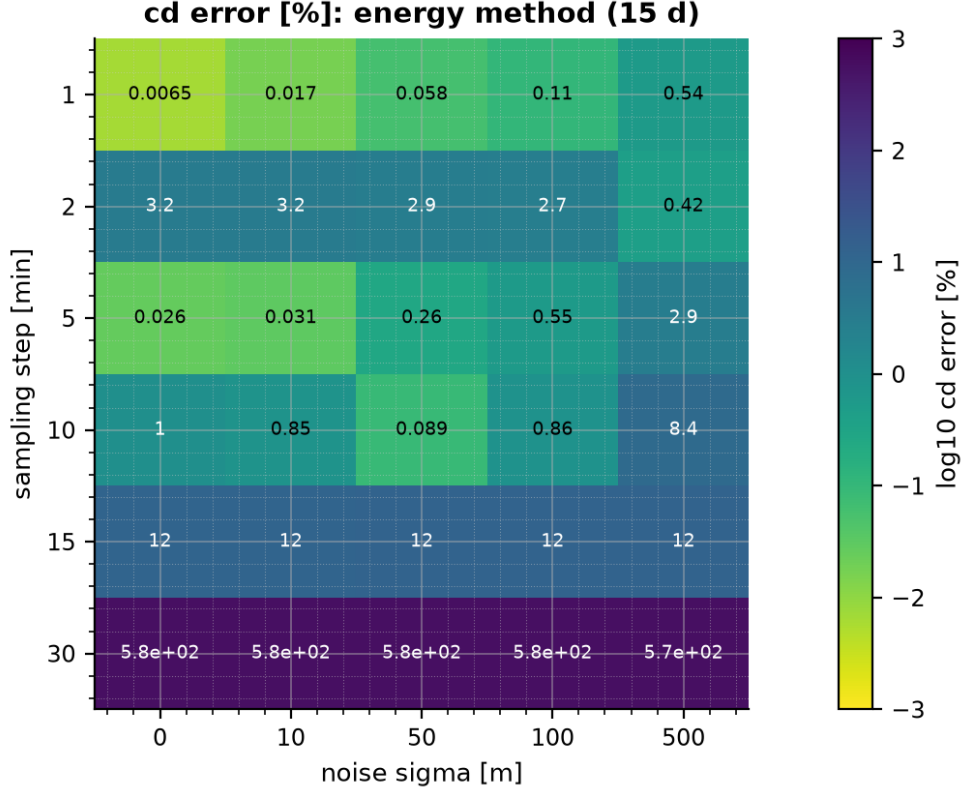


Figure 29: The same map for the energy method: good everywhere except at the coarsest steps.

Stage 1 ranking

For c_d : (1) EKF on coarse data and the energy method on fine data (0.01–0.04%); (2) the mean-SMA coast regression inside the pipeline (0.03–0.08% at every step); (3) shooting (0.05%, 100× slower); (4) spectroscopy (0.5%, fastest, no orbit model); (5) FD regression (6% at 10 m noise). FD is still the right tool for the many-term joint fit when the data are clean and fine.

5 Stage 2: detecting burns with a Gaussian mixture model (steps 6–7)

5.1 The feature: the unmodelled acceleration

From the measured states, the velocity derivative minus the known physics leaves only what is not modelled:

$$\mathbf{a}_u = \dot{\mathbf{v}}_{\text{measured}} - \left(-\frac{\mu \mathbf{r}}{r^3} + \mathbf{a}_{J_2} \right) \approx \mathbf{a}_{\text{drag}} + \mathbf{a}_{\text{thrust}} + \text{noise}. \quad (7)$$

The derivative is a Savitzky–Golay filter [10] (least-squares polynomial of order 5 through 61 samples). Two features are used: $x_1 = \log |\mathbf{a}_u|$ (how much unmodelled acceleration) and $x_2 = \mathbf{a}_u \cdot \hat{\mathbf{v}} / 10^{-4}$ (thrust pushes forwards; noise has no direction).

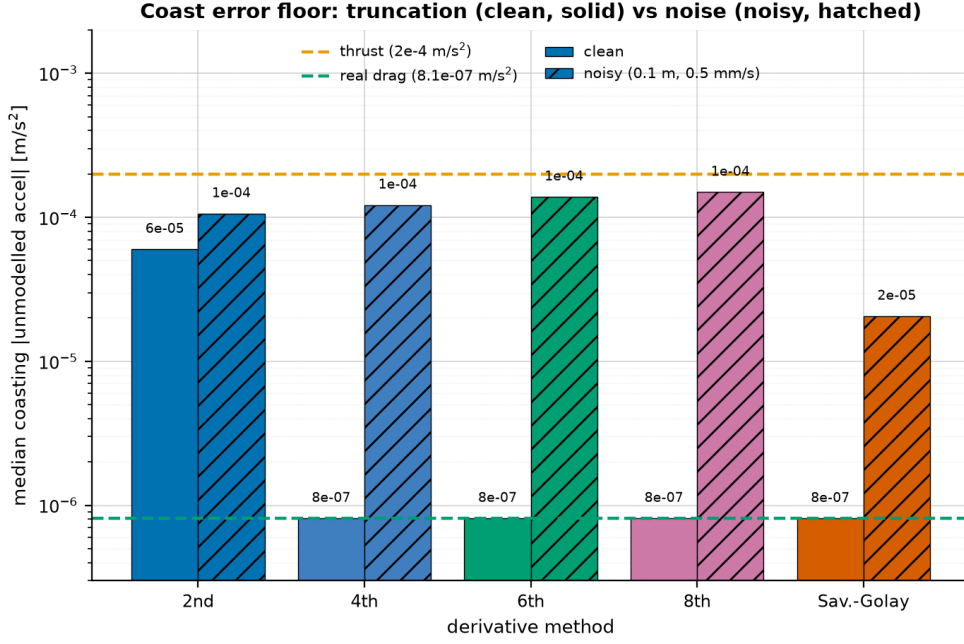


Figure 30: Coast residual floor of each derivative method on clean data: the 2nd-order stencil leaves a truncation error of $6 \times 10^{-5} \text{ m/s}^2$, only 4 times below the thrust; high-order stencils and Savitzky–Golay reach the real drag level.

5.2 The mixture and how it is fitted

A GMM models the feature density as a weighted sum of K Gaussians [11, 27],

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k), \quad (8)$$

fitted by expectation–maximisation on 20 000 random TRAIN samples (three restarts). The E-step computes each sample’s *responsibility* $\gamma_{ik} = \pi_k \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_k, \Sigma_k) / \sum_j \pi_j \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j, \Sigma_j)$; the M-step re-estimates $\pi_k = \frac{1}{n} \sum_i \gamma_{ik}$, $\boldsymbol{\mu}_k$ and Σ_k as responsibility-weighted means and covariances. The mixture is unsupervised: it never sees the true labels. A component is called **burn** if its mean x_1 lies more than three standard deviations of the heaviest (coast) component above the coast mean — thrust can only add acceleration. A sample is a burn if its most likely component is a burn component, and $P(\text{burn} \mid \mathbf{x}) = \sum_{k \in \text{burn}} \gamma_k$.

Table 2: The four fitted components (6-s data, reference noise). Feature 1 is $\log |\mathbf{a}_u|$ in $\log \text{m/s}^2$; feature 2 is the along-track acceleration in units of 10^{-4} m/s^2 .

Component	weight π_k	mean x_1	sd x_1	mean x_2	role
1	0.4248	−12.56	0.26	−0.009	coast (bulk)
2	0.1527	−13.22	0.46	−0.003	coast (low-residual tail)
3	0.0039	−8.55	0.11	+1.95	burn
4	0.4186	−12.07	0.24	−0.010	coast (upper part)

Calculation check. The burn weight should equal the fraction of time spent burning: $1368 \text{ s} / (3.9051 \times 86400 \text{ s}) = 0.0041$, against $\pi_3 = 0.0039$ (the samples on the ramps are shared with the coast). Its mean along-track acceleration, $1.95 \times 10^{-4} \text{ m/s}^2$, is the thrust (2×10^{-4}) less the smoothing of the plateau edges, and $e^{-8.55} = 1.9 \times 10^{-4} \text{ m/s}^2$ is the same number in feature 1. The coast components sit at $e^{-12.6} \approx 3 \times 10^{-6} \text{ m/s}^2$: drag (8×10^{-7}) plus the derivative noise.

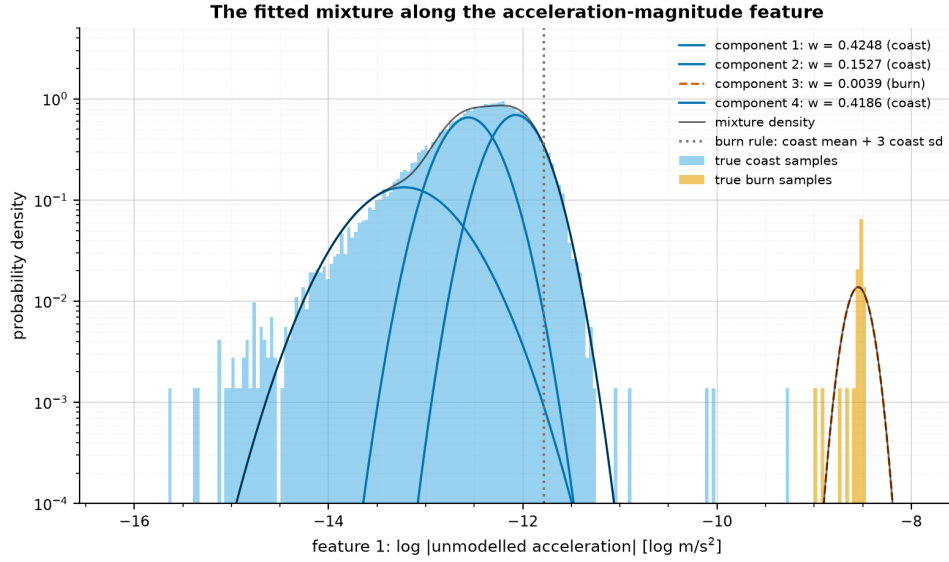


Figure 31: The fitted mixture along feature 1 (log scale): three coast components (blue) model the skewed coast cloud, one narrow component (orange, dashed) the burns. The dotted line is the burn rule; the two clouds are separated by about four decades.

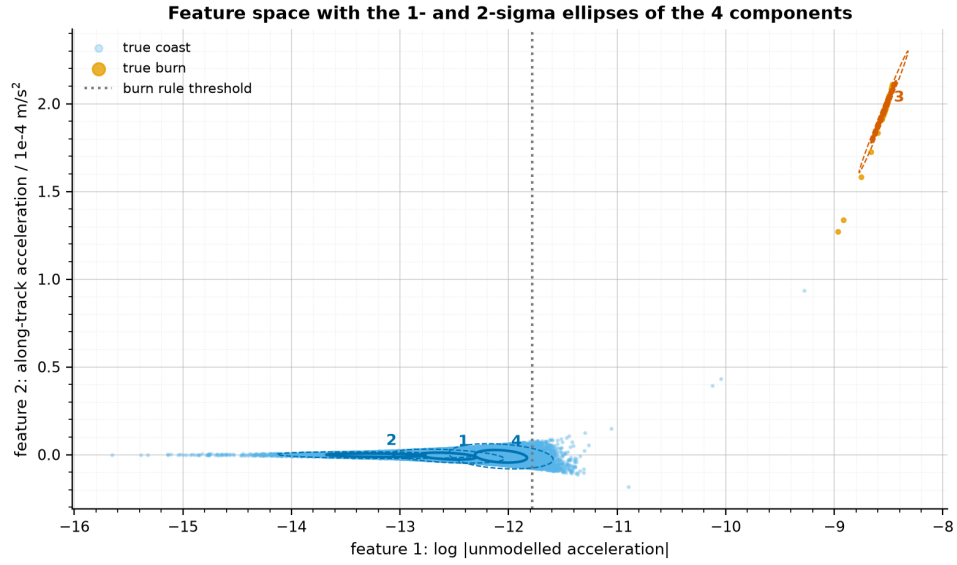


Figure 32: The same mixture in the two-dimensional feature space with the 1- and 2- σ ellipses. Burns sit up and to the right (large, forward acceleration); the few points between the clouds are the thrust ramps.

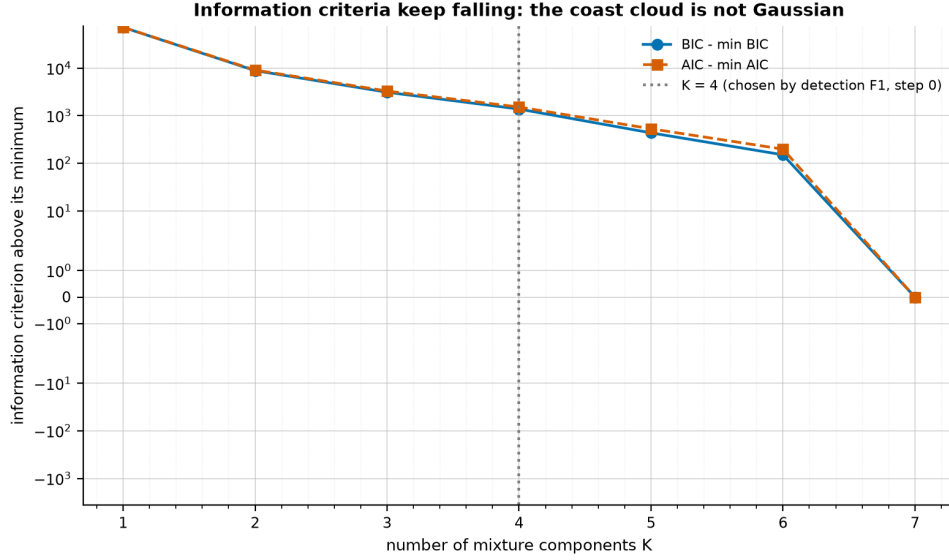


Figure 33: BIC and AIC against the number of components. They keep falling, because the coast cloud is skewed (a log of a noise magnitude) and more Gaussians always describe it better; the detection does not need that detail, so K is chosen by the Brier score of $P(\text{burn})$ (step 0), not by BIC [12].

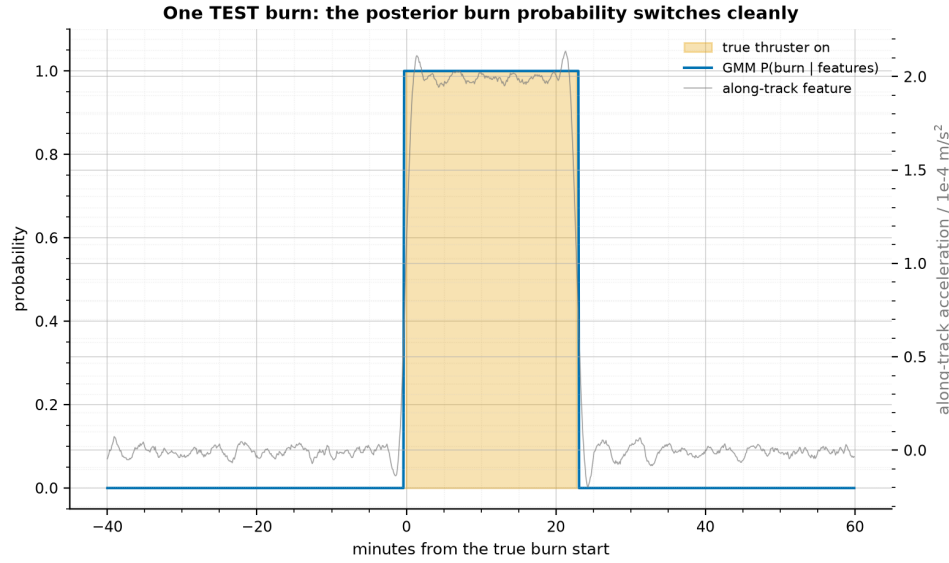


Figure 34: One TEST burn: $P(\text{burn} | \mathbf{x})$ (blue) jumps from 0 to 1 within the thrust ramp and back at the end; the along-track feature (grey) shows the $2 \times 10^{-4} \text{ m/s}^2$ plateau.

Phase space, days 40-48 (TEST), coloured by the GMM weight $P(\text{burn})$:
the coast line (dark) and the burn line (bright) are told apart sample by sample

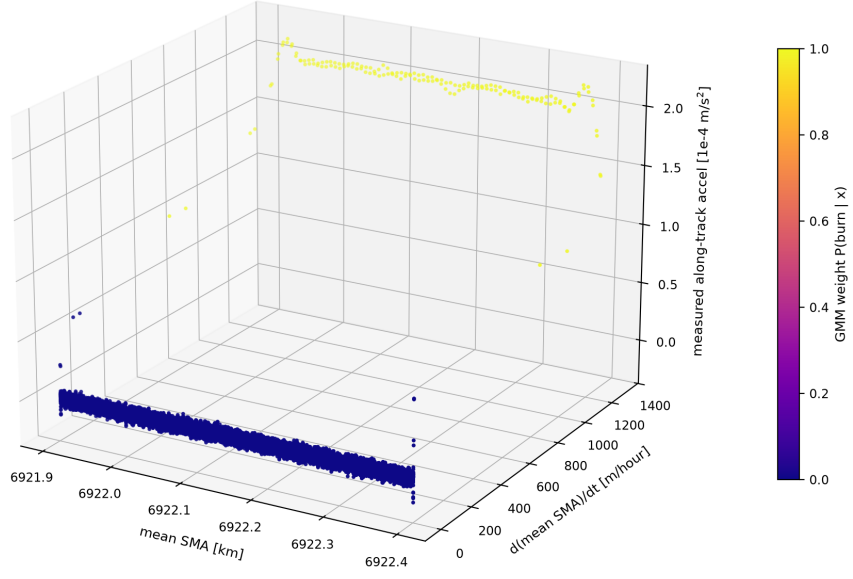


Figure 35: Phase space (mean SMA, its rate, along-track acceleration) over two TEST cycles, each sample coloured by its GMM burn probability.

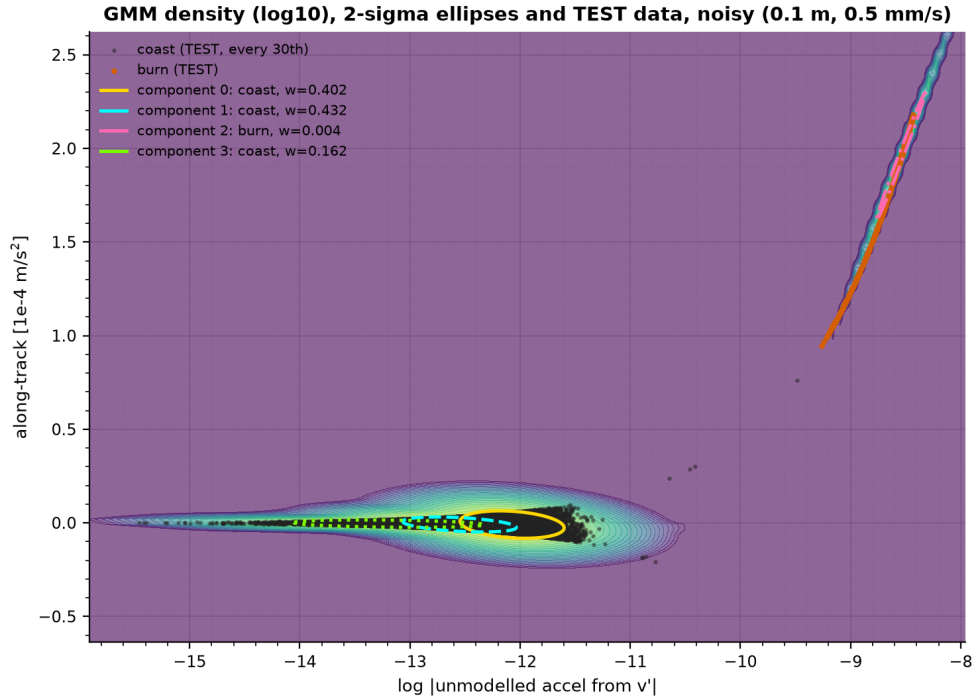


Figure 36: GMM density contours and component ellipses in the $(\log |\dot{\mathbf{v}}| \text{ residual, along-track})$ plane, from the workflow `gmm` stage.

5.3 From samples to events

A burn is about 230 samples. Flags are grouped into **events**: gaps of at most two samples are filled and runs shorter than one minute dropped. Scoring is per event: a true burn is found if a detected

event overlaps it within ± 60 s; precision = real events / detected events, recall = found burns / burns, F1 their harmonic mean. Over the TEST days at the reference noise the GMM classifies 5.18 million samples with 21 034 true positives, no false negatives and 499 false-positive samples, all on the burn ramps — they merge into the 92 events, so the event score is perfect.

5.4 Comparison with other detectors

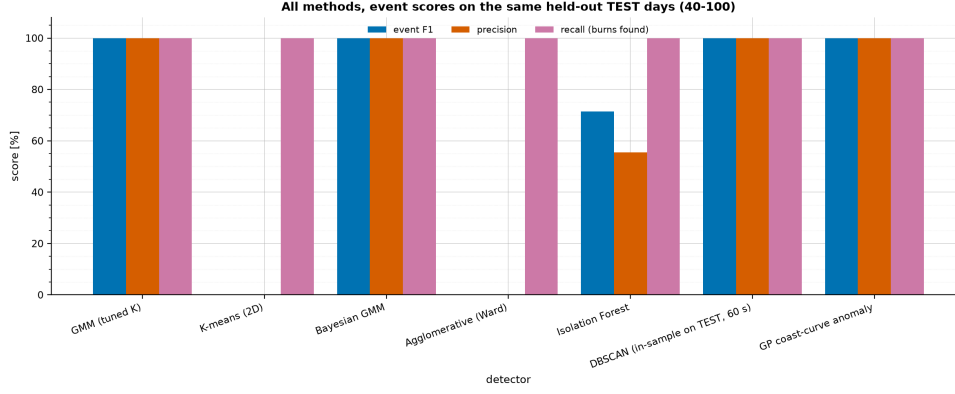


Figure 37: Event F1 of each detector on the 92 TEST burns. GMM, Bayesian GMM and a MAD threshold: 1.0; Isolation Forest [19] 0.85; k -means 0.001 (its two clusters split the coast cloud, 136 662 false events).

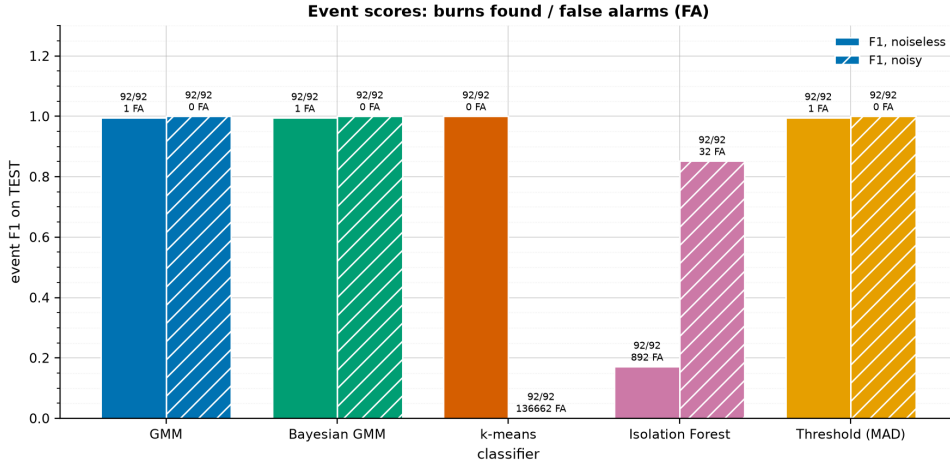


Figure 38: Precision, recall and F1 of the event scoring per method.

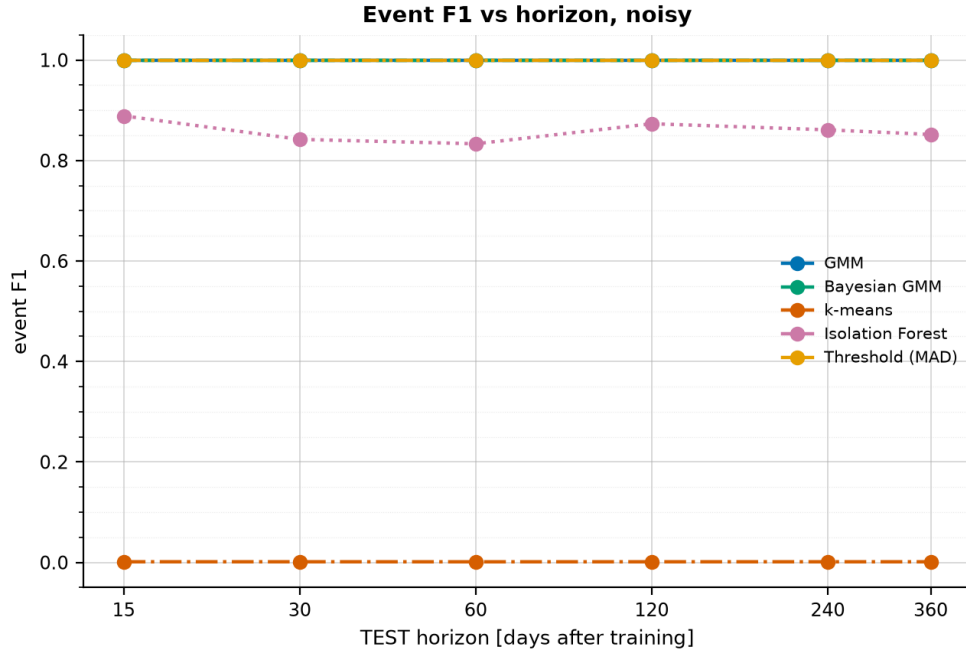


Figure 39: GMM event F1 against the test horizon: 1.0 from 15 to 360 days — a mixture fitted on 40 days does not drift.

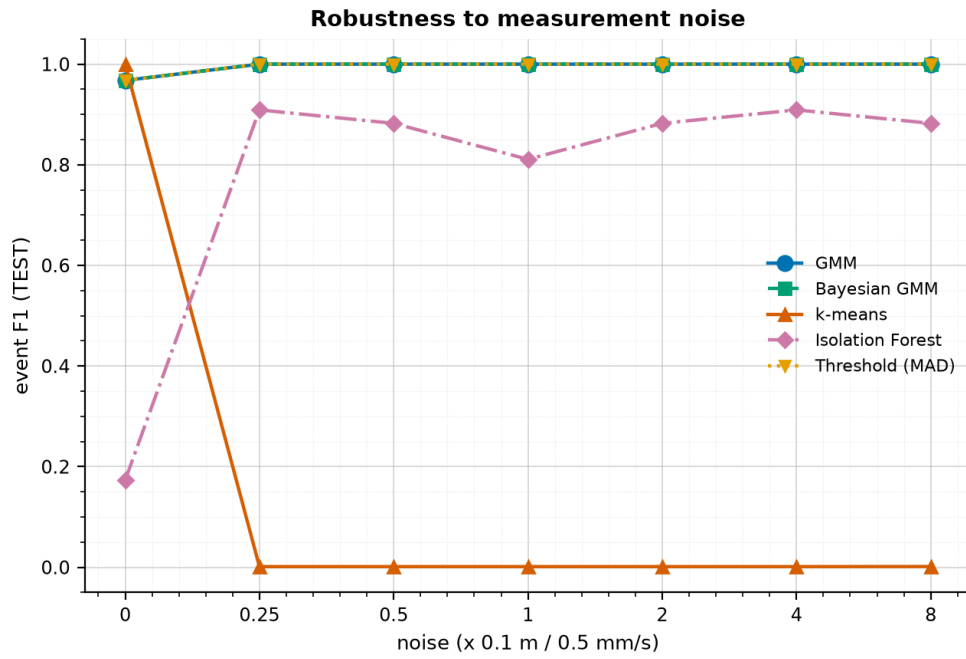


Figure 40: Event F1 against the noise level (multiples of the reference): GMM, Bayesian GMM and threshold stay at 1.0 from 0.25 to 8 times the reference noise.

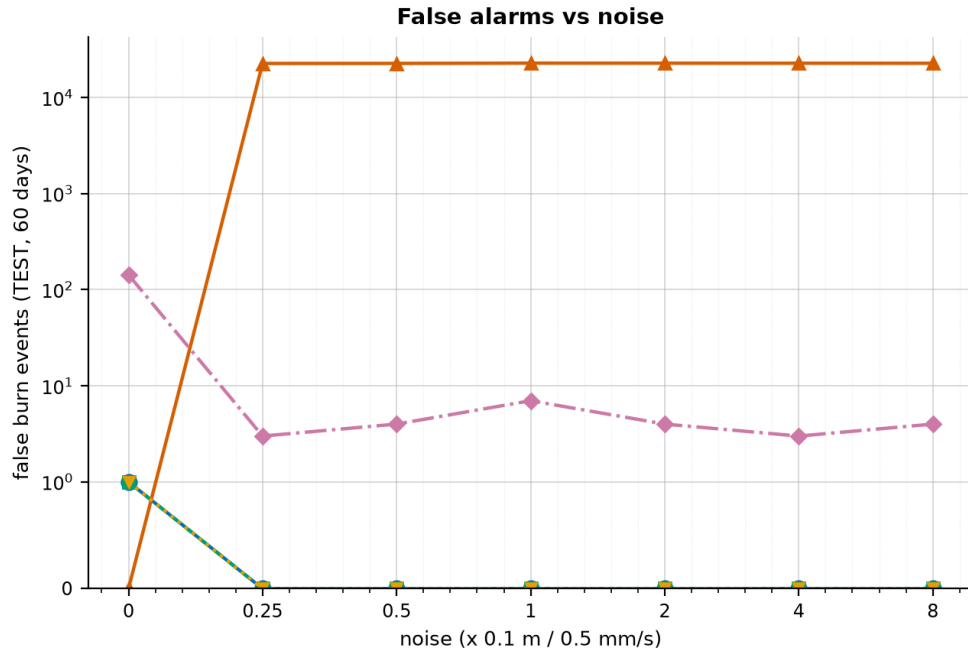


Figure 41: False burn events against the noise level (60 TEST days): none for the GMM, thousands for k -means.



Figure 42: F1 against the number of training samples: the GMM needs about 1000.

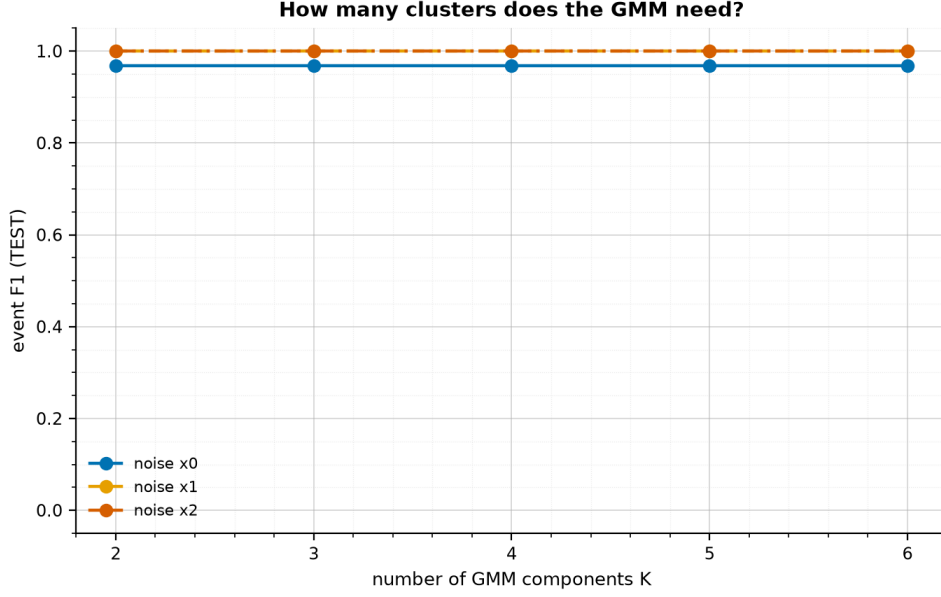


Figure 43: F1 against the number of components: between 2 and 6 it hardly matters at these noise levels.

Sampling robustness. The Savitzky–Golay derivative works up to a **2-min** step: beyond, a polynomial through the minimum five samples spans a large arc of the curved orbit and its truncation error exceeds the thrust. A **one-step propagation** feature — propagate each measured state one step with gravity + J_2 (RK4) and divide the velocity mismatch at the next measurement by the step, $\mathbf{a} = (\mathbf{v}_{k+1} - \Phi_h(\mathbf{r}_k, \mathbf{v}_k)_v)/h$ — has noise $\propto 1/h$ and works from about 1 to 20 min. Switching at 2 min gives $F1 \geq 0.9$ from 6 s to 20 min up to twice the reference noise. At four times the noise there is a gap: the derivative fails from 0.5 min ($F1$ 0.6, then 0 at 1 min) and the propagation feature only recovers at 2 min. Above one burn length (23 min) a burn is diluted into a single interval and $F1$ falls to 0.85 at 30 min and 0.5 at 60 min. In the full chain (Section 8) the GMM scores $F1 = 1.0$ at every step from 6 s to 10 min.

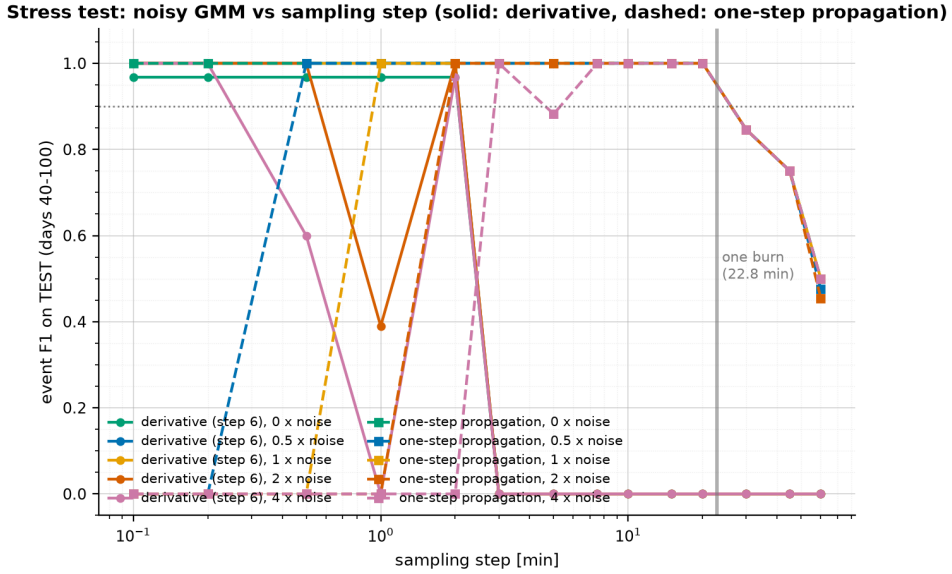


Figure 44: Stress test: event F1 against the sampling step for the two features and several noise levels.

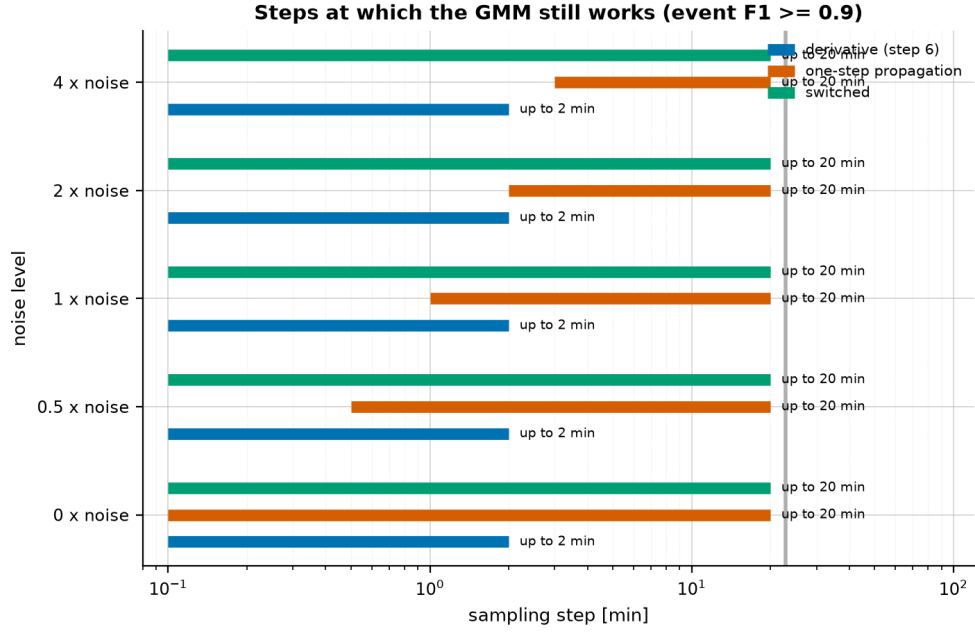


Figure 45: Sampling steps at which the GMM still works (event $F1 \geq 0.9$) for each noise level and feature.

Stage 2 ranking

(1) GMM and Bayesian GMM ($F1 = 1$, unsupervised, probabilities, no threshold to set); (2) MAD threshold ($F1 = 1$, but a threshold to choose); (3) GP coast-curve anomaly and DBSCAN [20] ($F1 = 1$ on held-out data, slower); (4) Isolation Forest (0.71–0.85); (5) k -means (fails with noise). For the derivative: Savitzky–Golay up to 2 min, one-step propagation from 2 to 20 min; plain central differences only on clean data.

6 Stage 3a: the thrust law from repetition (step 8)

A station-keeping engine fires the same burn again and again. Repetition can be used three ways.

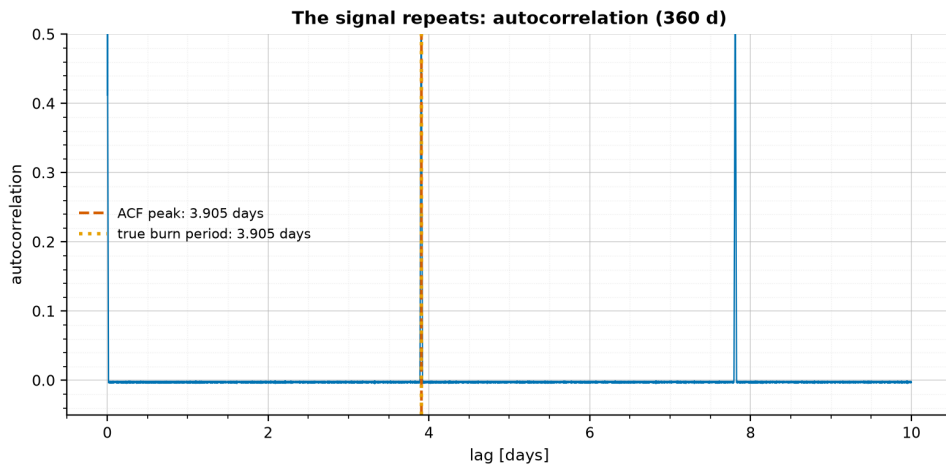


Figure 46: Autocorrelation of the noisy along-track acceleration: the first peak is at 3.9051 days, the true burn period.

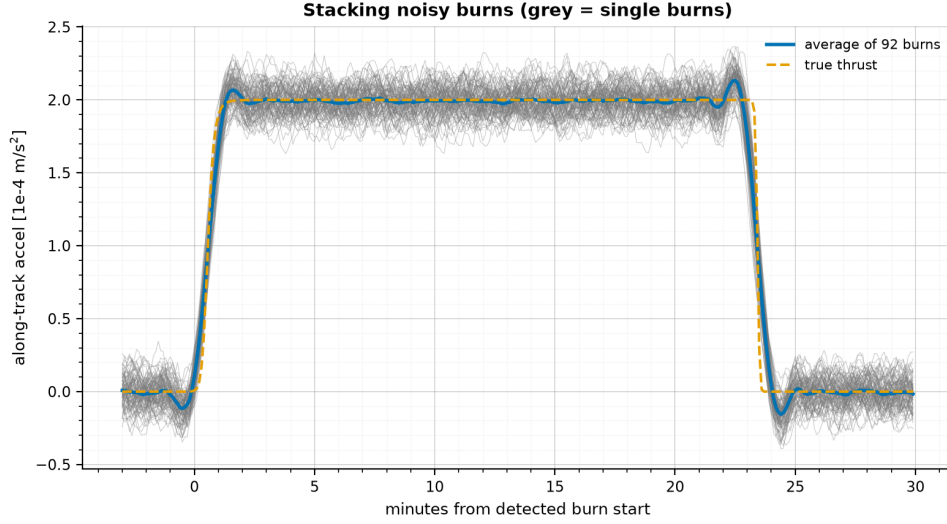


Figure 47: All detected burns aligned on their detected start: the individual noisy profiles and their average.

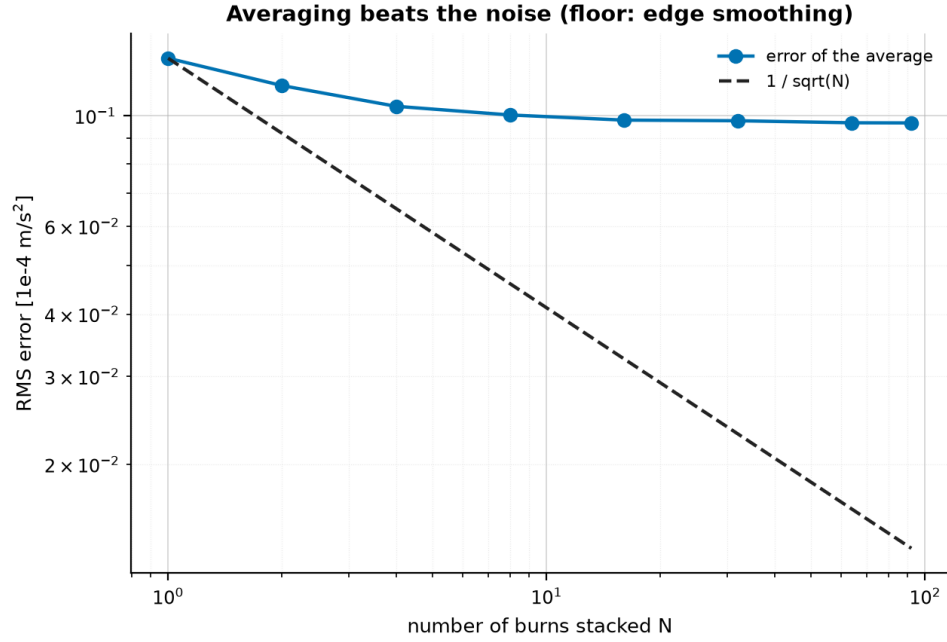


Figure 48: RMS error of the stacked average against the number of burns: it falls as $1/\sqrt{N}$ until it reaches the floor set by the smoothing of the edges.

Calculation check. Removing the floor (9.67×10^{-6}) in quadrature, the noise part of the error is 8.3×10^{-6} for one burn and 3.7×10^{-6} for four: the $1/\sqrt{N}$ law predicts 4.2×10^{-6} (11% off for a 4-sample statistic).

Gaussian-process thrust law. A GP [17] with an RBF + white-noise kernel is fitted through the stacked samples, giving a smooth law $a(t - t_{\text{start}})$ with an uncertainty band:

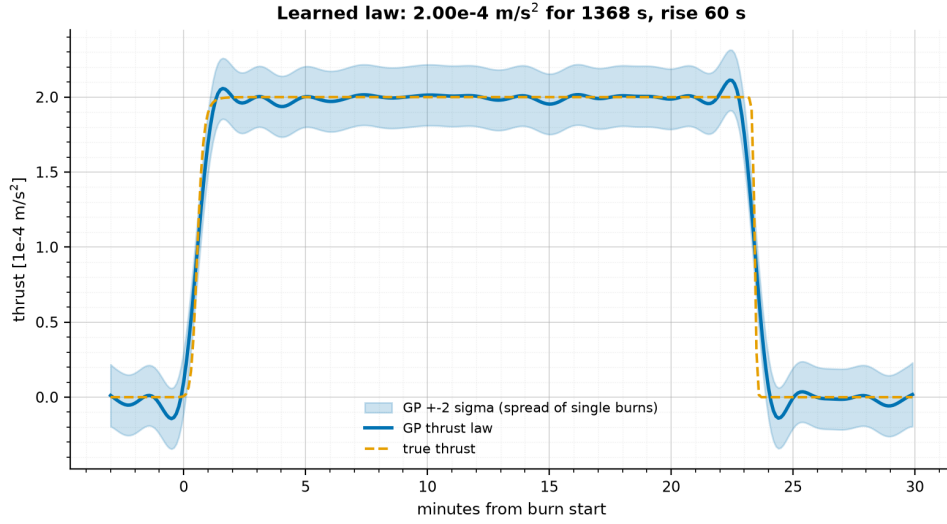


Figure 49: GP thrust law from 92 stacked burns: plateau $2.00 \times 10^{-4} \text{ m/s}^2$ (0.999 of the truth), duration 1368 s (exact), rise time 60 s (truth 30 s: the derivative filter smooths the edge).

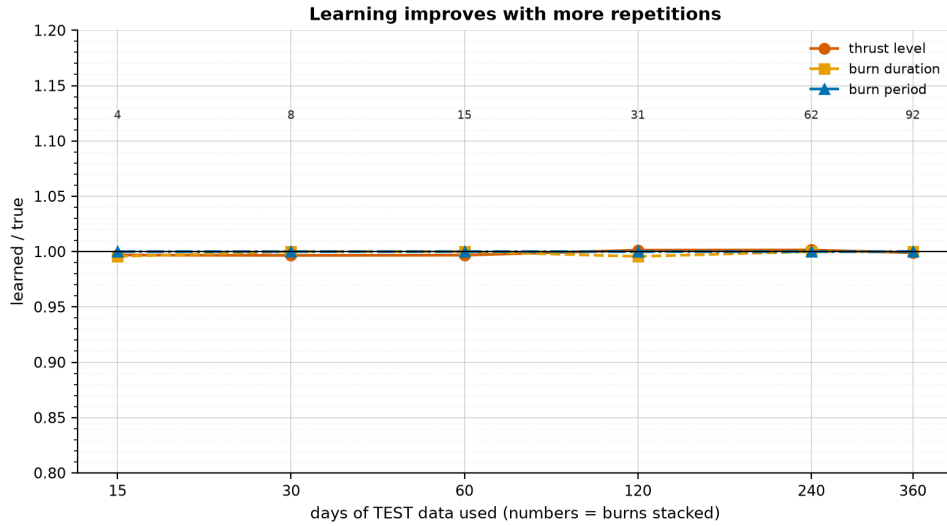


Figure 50: Thrust-law numbers learned from only the first H days: with 4 burns (15 days) the plateau is already within 0.3%.

Sampling robustness. The stacking derivative keeps its window in *time* (246–300 s), but it needs at least five samples, so at a 2-min step it widens to 600 s. Plateau / true: 0.995, 0.999, 0.995, 0.995, 0.993 at 6, 12, 30, 60 and 120 s; duration / true: 1.000, 0.991, 0.987, 0.965, 0.877; the rise time grows from 60 to 120 s. The level is robust; the burn *shape* needs steps of 30 s or less.

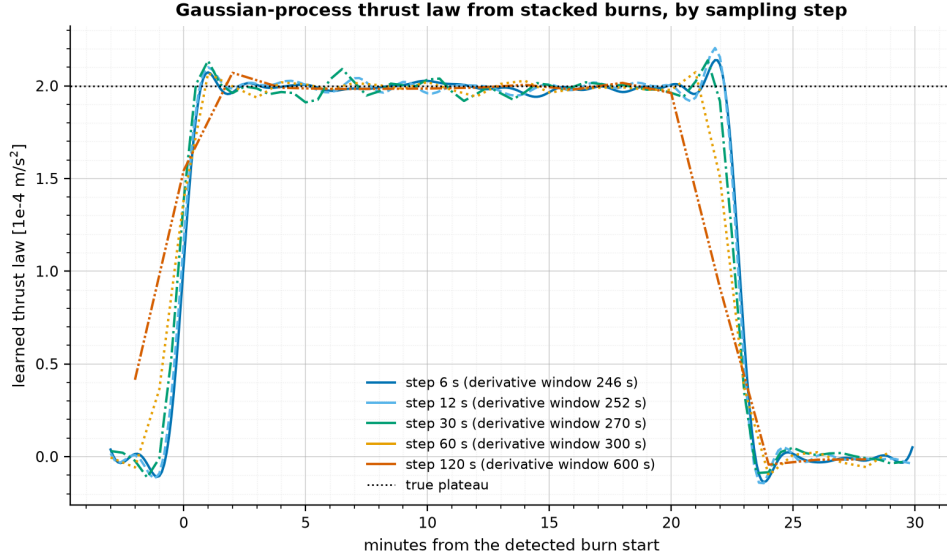


Figure 51: GP thrust law learned at each sampling step.

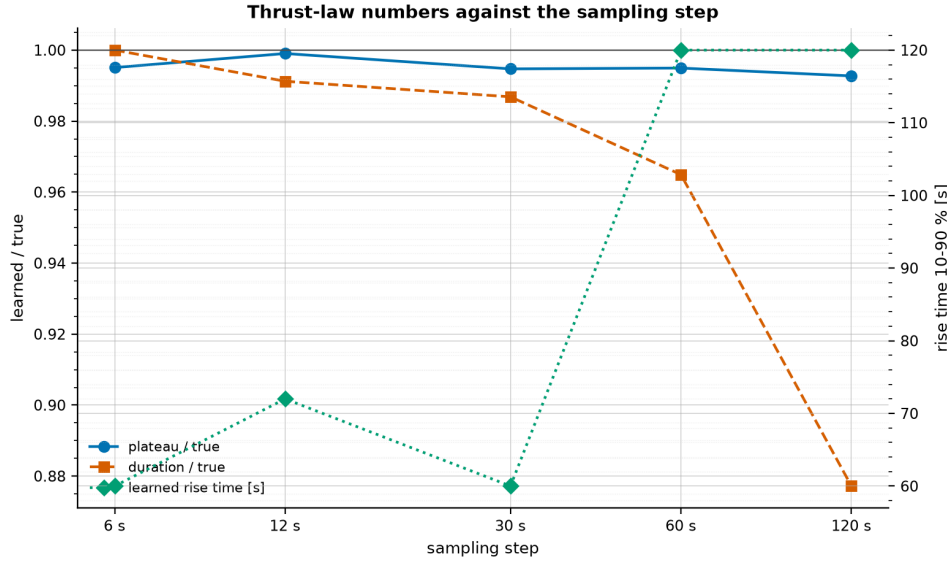


Figure 52: Plateau, duration and rise time of the GP law against the sampling step.

7 Stage 3b: SINDy and BINDy — the natural law, the thrust law and the switching law

7.1 SINDy and BINDy

SINDy [13] writes a rate as a sparse combination of candidate functions, $\dot{a} = \Theta(a, s)\xi$, and finds ξ by sequentially thresholded least squares (STLSQ): solve, zero the small coefficients, refit, repeat. Our BINDy is the Bayesian counterpart by automatic relevance determination [14]: each coefficient gets a Gaussian prior whose precision is learned, so irrelevant terms are pruned and every kept terms has a posterior standard deviation (in the spirit of [15, 16]). The input is the mean SMA (from the measured states), its smoothed derivative, and the GMM burn flags.

The physics library is singular. With drag shape $D = (2a^2/\mu)|\mathbf{v}|^3$ and thrust shape $B = (2a^2/\mu)|\mathbf{v}|$, $\dot{a} = -c_d D + T\theta B$. Over a 500m band D and B are constant to a few ppm and correlated at 0.997, so $[D, B, 1]$ cannot be separated. Step 9 therefore uses cluster indicators, $\Theta = [1 - s, s, a - a_{\text{ref}}, s(a - a_{\text{ref}}), (a - a_{\text{ref}})^2]$; STLSQ keeps only the two cluster terms, from which $\hat{c}_d = 1.0005 c_d$ and $\hat{T} = 0.9997 T$.

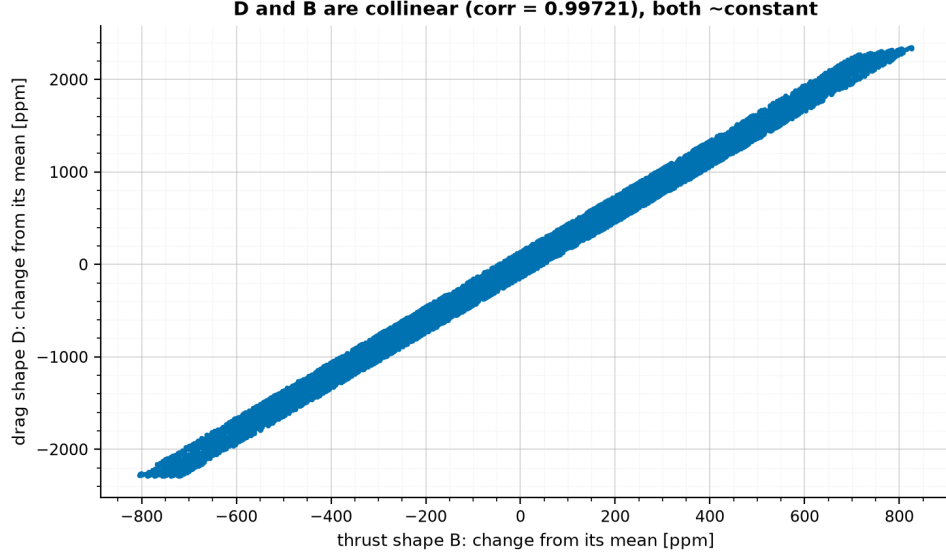


Figure 53: The drag shape, the thrust shape and the constant move together over the band: a physics library is singular, a cluster-indicator library is not.

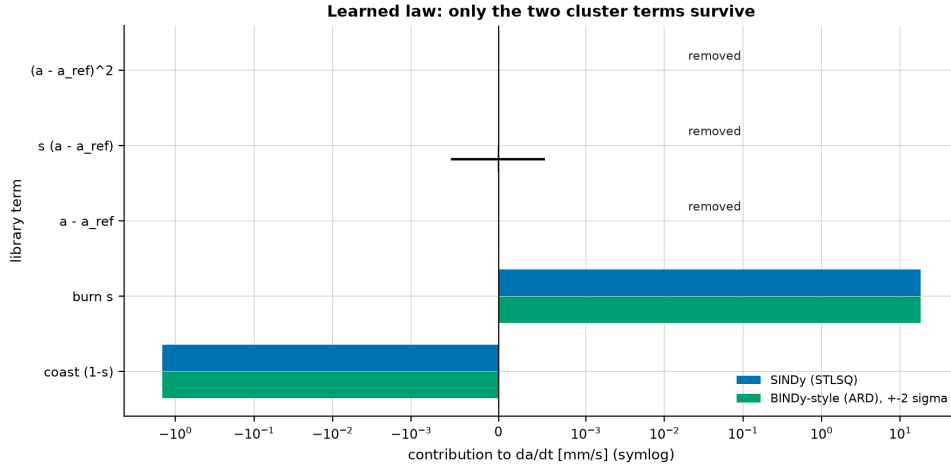


Figure 54: SINDy and BINDy coefficients of the cluster-indicator library with BINDy's $\pm 2\sigma$: only the coast and burn terms survive.

7.2 The switching law: memory is essential

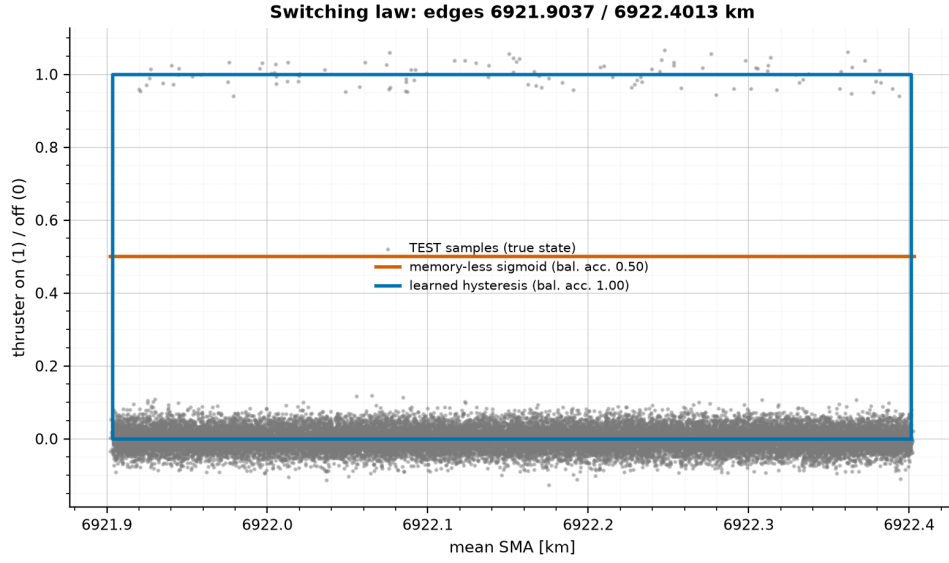


Figure 55: Burn flags against mean SMA. A memory-less sigmoid $s = f(a)$ reaches a balanced accuracy of 0.50 (chance); the hysteresis with the learned edges (6921.9037 / 6922.4013 km) reaches 1.00.

7.3 Four ways to split the learning problem (step 19)

The dynamics are the sum of a natural part and a thrust part plus the switching law. In operations one part is often known: the natural dynamics from a catalogue force model, or — for one's own satellite — the thruster.

- A. Whole system** one regression of $\dot{a}$ on $[-D, sB]$ over *all* TRAIN samples.
- B. Thrust law only (natural known)** c_d given; $\dot{a} + c_d D = T sB$ gives T .
- C. Natural law only (thrust known)** T , the throttle profile and the edges given; $\dot{a} - T\theta B = -c_d D$.
- D. Both, separately** c_d from clean coast arcs only, then T from clean burn arcs given c_d .

Table 3: Physics recovered and year-ahead forecast error of the four splits (TRAIN days 0–40).

Scenario	c_d/true (BINDy $\pm 1\sigma$)	T/true	period [d]	onset error at 360 d
A whole system	0.9922 ($\pm 1.4\%$)	0.924	3.9351	1668 min
B thrust only (natural known)	given	0.924	3.9047	26 min
C natural only (thrust known)	1.0015 ($\pm 1.0\%$)	given	3.9006	302 min
D both, separately	1.0006 ($\pm 0.6\%$)	0.9995	3.9011	262 min
truth	1	1	3.9051	—

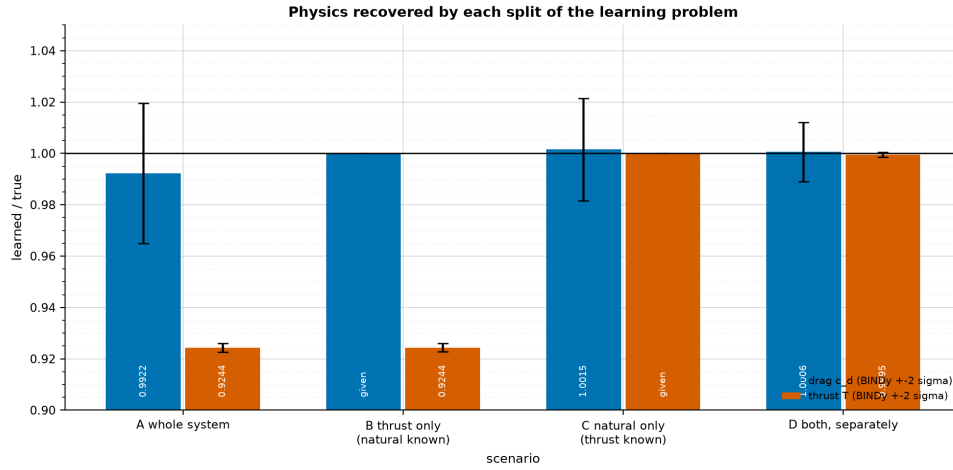


Figure 56: c_d and T recovered by each split, with BINDy $\pm 2\sigma$. Only the joint regression (A) is biased.

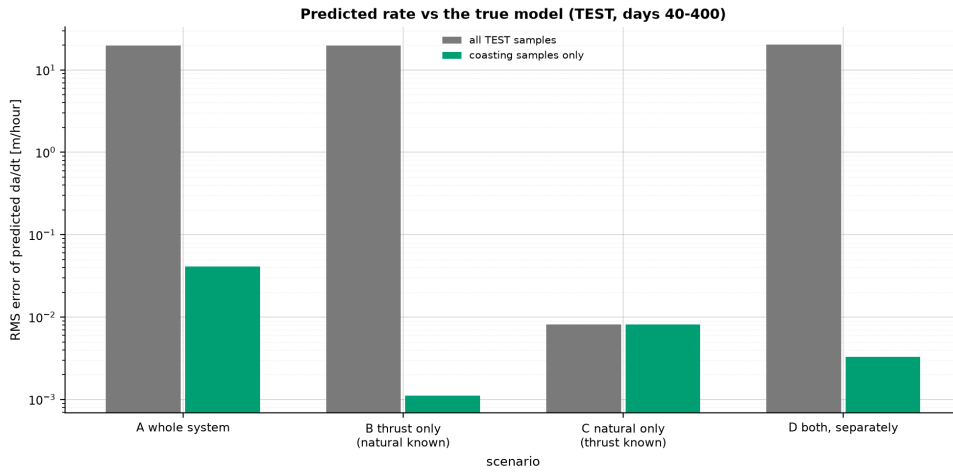


Figure 57: RMS error of the predicted $\dot{a}$ against the true model, on all TEST samples and on coasting samples only.

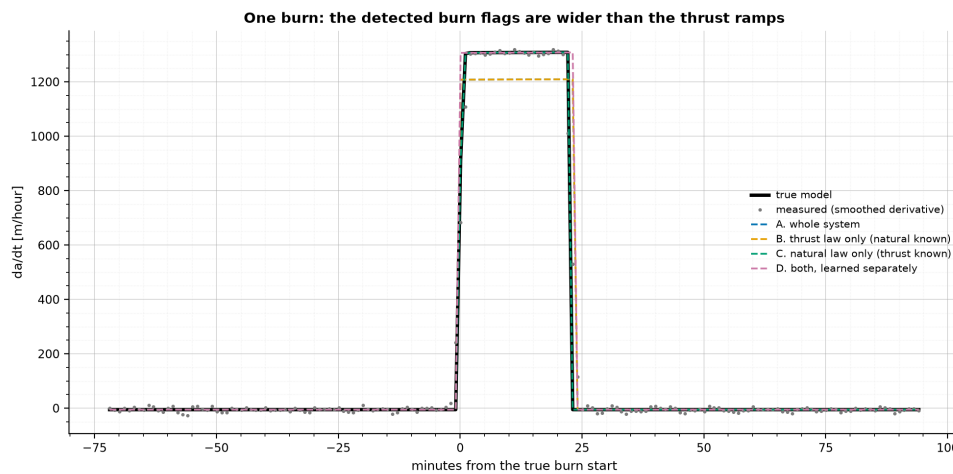


Figure 58: One TEST burn: the true $\dot{a}$ and what each split predicts. The detected flags are wider than the thrust ramps, which is what biases the joint regression.

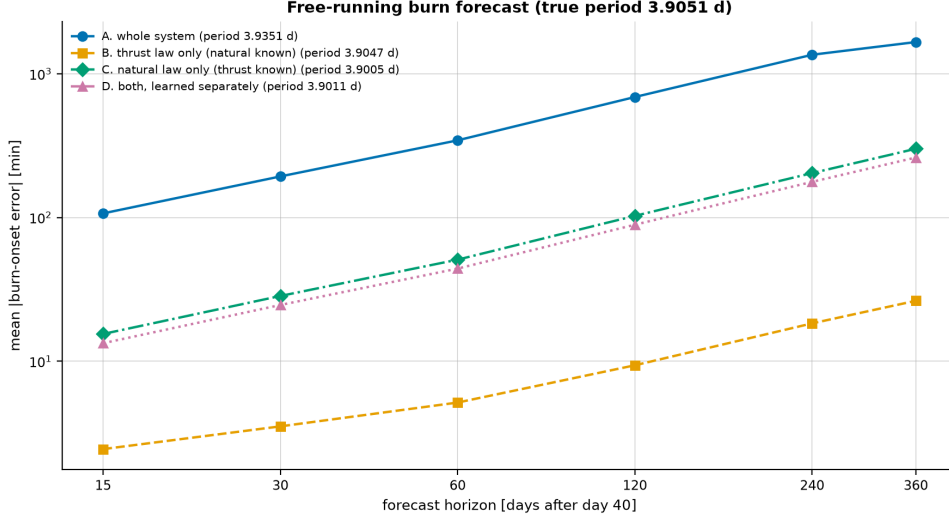


Figure 59: Free-running burn forecast error against horizon for the four splits.

- **The joint regression is biased:** samples with partial thrust on the ramps are labelled burn or coast, so least squares lowers T by 7.6% and pushes part of the leak into the drag (-0.8%). Learning on clean arcs (D) removes both biases.
- **The natural law governs the forecast:** a coast is 99.6% of the cycle, so knowing c_d (B) gives 26 min after a year even with T 7.6% low — that only lengthens each 23-min burn by about 2 min.

Calculation check. A relative period error e shifts the n -th burn by neP , so over a horizon H the mean |onset error| is $\frac{1}{2}He$. For C: $e = |3.9006/3.9051 - 1| = 1.15 \times 10^{-3}$, $\frac{1}{2} \cdot 360 \cdot 1440 \cdot e = 299$ min vs 302 min measured (0.9%); for D: 260 vs 262 min (0.9%). For A the formula gives 1995 min against 1668: the error is saturating as the predicted burns slip by more than half a period and start matching the wrong burn.

7.4 The natural law when there is no deadband

Without the controller the orbit decays freely by 51 km in 400 days. Eight libraries were fitted on 40 days of the free decay and tested on the remaining 360 (Table 4).

Table 4: Learning the natural (drag) law on the free decay.

Library	test RMS of $\dot{a}$ [m/h]	SMA error after 360 d	recovered c_d
constant / linear / quadratic*	0.012	84 m	—
$\sqrt{a}$ (true form)	0.0006	1 m	1.0001
physics shape $D(a)$ (true form)	0.0006	1 m	1.0001
exponential density $e^{-(a-a_0)/H}$	3.55	51 km	—
power laws $a^{-1}, a^{1/2}, a^2$	0.14	0.9 km	—
kitchen sink	0.14	0.9 km	—

* STLSQ prunes the linear and quadratic terms, leaving a constant rate.

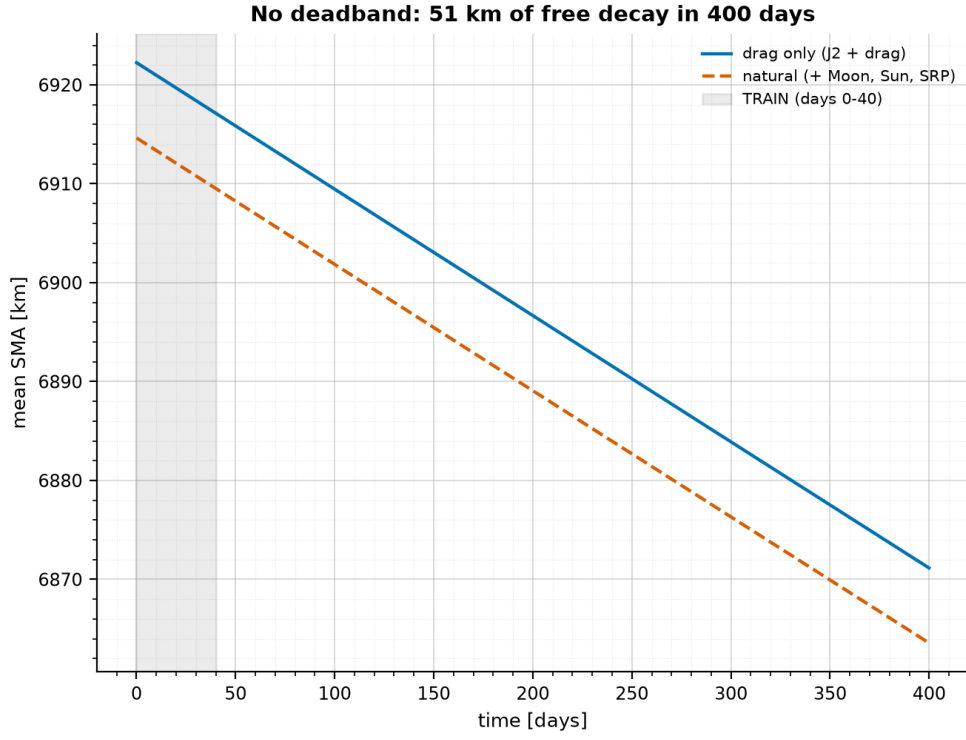


Figure 60: The free decay: mean SMA of the drag-only and natural runs (no deadband).

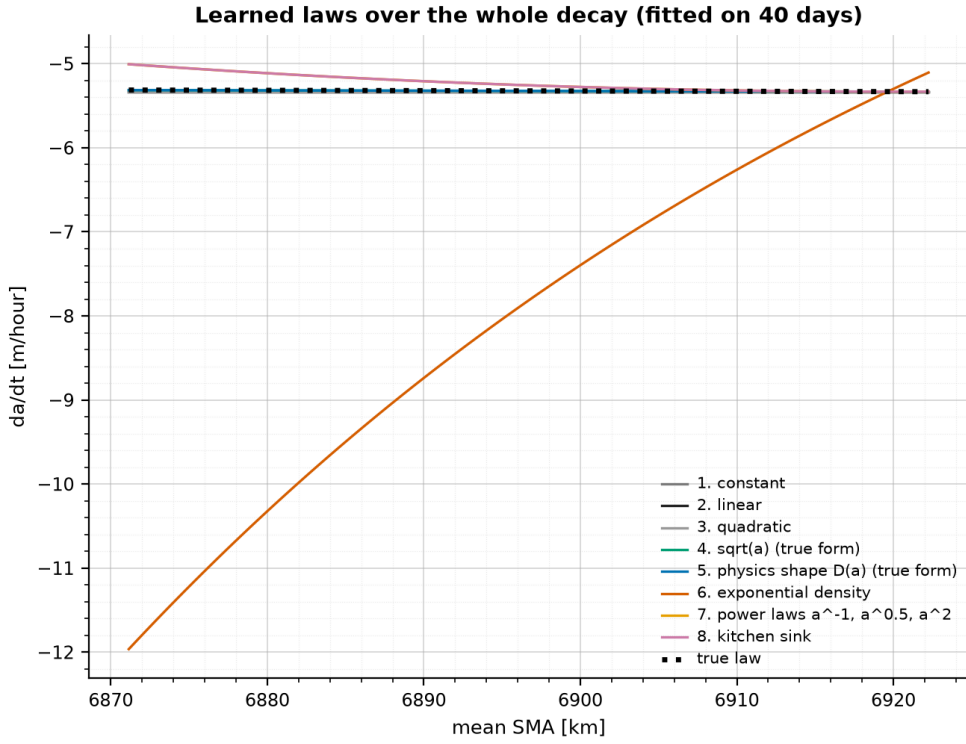


Figure 61: The learned rate laws over the whole decay (fitted on the first 40 days), against the true law $-2c_d\sqrt{\mu a}$.

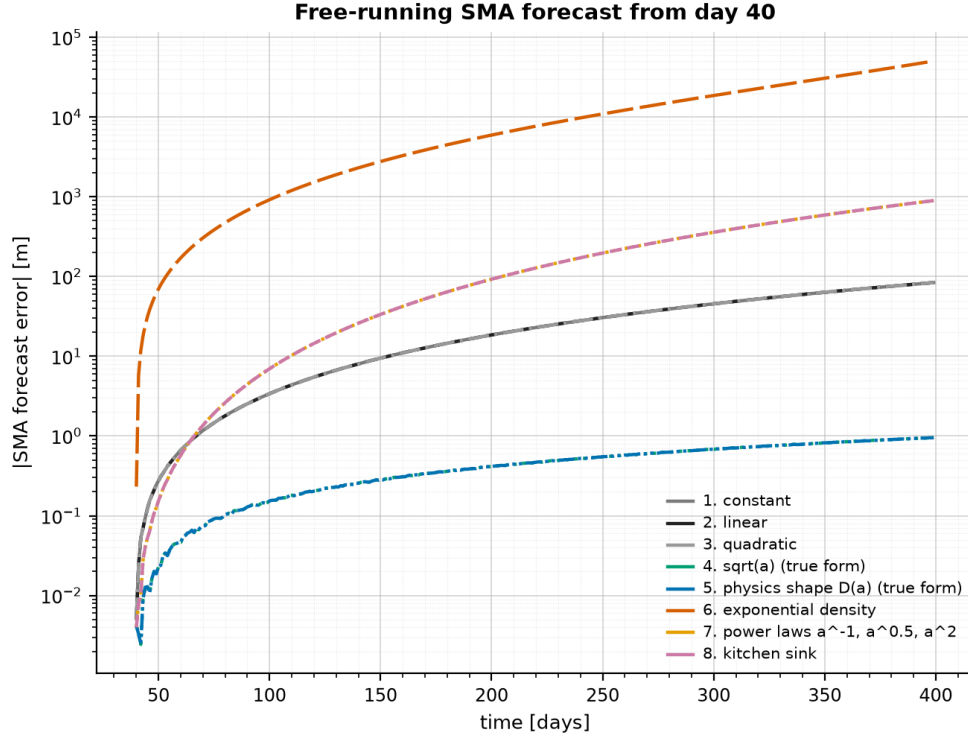


Figure 62: Free-running SMA forecast error of each library.

The true form recovers c_d to 0.01% and the SMA a year ahead to 1 m. The exponential-density library encodes physics absent from this simulator (density rising as the orbit falls) and extrapolates 51 km off: a plausible library is not automatically right, and free-running forecasts, not training fit, decide.

7.5 The equation zoo (step 18)

Ten libraries per data set (cluster constants, polynomials, exponential density, time polynomials, Fourier, a space-weather proxy, proxy $\times$ density, a library without a cluster indicator, a kitchen sink; for GEO harmonics in longitude) were compared on the full-physics, space-weather and GEO data.

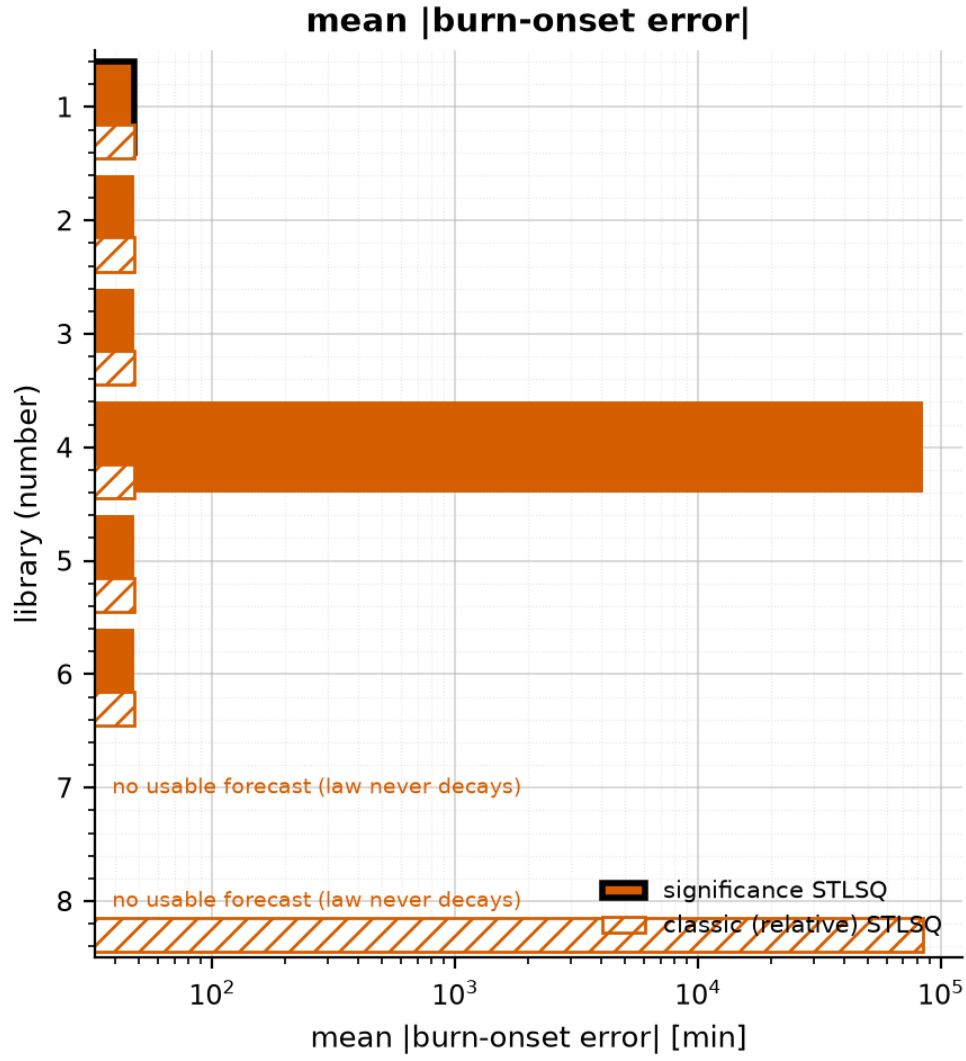


Figure 63: Equation zoo on the full-physics data: test error of each library. Every richer library is pruned back to the same two-term law.

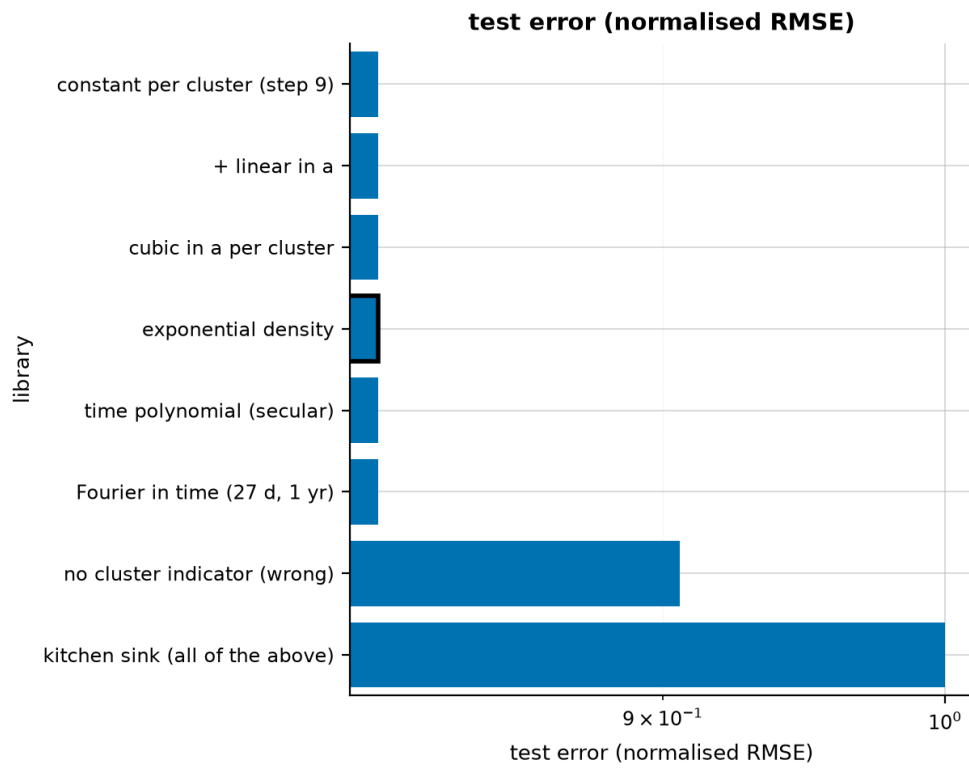


Figure 64: Equation zoo, full physics: normalised test error of the predicted rate per library.

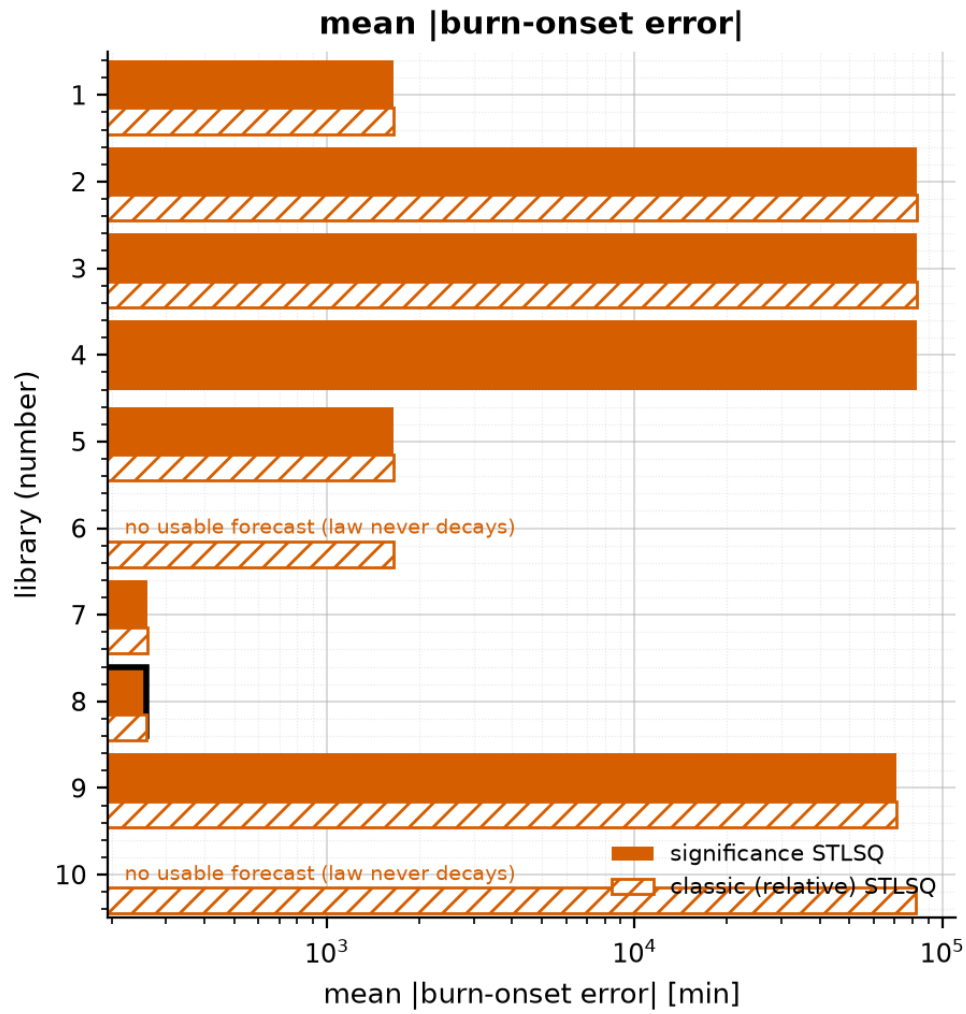


Figure 65: Equation zoo with space-weather drag: only the proxy libraries forecast well (about 260 vs 1640 min).

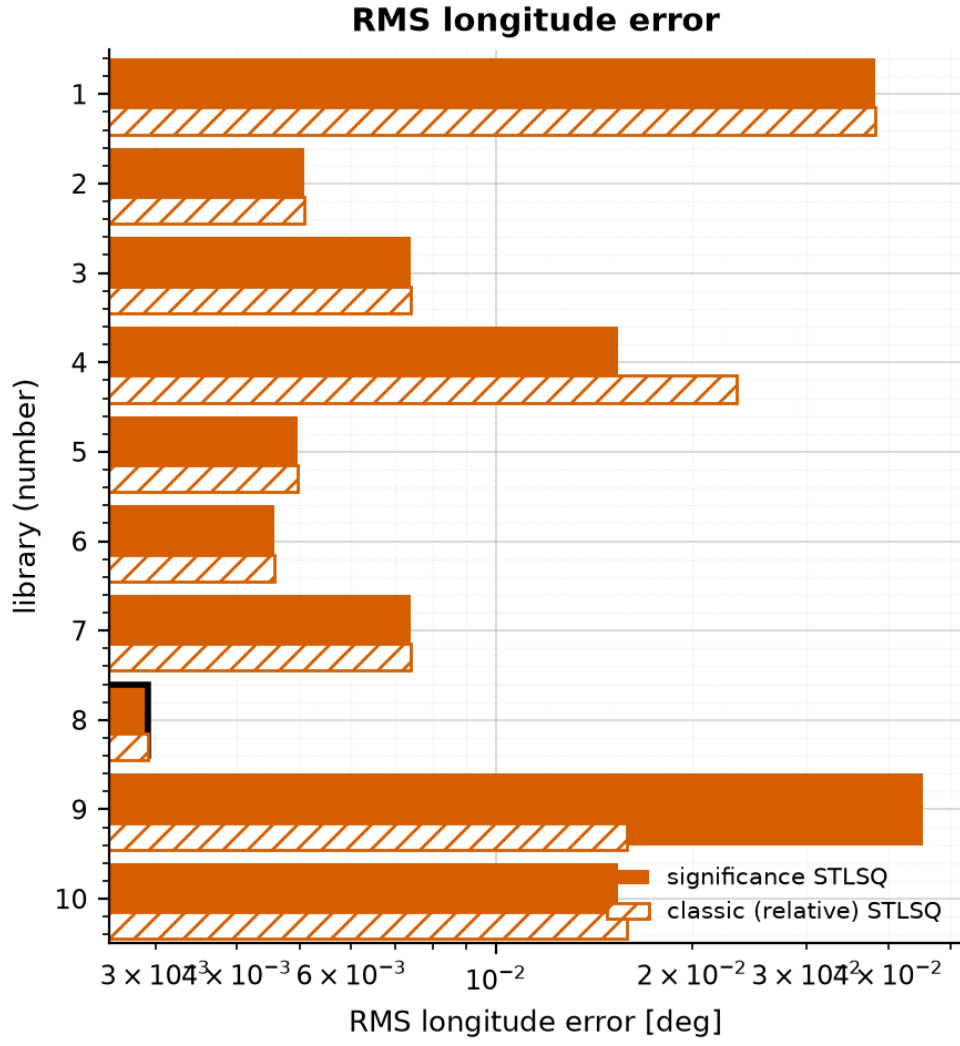


Figure 66: Equation zoo for GEO: the true $\sin 2\lambda$, $\cos 2\lambda$ form forecasts best.

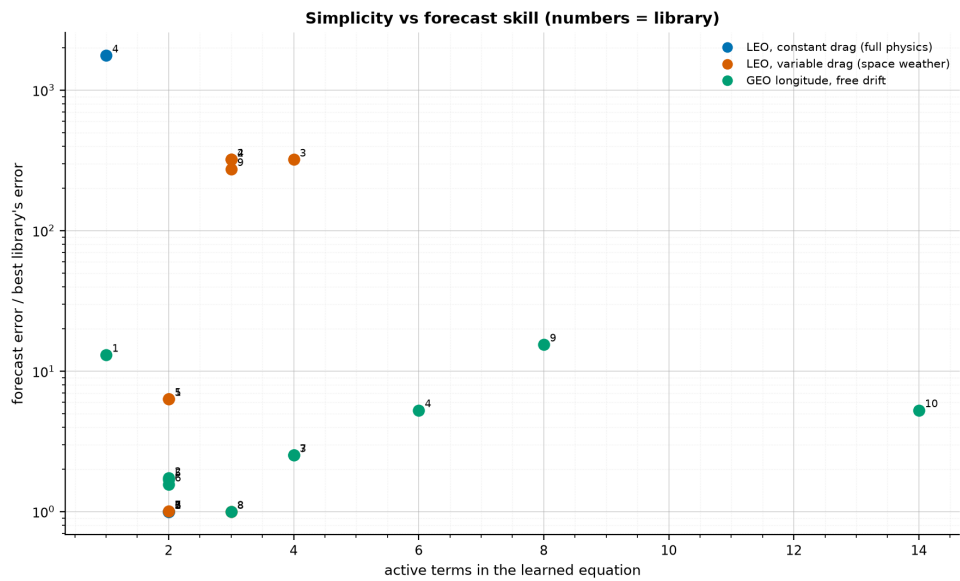


Figure 67: Complexity against error: BIC can prefer an over-fitted library that forecasts worse, so free-running forecasts are the decisive test.

8 Stage 4: forecasting and the whole chain end to end (steps 9 and 20)

8.1 The forecasters

From the end of the last TRAIN burn, with no further data:

Law forecast (SINDy split, joint, BINDy) constant-rate hysteresis: coast time = band / coast rate, burn time = band / net burn rate, with the shapes averaged over TRAIN; onsets every period.

GP coast curve a GP of SMA against time since the last burn ended; a coast ends where the curve reaches the learned lower edge.

Periodic baseline the mean interval between the detected TRAIN burns.

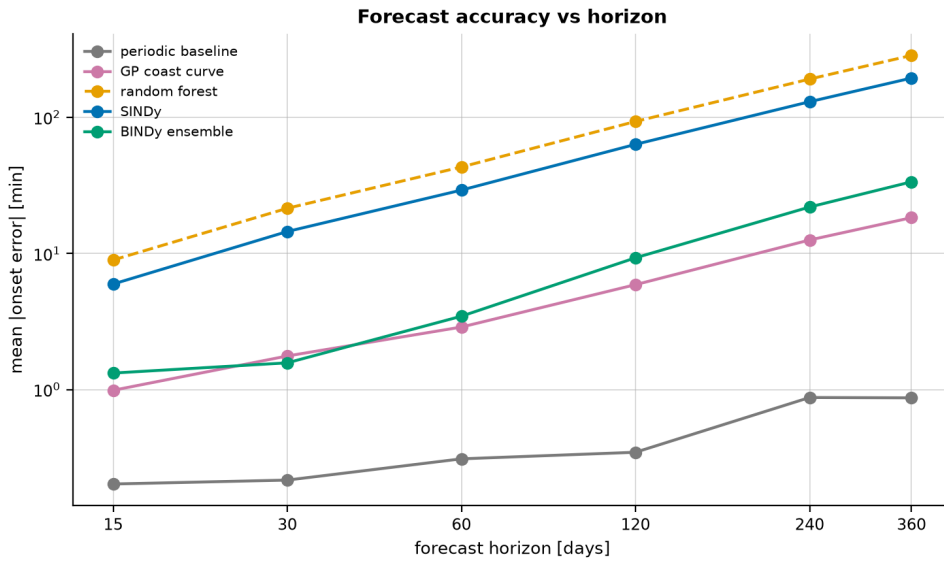


Figure 68: Step 9 forecasters against horizon (true labels for the step 9 laws): periodic 0.9 min, GP coast curve 18 min, BINDy ensemble 34 min, SINDy 195 min and a random forest [21] 286 min at 360 days.

8.2 Stage accuracy and the error budget

The chain is run with everything *learned* (GMM labels, learned c_d , T and edges); then one stage at a time is replaced by the truth. For the edges and rates the “truth” is the effective constant-rate law that reproduces the full run exactly (c_d 0.036% and T 0.16% above the physical values: replacing $D(a, v)$ and $B(a, v)$ by their TRAIN means is a model-form error of the constant-rate law).

Table 5: Accuracy of each stage (6-s data, reference noise, split law D).

Stage	Output	Accuracy
1 natural dynamics	c_d	+0.029% (BINDy $\pm 0.18\%$ at 1σ)
2 detection	burn events	F1 = 1.000 (92/92, 0 false)
3 thrust law	T	-0.005% (BINDy $\pm 0.015\%$)
3 switching law	edges	lower -2.2 m, upper -2.4 m
4 forecast	onsets	2.6 / 22 / 66 min at 15 / 120 / 360 days

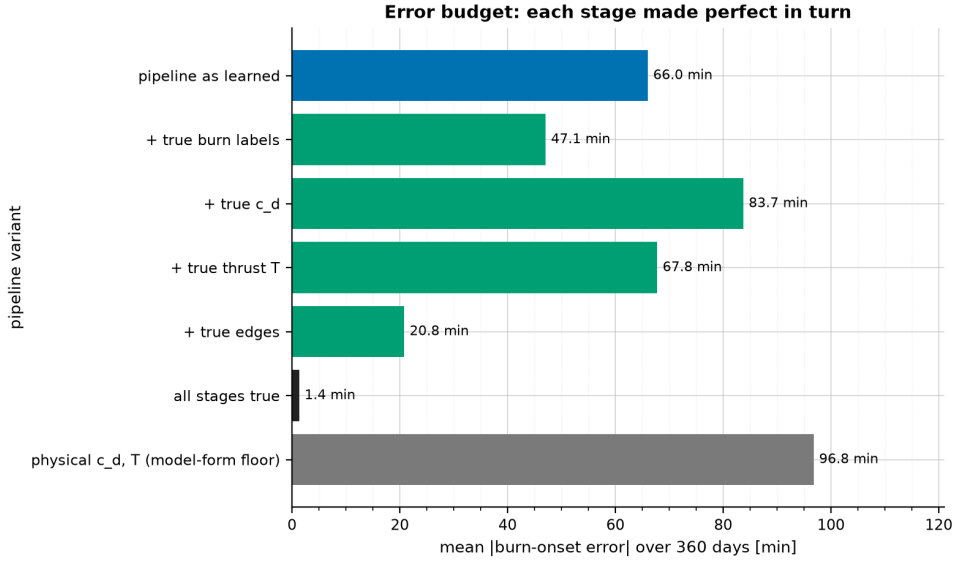


Figure 69: Error budget at 360 days: forecast error with each stage made perfect. With everything true the error is 1.4min (the sanity check); true edges remove two thirds of the error; true labels a third; a true c_d alone makes it *worse*, because the small learned c_d error was partly cancelling the edge error.

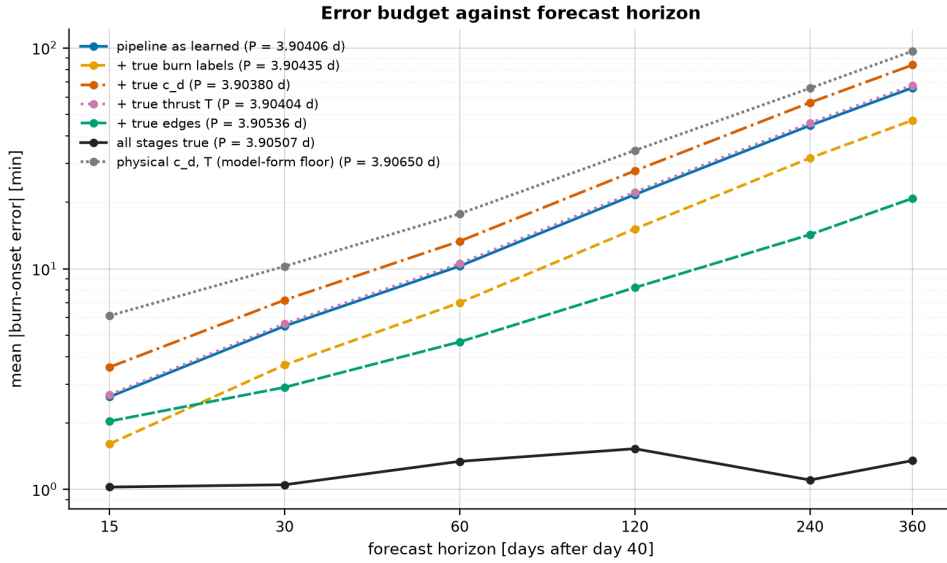


Figure 70: The same error budget against horizon. The error grows linearly (a period error) for every variant except the all-true one.

The edges are learned to about 2 m, but both are 2 m *low*, so the band width is right to 0.2 m and the dominant effect is the phase of the first coast; a 2 m edge error is 0.4% of the band and 22 min of coast.

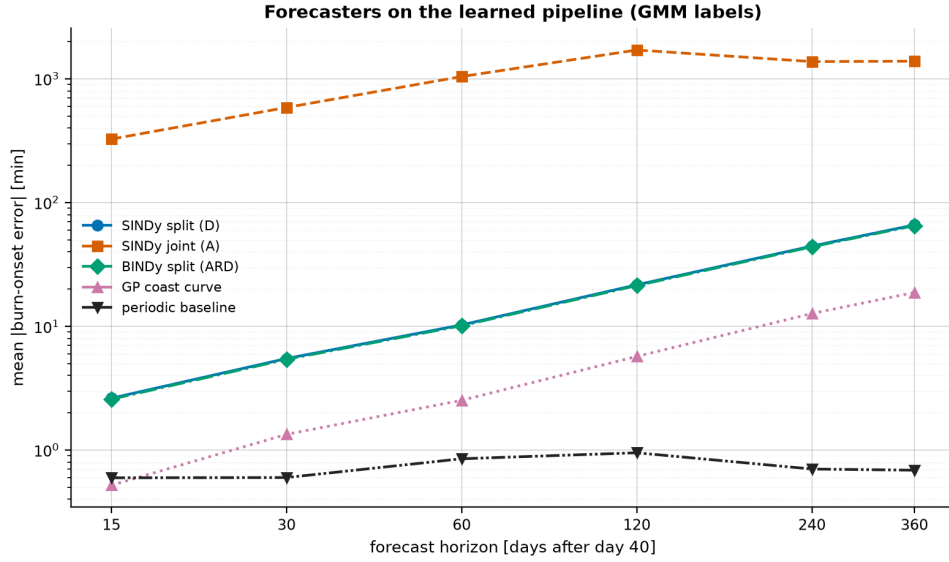


Figure 71: All forecasters on the fully learned chain (GMM labels). The joint regression (A) drifts by 23 hours; the split law and BINDy reach 65–66 min, the GP coast curve 19 min, the periodic baseline 0.7 min (constant drag only).

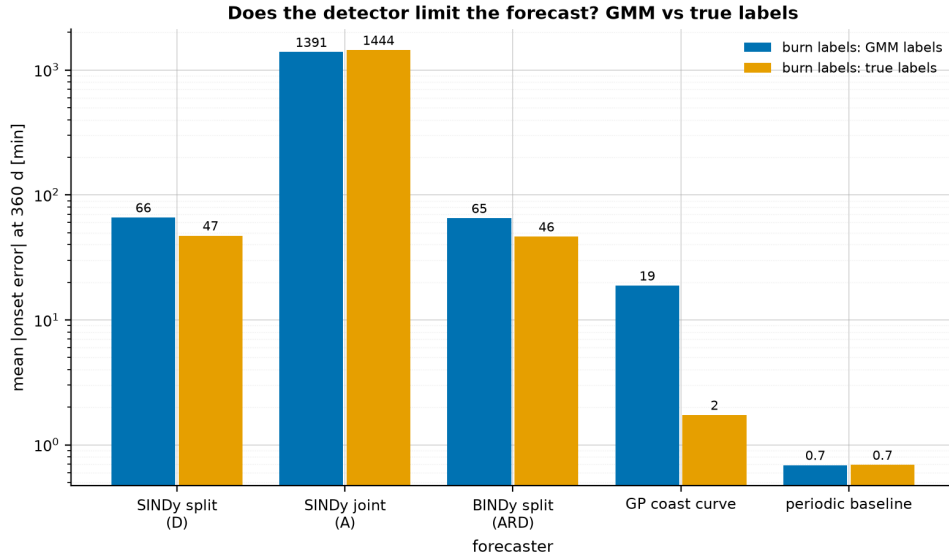


Figure 72: The same forecasters with GMM labels and with true labels. The GMM costs 19 min (split law) and 17 min (GP) at 360 days; the joint regression is biased either way.

8.3 Sampling robustness of the whole chain

The chain was re-run on the same 400 days sampled every 6 s, 30 s, 1, 2, 5 and 10 min (Table 6).

Table 6: The whole chain against the sampling step (law D = SINDy split; GP = GP coast curve).

Step	Feature	F1	c_d err. [%]	T err. [%]	lower edge [m]	upper edge [m]	law (D) 360 d [min]	GP 360 d [min]
6 s	S-G derivative	1.000	+0.029	-0.00	-2.2	-2.4	66	19
30 s	S-G derivative	1.000	+0.065	+0.02	-2.3	-2.6	226	11
1 min	S-G derivative	1.000	+0.074	-0.07	-2.3	-2.6	238	150
2 min	propagation	1.000	+0.007	+0.04	-3.3	-2.7	398	479
5 min	propagation	1.000	+0.071	-4.81	+5.2	-3.1	1455	1361
10 min	propagation	1.000	+0.077	-8.65	+13.9	-4.4	1371	1352

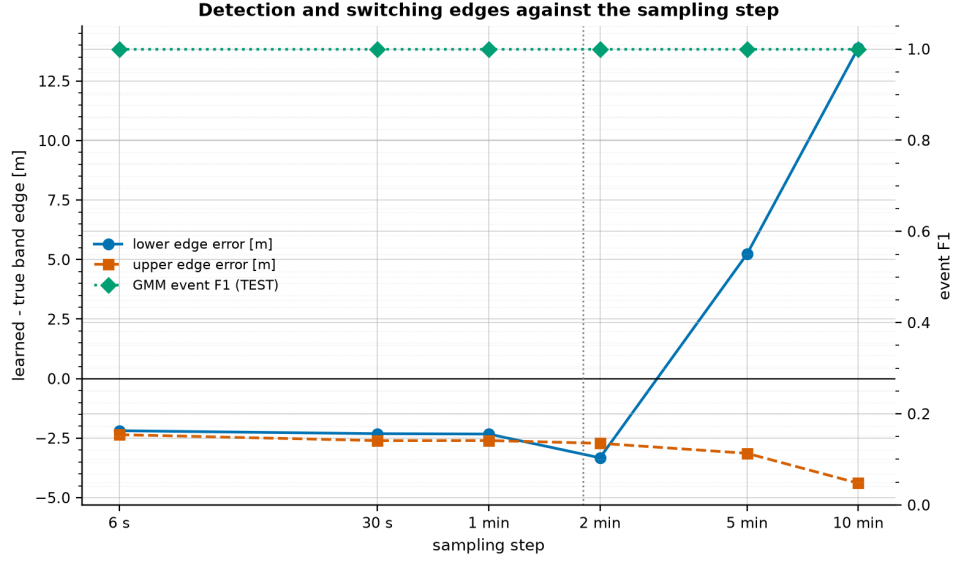


Figure 73: GMM event F1 (right axis) and the learned band edges (left) against the sampling step. Detection stays perfect; the edges drift from 5 min because only a few samples fall on each 23-min burn.

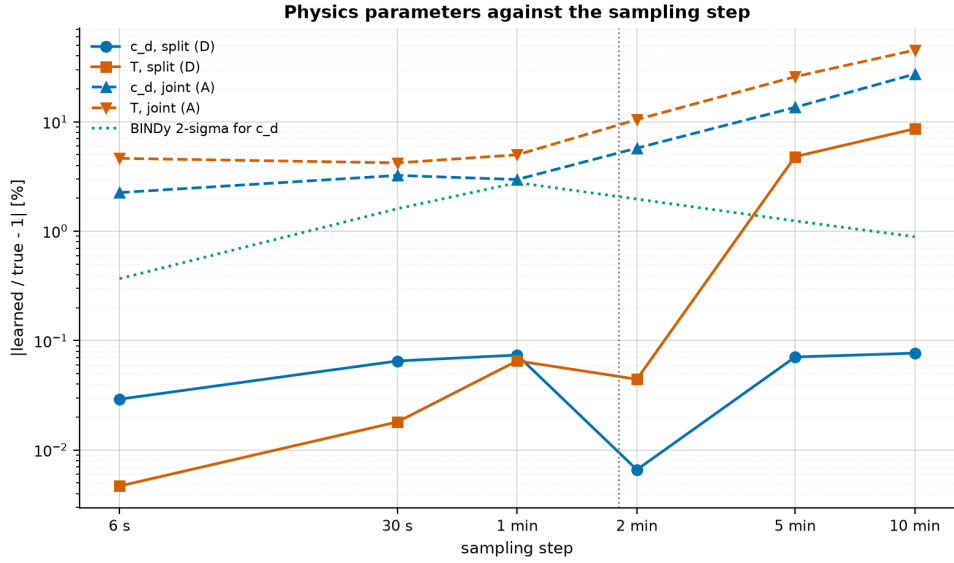


Figure 74: c_d and T errors against the sampling step. c_d stays within 0.08% at every step; T needs clean burn samples, which disappear above 2 min, when it falls back on the SMA gained per burn divided by the sampled burn duration (uncertain by $\pm$ one step).

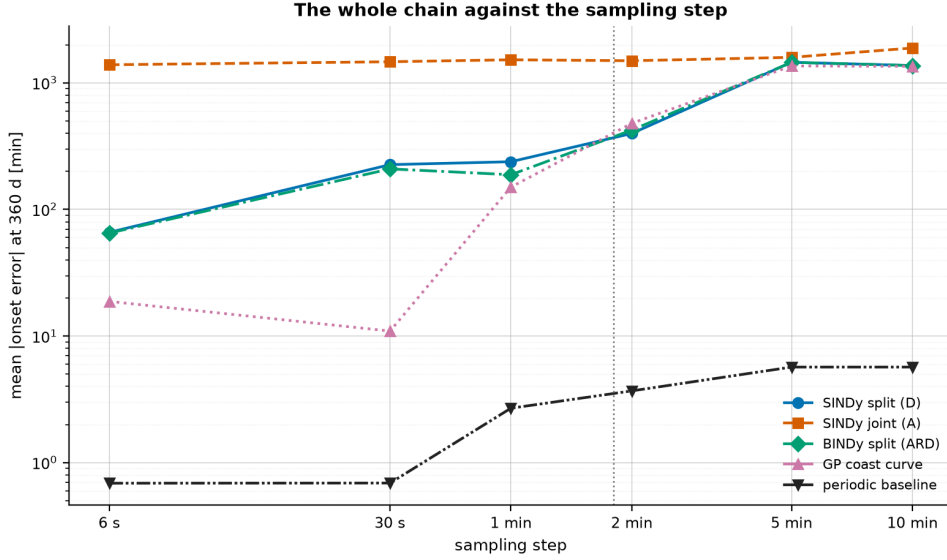


Figure 75: Year-ahead forecast error of every forecaster against the sampling step.

What limits the forecast

With constant drag the learned law is limited by the *edges* and the *coast rate*: 0.03% in c_d or 2 m in an edge each give tens of minutes after a year. Coarser sampling hurts through the edges (burns sampled by a few points), not through c_d . The periodic schedule is excellent here only because drag is constant; the next section removes that assumption.

9 Towards operations: extensions to the brief (steps 10–17)

9.1 A calibrated mean-element model

Hundreds of scenarios need a fast model. The one the controller follows is

$$\dot{a} = -k w(t) e^{-(a-a_{\text{ref}})/H} + T s, \quad s : 0 \rightarrow 1 \text{ at } a \leq a_L, 1 \rightarrow 0 \text{ at } a \geq a_U, \quad (9)$$

with $k = 5.335 \text{ m/h}$, $T = 1311 \text{ m/h net}$, $H = 60 \text{ km}$ and the space-weather factor $w(t)$, calibrated on the full run (burn period within 0.01%).

9.2 Step 10: space-weather drag

$w(t)$ combines a 27-day solar-rotation wave, a solar-cycle trend and eight storms (density $\times 2-4$); an observable daily proxy (F10.7/Ap-like [25]) has 5% noise.

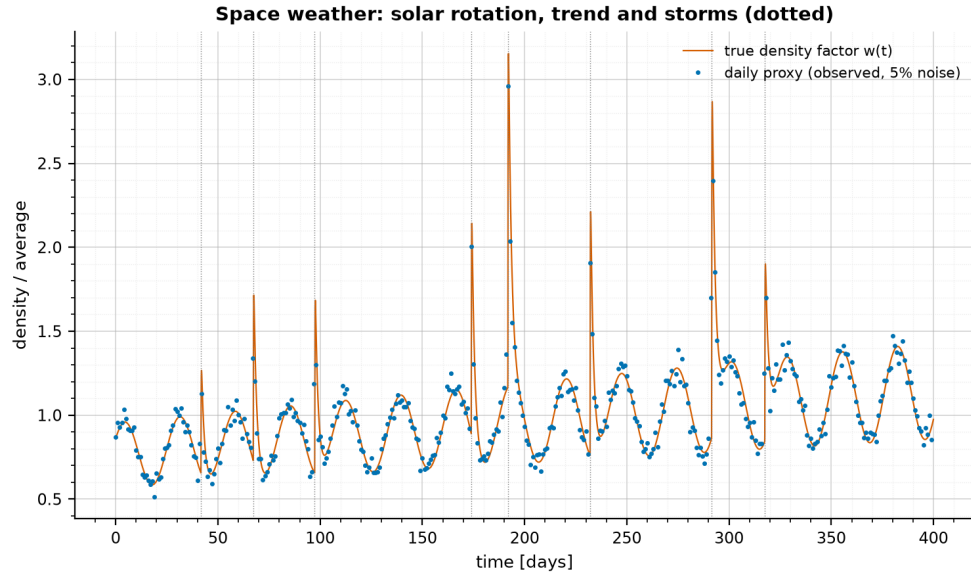


Figure 76: The density factor $w(t)$ and its noisy proxy over 400 days.

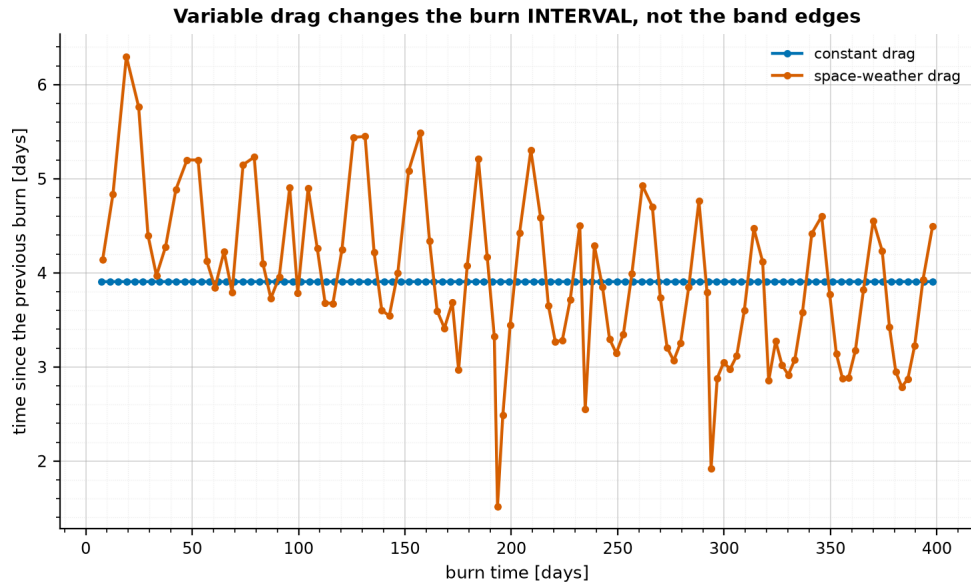


Figure 77: Time between burns against burn time: variable drag changes the interval, not the band edges.

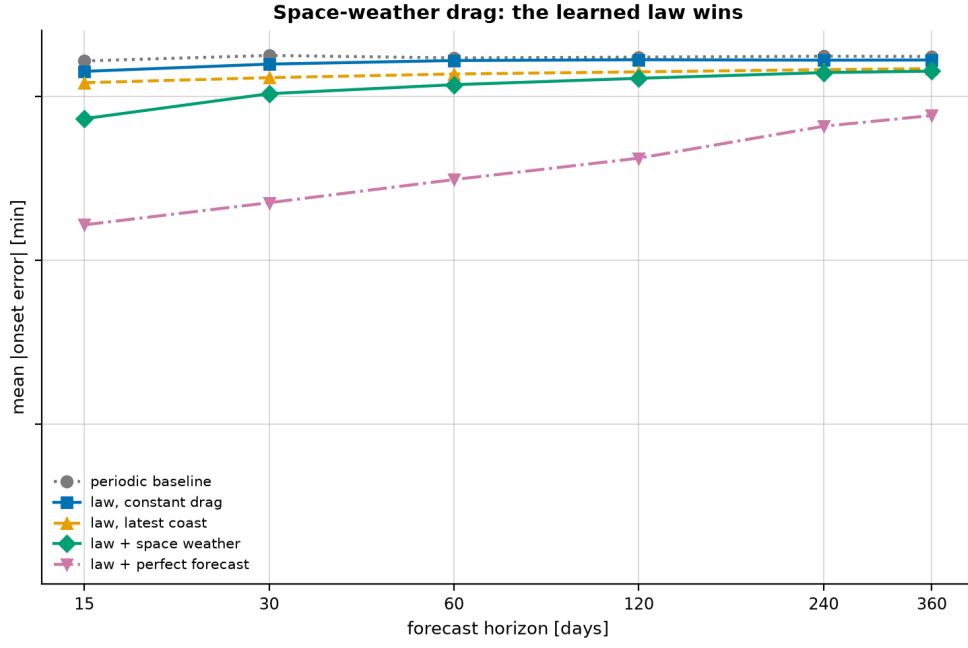


Figure 78: Forecast error against horizon with space-weather drag.

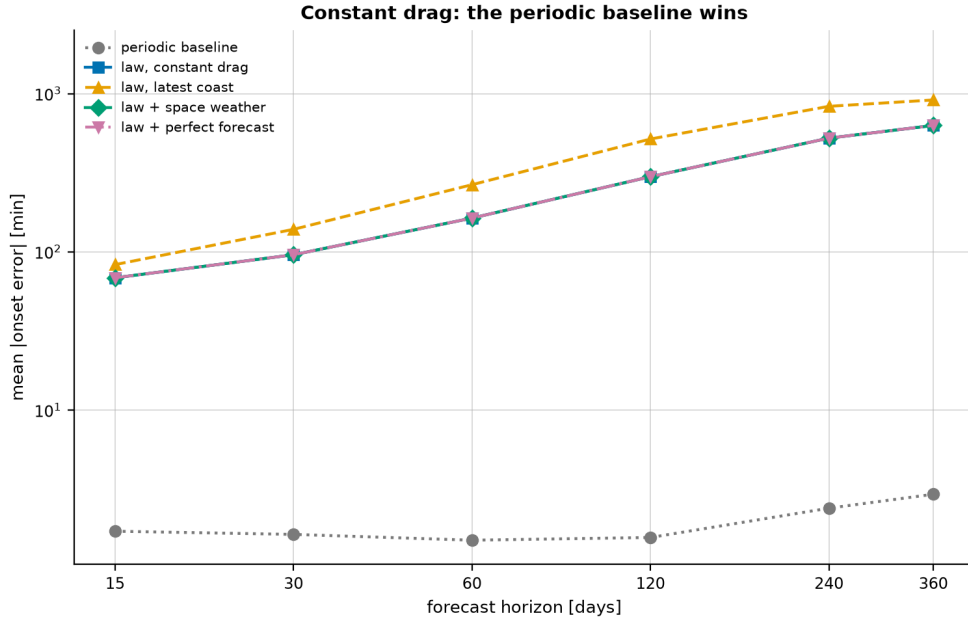


Figure 79: The same forecasters with constant drag: the periodic schedule wins.

Table 7: Mean |onset error| [min] with constant and with space-weather drag.

Forecaster	const. drag 15 d	const. drag 360 d	space weather 15 d	space weather 120 d
periodic baseline	2	3	1642	1732
law, constant drag	68	630	1418	1670
law + space-weather proxy	68	630	730	1286
law + perfect forecast	68	630	164	419

Variable drag changes the interval between burns but not the band edges: the periodic schedule loses its phase, while the learned law tracks it as well as drag can be forecast.

Sampling robustness. With SMA measurements every 10 min, 30 min, 1 h, 2 h or 4 h, all 102 burns are still detected (the burn is a 500 m step), and the space-weather law stays at 730–861 min at 15 days against 1642–1703 min for the periodic schedule (Figure 80). Only the perfect-forecast bound degrades with cadence (164 min at 10 min to 353 min at 4 h), because the starting SMA of each forecast is fitted from fewer samples.

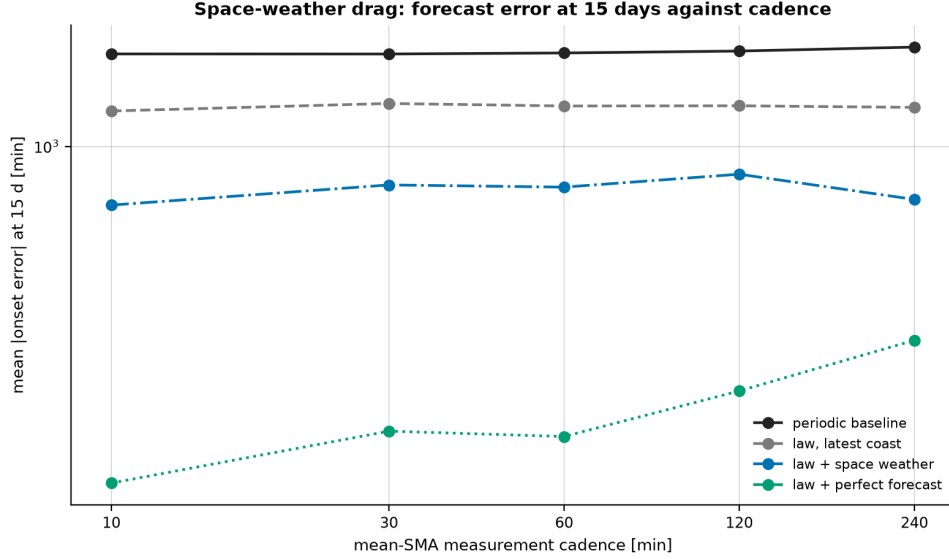


Figure 80: Space-weather forecasters at 15 days against the measurement cadence.

9.3 Step 11: realistic tracking passes

Eight-minute passes of 12-s positions (10 m to 1 km noise), 1–8 per day, are fitted by batch least-squares orbit determination; burns are found in the gaps from the SMA jump minus the drag decay.

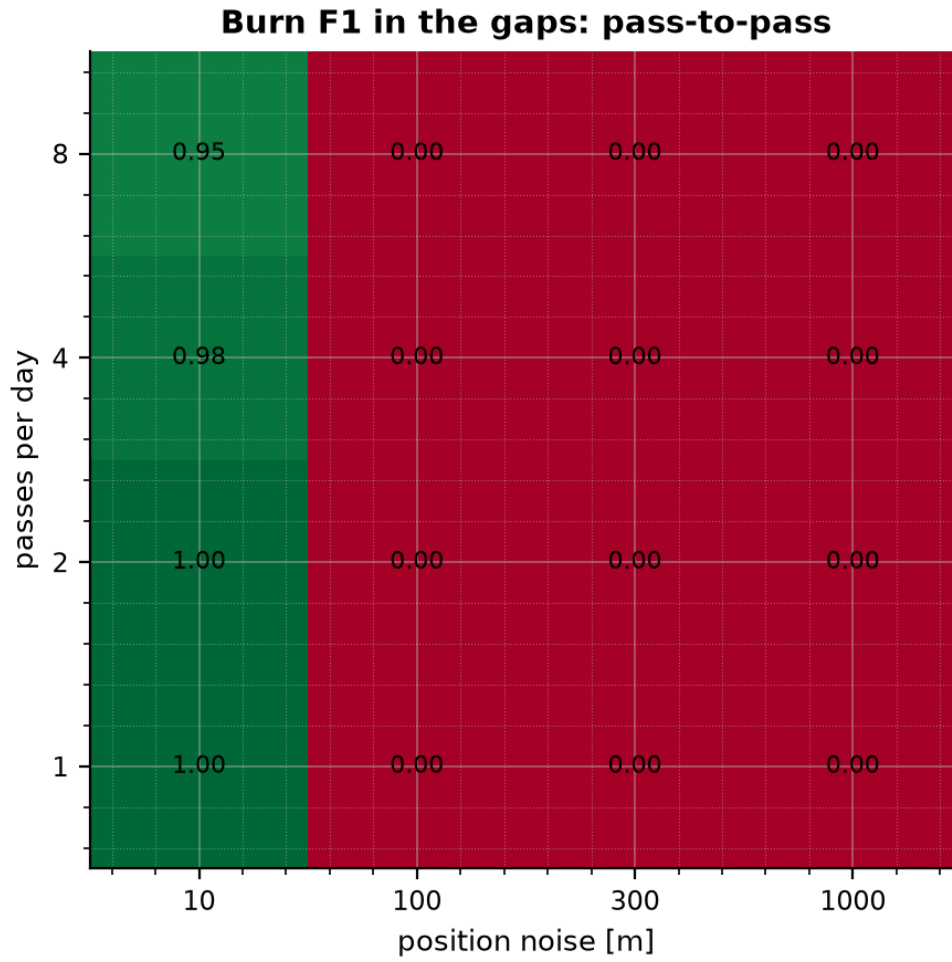


Figure 81: Burn-detection F1 over passes per day $\times$ position noise. At 10 m $F1 \approx 1$ for 1–8 passes a day; at 100 m only a multi-pass test finds burns (0.47 at 8 a day); at 300 m the 500 m burn is buried.

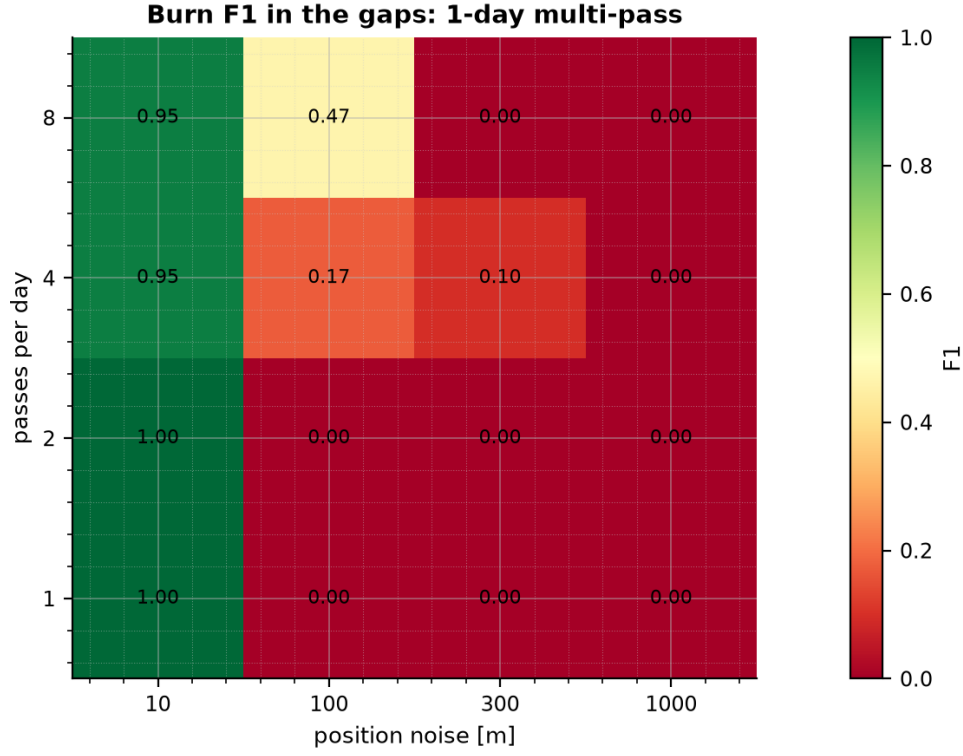


Figure 82: The same grid for the 1-day multi-pass test, which averages all passes a day before and after.

Calculation check. One pass gives the SMA to about $2\sigma_r$ (19 m at 10 m noise); a gap test with threshold $\approx 3\sqrt{2}\times$ that is 54 m, ten times below the 475 m jump — detection is easy. At 100 m the floor is 320 m, comparable with the jump, as observed.

9.4 Step 12: conjunction screening

An SMA error δa changes the mean motion by $\delta n = -\frac{3}{2}(n/a)\delta a$, so the along-track error grows as $\delta s = \frac{3}{2}n \delta a t$. Screening uses $P_c \approx \pi R^2 \mathcal{N}(\text{miss}; 0, \Sigma)$, $R = 20$ m.

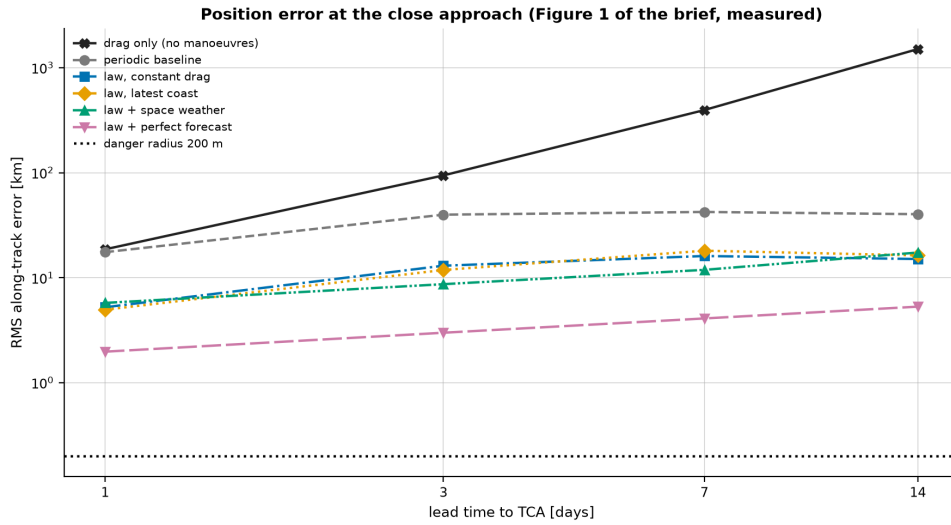


Figure 83: RMS along-track error against prediction time: drag-only 19 km at 1 day and 396 km at 7 days; learned law with space weather 5.8 / 11.9 km; perfect forecast 2.0 / 4.1 km.

Calculation check. Missing one full-band burn ($\delta a = 0.498$ km) gives $\frac{3}{2}n \delta a = 2.95$ km/h, 495 km in a week — the brief’s “hundreds of km a week”. The simulated drag-only RMS (396 km) is below this bound, as it must be: the burn falls at a random point of the week.

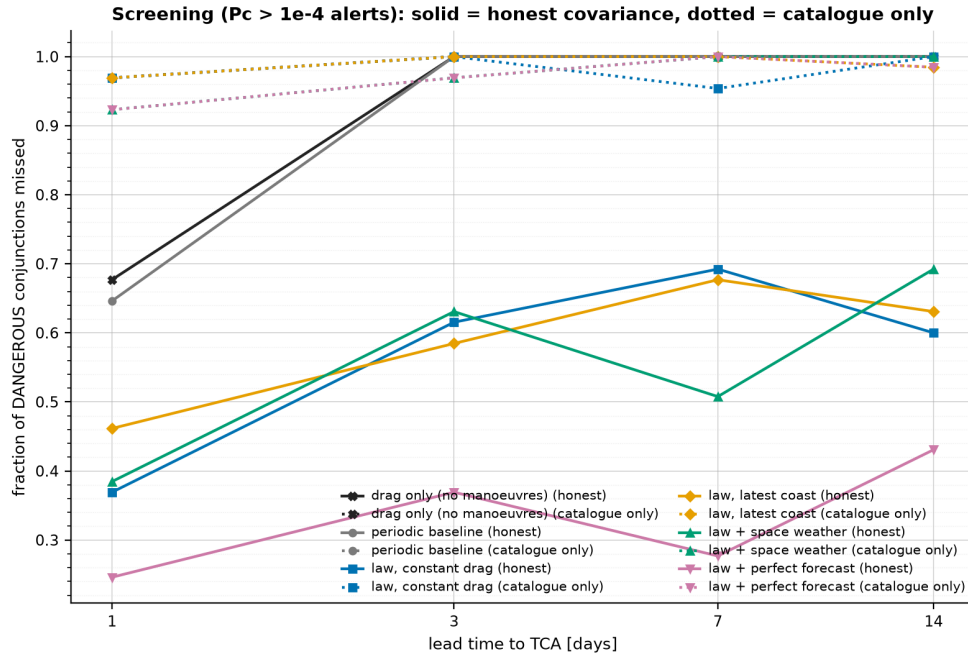


Figure 84: Fraction of dangerous conjunctions missed by screening at 1 day: 68% drag-only, 25% with the best manoeuvre model and an honest covariance.

9.5 Step 13: sensor tasking and track association

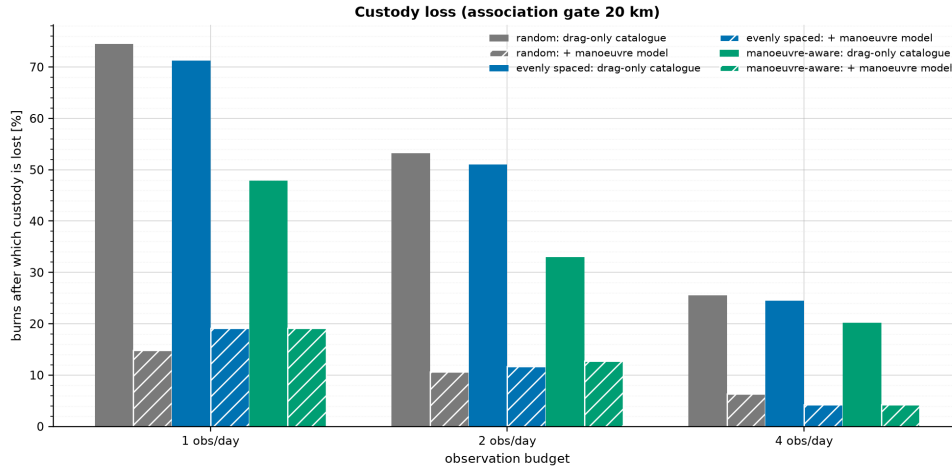


Figure 85: Custody lost after a burn with a 20 km association gate: 74% with one random observation a day, 48% if it is scheduled after the predicted burn, 15–19% if the catalogue also carries the predicted burn; 20% and 4% with four observations a day.

9.6 Step 14: several control laws

Orbit raising, the deadband, a band change at day 150 and six collision-avoidance burns on one satellite.

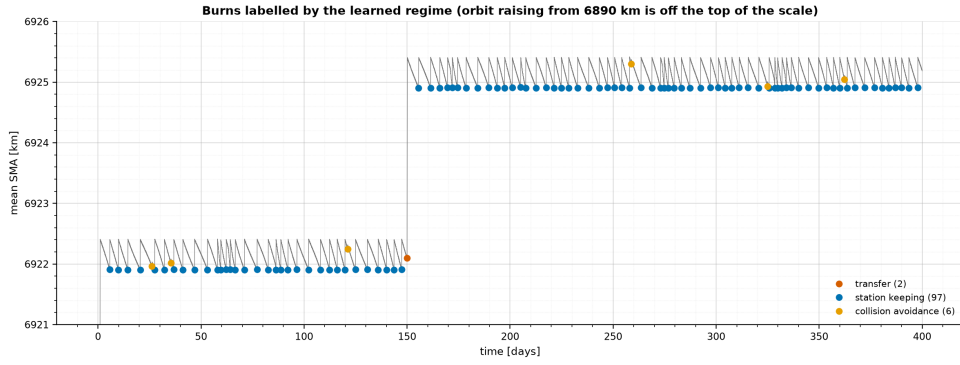


Figure 86: The regime timeline: every one of 105 burns is labelled correctly by rules (104/105 by an unsupervised GMM), and the band change is found as a changepoint at day 150.0 [22].

9.7 Step 15: a fleet of 40

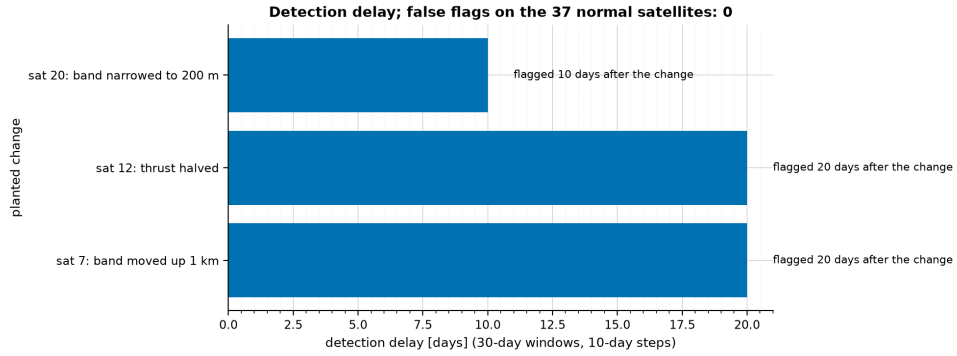


Figure 87: Pattern-of-life change detection in a fleet of 40: the three planted changes (band raised, thrust halved, band narrowed) are flagged 10–20 days later, with no false flags.

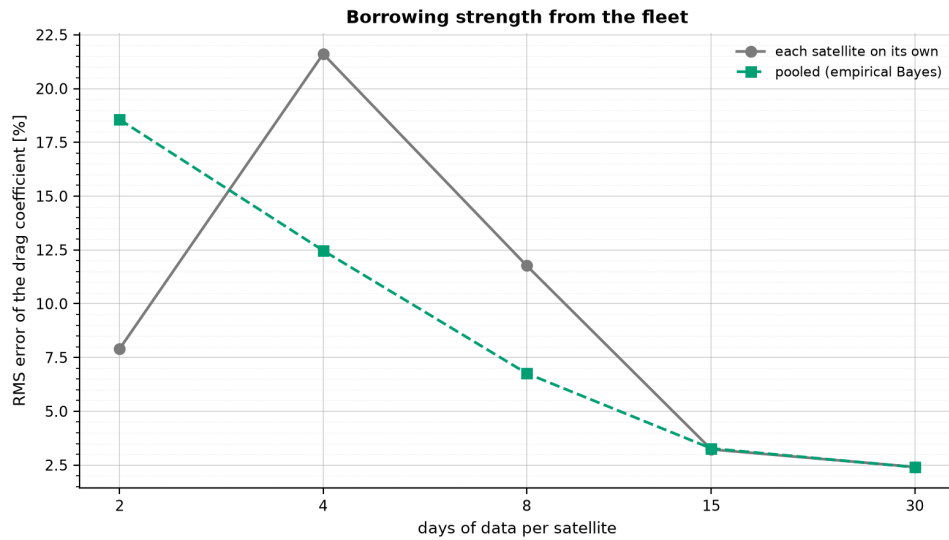


Figure 88: Empirical-Bayes pooling of the drag rate: with 4 days of data the error falls from 21.6% to 12.5%, with 8 days from 11.8% to 6.8%.

9.8 Step 16: calibrated intervals

Split conformal prediction [23, 24]: the calibration errors (origins on days 40–200) give, per lead-time bucket, the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile q , and the interval is $\pm q$.

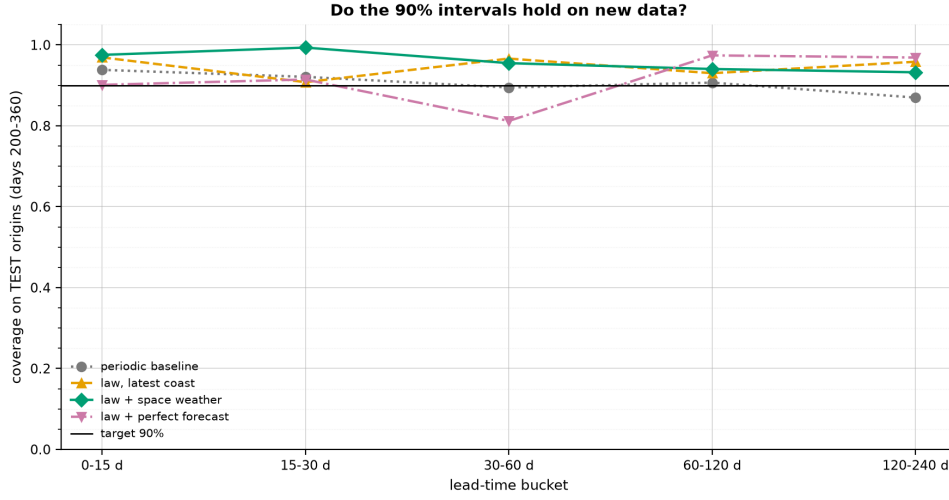


Figure 89: Coverage of the 90% intervals on new origins (days 200–360): 0.93–0.99 for the space-weather law, above the $1 - \alpha$ guarantee.

9.9 Step 17: geostationary station keeping

The GEO longitude obeys $\ddot{\lambda} = -A \sin 2(\lambda - \lambda_s)$. SINDy recovers $A = 0.00171^\circ/\text{day}^2$ (true 0.0017) and $\lambda_s = 74.98^\circ$ (true 75.07°) from a free-drift phase.

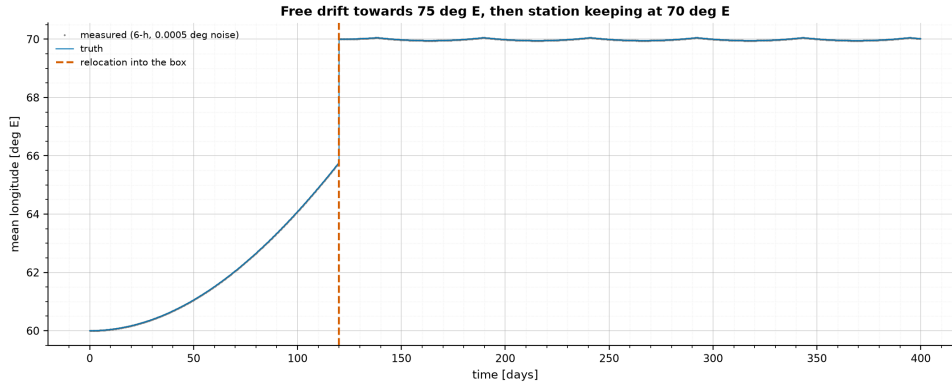


Figure 90: GEO east-west station keeping: longitude inside the $\pm 0.05^\circ$ box.

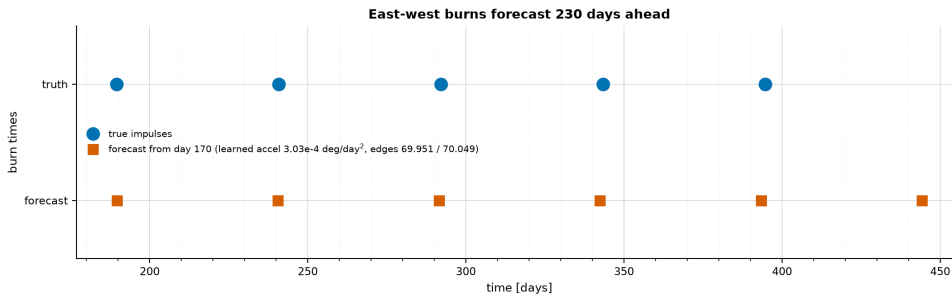


Figure 91: East-west burn forecast errors: 0.09–1.34 days up to 225 days ahead.

10 Ranking the methods

Each method is scored 1–5 on accuracy (at the reference noise and the best setting), robustness (to noise and to sampling), cost, and whether it reports its own uncertainty; the scores are judgements made from the numbers in this report, and the rank is by the sum within each stage.

Table 8: Methods ranked within each stage of the chain (A = accuracy, R = robustness, C = cost, U = uncertainty).

Method	Evidence	A	R	C	U	Σ
<i>Stage 1: drag coefficient</i>						
1 EKF	0.01% at 30 d on 10-min data	5	4	3	5	17
2 energy method / mean-SMA coast fit	0.03%, 0.03–0.08% at every step	5	4	5	2	16
3 shooting	0.04–0.05%	5	3	1	3	12
4 force spectroscopy	drag 0.5%, Sun not separable	3	2	5	1	11
5 FD regression	6% at 10 m noise	2	1	5	2	10
<i>Stage 2: burn detection</i>						
1 GMM / Bayesian GMM	F1 = 1 at 0.25–8 $\times$ noise (6 s); F1 $\geq$ 0.9 from 6 s to 20 min up to 2 $\times$ noise	5	4	4	4	17
2 MAD threshold	F1 = 1, threshold to choose	5	4	5	1	15
3 GP coast anomaly / DBSCAN	F1 = 1 held out	5	3	2	3	13
4 Isolation Forest	0.71–0.85	3	3	4	1	11
5 k -means	0.001 with noise	1	1	5	1	8
<i>Stage 3: thrust law</i>						
1 SINDy/BINDy on clean burn arcs (D)	T to 0.005%, $\pm 0.015\%$	5	3	5	4	17
2 GP through stacked burns	plateau 0.999, duration exact, shape	4	4	3	5	16
3 joint regression (A)	T -4.6 to -7.6%	2	2	5	3	12
<i>Stage 3: natural law and switching</i>						
1 split learning (D) + hysteresis edges	c_d 0.03%, edges 2 m, balanced accuracy 1.0	5	4	5	4	18
2 natural law given (B)	26 min at 360 d	5	4	5	2	16
3 thrust given (C)	302 min at 360 d	4	4	5	3	16
4 joint (A)	1668 min	2	2	5	3	12
5 memory-less switching $s = f(a)$	balanced accuracy 0.50	1	1	5	1	8
<i>Stage 4: forecasting</i>						
1 law + space-weather proxy	730 min vs 1642 periodic (variable drag)	4	4	4	4	16
2 SINDy/BINDy split law (constant rate)	66 min at 360 d, accepts a drag proxy	3	3	5	3	14
3 GP coast curve	19 min (constant drag), weak at ≥ 1 -min steps	4	2	3	4	13
4 periodic schedule	0.7 min constant drag, 1642 min variable	3	1	5	1	10
5 random forest / joint law	286 / 1391 min	1	2	4	1	8

Notes on the forecasting rank. The periodic schedule is the most accurate when drag is constant but the least robust: under space weather it is the worst, which is the realistic case, so its robustness score is 1. The GP coast curve beats the constant-rate law on accuracy with fine data but degrades from 1-min sampling; the law is the form that accepts a space-weather proxy, which is what makes the top-ranked forecaster.

Pipeline recommendation. Savitzky–Golay (or one-step propagation above 2 min) $\rightarrow$ GMM detection $\rightarrow$ drag on clean coast arcs (mean SMA, or an EKF on sparse data) $\rightarrow$ thrust on clean burn arcs (with a GP for the shape) $\rightarrow$ hysteresis edges $\rightarrow$ a law forecast driven by a space-weather proxy, with conformal intervals.

11 Checking the mathematics

Every headline number was recomputed from a closed-form expression and compared with the simulation or the analysis (Table 9, `deadband/report_checks.py`). All agree within 1% or obey their bound; the residual differences have known causes (first-order J_2 theory, the 30-s throttle ramps, the 5-m smooth switch).

Table 9: Closed-form checks. “bound” rows test an inequality rather than an equality.

Check	Formula	Unit	Formula	Measured	Diff.
coast decay rate	$\dot{a} = -2c_d\sqrt{\mu a}$	m/h	5.333	5.335	0.05 % ✓
J2 nodal regression	$\dot{\Omega} = -\frac{3}{2}nJ_2(R_e/a)^2 \cos i$	deg/day	-4.503	-4.505	0.04 % ✓
osculating SMA swing (J2)	$3J_2R_e^2 \sin^2 i/a$	km	12.17	12.22	0.37 % ✓
J2 Nyquist sampling limit	$T_{\text{orb}}/4$	min	23.88	23.88	0.02 % ✓
delta-v per burn	$\Delta v = v \Delta a/(2a)$	m/s	0.2741	0.273	0.40 % ✓
SMA climb rate in a burn	$\dot{a} = 2aT/v - k$	m/h	1.308×10^3	1.311×10^3	0.17 % ✓
burn duration	$t_b = \Delta v/T$	s	1.370×10^3	1.368×10^3	0.17 % ✓
burn period	$P = w/k + t_b$	days	3.905	3.905	0.00 % ✓
band width	$a_{\text{up}} - a_{\text{low}}$	m	500	498	0.40 % ✓
yearly delta-v budget	$N \Delta v$	m/s	28.07	27.84	0.82 % ✓
missed-burn along-track drift (7 d)	$\frac{3}{2}n \Delta a t$	km	495.3	396	bound $\leq$ ✓
forecast error, scenario C (360 d)	$\frac{1}{2}H \Delta P/P $	min	298.9	301.5	0.86 % ✓
forecast error, scenario D (360 d)	$\frac{1}{2}H \Delta P/P $	min	259.8	262	0.85 % ✓
stacking noise (1 $\rightarrow$ 4 burns)	$\sigma/\sqrt{N}$	m/s ²	4.367×10^{-6}	3.936×10^{-6}	10.96 % ✓
FD c_d error linear in noise (1 $\rightarrow$ 10 m)	$\propto \sigma_r$	%	14.14	14.16	0.12 % ✓
conformal coverage	$\geq 1 - \alpha$		0.9	0.9593	bound $\geq$ ✓

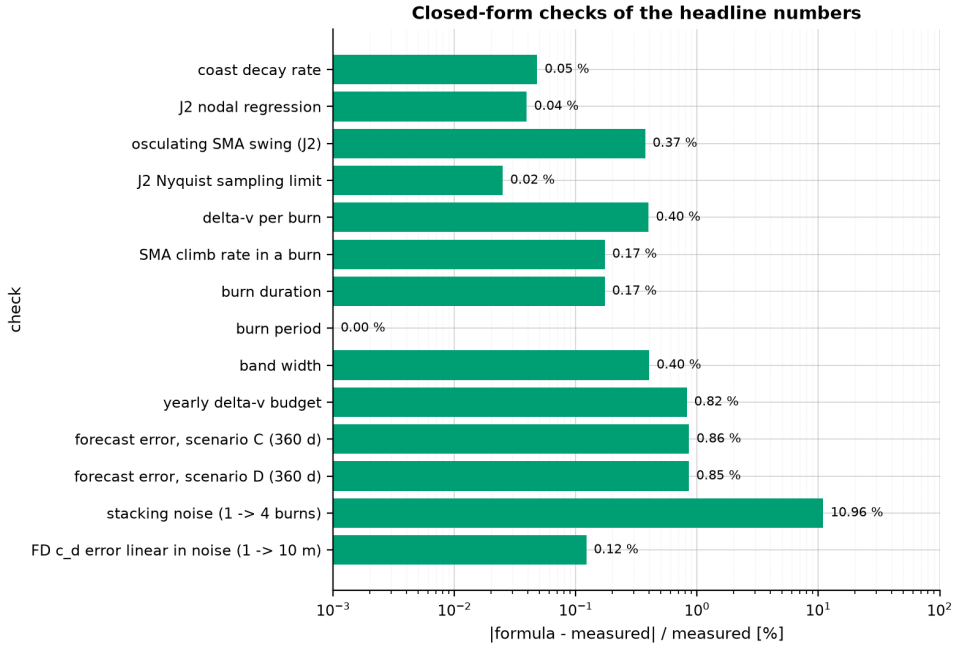


Figure 92: Relative difference between each formula and the measured value (log scale).

12 How well the project answers the brief

Table 10: Brief requirements and the evidence in this report.

Brief requirement	Evidence	Section
Figure 1 (deadband, miss distance)	reproduced; drift bound 495 km/week checked	2, 9
Detect past manoeuvres	92/92 burns, no false events, 6 s–10 min, 0.25–8× noise; tracking passes at 10 m	5, 9
Reconstruct the control law	edges to 2 m, c_d and T to <0.03%, thrust profile, hysteresis, several regimes, GEO	6, 7
Predict future manoeuvres	yes, with error budget; beats a periodic schedule under space weather; conformal intervals	8, 9
Screening / association / tasking	missed conjunctions 68→25%, lost custody 74→15%	9
Robustness	noise, data length, sampling step for every stage	all
Real data	OEM reader and tracking-pass simulation; no real ephemerides yet	9

12.1 Limitations

- Steps 1–9 and 19–20 use full state vectors with 0.1 m / 0.5 mm/s noise, far better than real tracking; step 11 shows how fast detection degrades with realistic passes.
- Constant drag in the 400-day run flattens the periodic schedule; step 10 corrects this with modelled, not measured, space weather.
- Steps 10–16 use the calibrated mean-element model, not full integration.
- No real ephemerides were analysed (an account with a data provider is needed); angles-only orbit determination is not implemented.
- The method ranking is a judgement from the numbers above, not an independent benchmark.

12.2 Future work

Run the chain on real Starlink ephemerides (CCSDS OEM [26]); replace the constant-rate law by the event-driven law of step 10 inside the end-to-end chain; learn the density model jointly with the control law; and couple the manoeuvre forecasts to a real conjunction screening and sensor-tasking loop.

Use of Generative AI Tools

Claude (Anthropic), used through Claude Code, assisted with writing and refactoring the Python code, running the studies, producing the figures and drafting this report. All methods, numbers and conclusions were checked against the code and its outputs (`outputs/results.json`), the unit tests and the closed-form checks of Section 11; responsibility for the content remains with the author.

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A Running the code

```
python main.py                # all 32 stages -> outputs/ (~40 min)
python main.py --list         # the stages, in run order
python main.py --stages step20 # only some stages
python -m unittest discover tests # unit tests
```

Library code is in `deadband/` (no plotting). Each stage in `stages/` draws its figures into `outputs/report/stepN_*/`, with single panels in `panels/`, and returns its numbers for `outputs/results.json`.

B Glossary

SMA

semi-major axis; *osculating* from the two-body energy, *mean* including the J_2 potential.

Deadband, hysteresis

the allowed SMA band; the controller’s memory of whether it is burning.

GMM, EM

Gaussian mixture model, expectation–maximisation; responsibility = posterior of a component.

Event F1

harmonic mean of event precision and recall.

SINDy / STLSQ

sparse identification of nonlinear dynamics / sequentially thresholded least squares.

BINDy / ARD

the Bayesian variant / automatic relevance determination.

GP Gaussian process; EKF = extended Kalman filter.

P_c collision probability; conformal prediction = distribution-free calibrated intervals.